



ISSN 2536-569X | eISSN 2536-5703

Journal of Anthropology of Sport and Physical Education

www.jaspe.ac.me



OCTOBER 2020

**VOL.4
No.4**





**Journal of Anthropology of Sport
and Physical Education**

Editor-in-Chief

Bojan Masanovic | University of Montenegro, Montenegro

Section Editors

Radenko Matic (Cultural Anthropology) | University of Novi Sad, Serbia
Kubilay Ocal (Global Anthropology) | Mugla Sitki Kocman University, Turkey
Dusan Stupar (Biological Anthropology) | Educons University, Serbia
Tatjana Jovovic (Linguistic Anthropology) | University of Montenegro, Montenegro
Nina Djukanovic (Medical Anthropology) | University of Belgrade, Serbia

Editorial Board

Fitim Arifi | University of Tetova, North Macedonia
Hassan Sedeghi | University Putra Malaysia, Malaysia
Ivana Cerkez Zovko | University of Mostar, Bosnia and Herzegovina
Izet Bajramovic | University of Sarajevo, Bosnia and Herzegovina
Juel Jarani | Sports university of Tirana, Albania
Luiz Fernando Rojo | Universidade Federal Fluminense, Brazil
Marin Corluca | University of Mostar, Bosnia and Herzegovina
Marko Aleksandrovic | University of Nis, Serbia
Sami Sermahaj | Universi College, Kosovo
Stefan Seman | University of Belgrade, Serbia
Taher Afsharnezhad | Shomal University, Iran
Tonci Bavcevic | University of Split, Croatia

Index Coverage

DOAJ; Index Copernicus; Google Scholar; Crossref; ROAD; Open Academic Journals Index

Proofreading Service

Kristina Perovic Mijatovic

Prepress

Milicko Ceranic

Print

Art Grafika | Niksic

Print run

1500

MontenegroSport



**JOURNAL OF ANTHROPOLOGY OF SPORT
AND PHYSICAL EDUCATION**
International Scientific Journal

Vol. 4(2020), No. 4 (1-77)

TABLE OF CONTENTS

Radenko M. Matic, Nebojsa Maksimovic

(Original Scientific Paper)

Characteristics of Organizational Learning Culture in Serbian Elite Team Sports..... 3-7

Werlayne Leite, Richard Pollard

(Original Scientific Paper)

Comparison of Home Advantage between Level 1 and Level 2 in Women's Football Leagues..... 9-13

Jovan Gardasevic, Dusko Bjelica, Ivan Vasiljevic, Marko Aleksandrovic, Tomislav Okicic

(Original Scientific Paper)

Water Polo Players of Montenegro and Serbia who participated in the FINA World Junior Water Polo Championship 2019 and Differences in their Body Composition 15-19

Amel Mekic, Erol Vrevic, Pavle Malovic, Safet Kapo, Husnija Kajmovic

(Original Scientific Paper)

Differences in the Quality of Movement Functionality between Judokas, Karatekas, and Non-Athletes 21-26

Ivana Cerkez Zovko, Milena Mitrovic, Marin Corluka

(Original Scientific Paper)

Comparative Analysis of Anthropometric Parameters as Obesity Indicators for Sixth Grade Boys from Different Regions in Montenegro 27-31

Izet Bajramovic, Srdjan Redzepagic, Dusko Bjelica, Dragan Krivokapic, Eldin Jeleskovic, Slavenko Likic

(Original Scientific Paper)

Level of Active Lifestyle and Exercise Approach among Sports-Active Female Students of The University of Sarajevo during the Covid-19 Pandemic..... 33-36

Emir Mustafovic, Denis Causevic, Nedim Covic, Mirza Ibrahimovic, Haris Alic,

Ensar Abazovic, Semir Masic

(Review Paper)

Talent Identification in Youth Football: A Systematic Review 37-43

Miljan Hadzovic, Ana Lilic, Nikola Prvulovic, Predrag Ilic, Mima Stankovic

(Review Paper)

Effects of the Aerobic Exercise Program with Music on the Body Composition and Subcutaneous Fat of Young Women: A Systematic Review..... 45-55

Milena Mitrovic (Review Paper)	
Special Editions of Full Prof. Dusko Bjelica, PhD: 1992-2020	57-61
Zainal Ilmi, Dio Caisar Darma, Musdalifah Azis (Short Report)	
Independence in Learning, Education Management, and Industry 4.0: Habitat Indonesia during COVID-19.....	63-66
Guidelines for the Authors.....	67 -77

Full-text available free of charge at <http://www.jaspe.ac.me/>

ORIGINAL SCIENTIFIC PAPER

Characteristics of Organizational Learning Culture in Serbian Elite Team Sports

Radenko M. Matic¹, Nebojsa Maksimovic¹

¹University of Novi Sad, Faculty of Sport and Physical Education, Novi Sad, Serbia

Abstract

Organizational culture represents employees' integrated values and business orientation of employees in the work environment, and conduct unwritten and written rules. This research aimed to evaluate the characteristic of organizational learning culture in Serbian team sports. The sample of respondents in this study consisted of 118 available managers of different levels in the organizational structure of sports organizations from four selected team sports (football, basketball, handball, and volleyball), which institutionally belong to the highest rank of the competition. At the individual level, the authors revealed the differences in favor of managers in volleyball clubs that come from better support, openness, and feedback from each other in the organization, all for their career development ($\chi^2=12,79$, $p<0.01$). From another side, the organizational level's key differences come from the availability of information to all, better resource management, initiative recognition, openness to cooperation, and leadership ($\chi^2=11,33$, $p<0.01$). Based on the obtained results, the authors discussed the theoretical and practical values of research.

Keywords: *Sports Organizations, Culture, Individual, Group, Team Level*

Introduction

Organizational culture is described as a feature or characteristic of an organization, or only as "the way things are organized in the organization" (Wright & Boswell, 2002), or the way decisions are made, goals are set, and employees are lead to the realization of these goals Stare (2012). Customs, habits, and beliefs are recognized as segments that affect all employees' daily functioning. Therefore, any non-fitting of an individual into the stated, appropriate patterns and standards of group behavior puts the individual in an unenviable position. This mismatch can be viewed in both directions: 1) how useful and efficient an individual feels in such an organization, and 2) how much the organization can count on a usable, motivated, and quality employee.

Examining organizational culture is a relatively young, scientific, interdisciplinary research area where psychologists, sociologists, economists, and anthropologists deal with equal interest. It is one of the most used management concepts and belongs to organizational behavior. For Britton (2002), organizational learning culture can be defined as "an organizational environment that enables, encourages, values and rewards using individual and collec-

tive learning." Similar to Britton, Pisano (1994) describes this process of acquiring knowledge and skills as a significant competitive advantage in the market. It is up to the organization's managers to "enable, support and accelerate the creation, dissemination, and use of knowledge in the organization" within the environment itself (Janićijević, 2008) and human resource management. In a way, this implies "learned patterns of behavior, opinions, feelings, and actions of a group, community or society, as well as their expressions in material objects that contribute to internal stability and the creation of characteristic features of a certain group of people" (Raduan, Naresh, Haslinda, & Goh, 2008).

Based on these mentioned facts, in the context of sports organizational culture, the effects of "motivational climate created by coach and teammates" can contribute to the satisfaction of athletes' basic psychological needs (Trbojević & Petrović, 2020), as one segment of athlete's motivation.

The importance of organizational culture has contributed to many authors dealing with this issue (Cole & Martin, 2018; Colyer, 2000; Skille & Chroni, 2018). Its development is related to individual Japanese companies' success in the 60s and 70s of the last

Correspondence:

**Montenegro
Sport**

R.M. Matic
University of Novi Sad, Faculty of Sport and Physical Education, Lovcenska 16, 21000 Novi Sad, Serbia
E-mail: radenkomatic@uns.ac.rs

century, which achieved notable business results, built on organizational culture. The consistent growth of research interest in this segment in management theory stems from numerous studies' results. Thus, some researches confirmed its positive relations for job satisfaction and work motivation (Okpara, & Wynn, 2008), employee commitment and cooperation (Singh, 2006), organizational efficiency, and improving organizational outcomes (Balthazard, Cooke & Potter 2006; Wilder, van den Berg, & Wiersma, 2012; Yang, 2003), and increasing innovation (Jaskyte, 2004). Organizational culture's determination of business success has led other organizations to deny their importance and pay more attention to these opportunities. During the acquisition of human resources or selection, organizational culture is an essential factor as an indicator of how much a particular individual will accept the challenges of the current organizational culture and achieve the organization's expected standards. Therefore, what gives it a significant role in management is its dynamism, and the possibility of transfer to individuals, groups, and teams.

Namely, Miyagawa, & Yoshida (2010) emphasize the "specific attitude" of managers and employees towards their work and the company from which their outstanding commitment stemmed. Their "specific relationship," which results in employees' better commitment, is precisely the "product" of a well-developed and nurtured organizational culture.

It encompasses everything that determines human behavior in the workplace when employees are not supervised (Lorch & Tierney, 2002). The authors emphasize here the deep, ingrained methods, procedures, and decisions used by employees daily, which are so stable and accepted that they would be applied without supervision and control. Denison (1996) considers in a similar way, which also emphasizes "the deep structure of the organization, which is rooted in the values, beliefs, and assumptions maintained by organizational members." From the definition itself, it can be seen that each individual must know how the organization "breathes" in the business system, to limit as much as possible the possibilities for ambiguous interpretation and different views on specific business tasks and disorganization. In his holistic approach, Hofstede (2005) calls organizational culture "collective mental software."

This paper aimed to determine the individual, group, and team level of organizational learning culture in Serbia's elite team sports.

Methods

The sample of respondents in this study consisted of 118 available managers of different levels in sports organizations' organizational structure from four selected collective sports (football, basketball, handball, and volleyball), which institutionally belong to the highest rank of competition. In this paper, the evaluation of

organizational learning culture prepared using the recommendations of Watkins & Marsick (1993, 1996). These authors divided the organizational learning culture into three levels: individual, team or group, and organizational.

The individual-level refers to how the organization creates continuous learning (e.g., Continuous Training Center, online learning) for each employee individually. The organization creates a climate of support for a developmental approach to learning by promoting employees' information and dialogue. Team or group level refers to how the organization encourages team learning and cooperation within the organization. Team learning is enhanced when teams master the skills of staffing, testing, experimenting, and creating a shared perspective. The organizational level means the organization's ability to "preserve" what has been learned and create an organization that understands to support appropriate leadership in the organization.

The questionnaire from this research was applied by Xie (2005) in researching the organizational culture of learning in sports organizations. Thus, the questionnaire included six items for evaluating the individual, four items for group, and twelve items for respondents' organizational learning culture.

Indicators of the organizational culture of learning at the individual, group, and team level were formed by positive answers for each level of the organizational culture of learning coded with 1, and negative with 0. Therefore, the maximum value of organizational culture at an individual was 6, at group level - 4 and organizational level - 12.

Kruskal Wallis and Man Whitney tests, in SPSS, version 24.0, were used to analyzing the differences.

Results

An overview of descriptive statistics of organizational learning culture is given according to aggregate indicators separately at all examined levels: individual, (Table 1) group, or team (Table 2) and organizational (Table 3).

None of the sports calculated that the sum of all answers was 0; that is, a deficient organizational culture level was not determined for an individual level.

According to the respondents, the level of organizational culture at the individual level (Table 1) shows that: 1) in over ½ sports organizations (50.9%), there are high conditions for this type of organizational culture, 2) 15.5% have above average with specific shortcomings, 3) 18, 2% average, and 4) 15.5% below average conditions.

Regarding the indicators of group level of organizational culture, the sum of all variables categorized the respondents into four categories: 1) the sum of all results 0 or all negative answers, 2) one positive and two negative answers, 3) two positive and one negative answer, 4) all positive responses.

Table 1. Individual-level - respondents from all sports

Level	Recoded Category	Frequency	%	Cumulative %
Low	1	2	1,8	1,8
	2	15	13,6	15,5
Medium	3	20	18,2	33,6
	4	10	9,1	42,7
High	5	7	6,4	49,1
	6	56	50,9	100
Total		110	100,0	

$$\chi^2=12,79^{**}$$

The obtained results in Table 2 show that about 70% of the respondents consider the sports organization's environment favor-

able for realizing the group or teams' goals. The chi-square analysis ($\chi^2 = 4.83$) did not show differences in sports at group level.

Table 2. Group level - respondents in all sports

Level	Category	N	%	Cumulative %
Low	1	6	5,7	5,7
Low	2	13	12,3	17,9
Medium	3	13	12,3	30,2
High	4	74	69,8	100,0
Total		106	100,0	

$$\chi^2=4,83$$

The cumulative percentage of an organizational learning culture (Table 3) indicates a high level of this type of culture in sports

organizations (over 60% has a high level), with the difference between sports being determined.

Table 3. Organizational level - respondents in all sports

Level	Category	N	%	Cumulative %
Low	2	2	1,9	1,9
	5	2	1,9	3,8
	6	6	5,7	9,4
Medium	7	8	7,5	17,0
	8	7	6,6	23,6
	9	15	14,2	37,7
High	10	7	6,6	44,3
	11	7	6,6	50,9
	12	52	49,1	100,0
Total		106	100,0	

$$\chi^2=11,33^{**}$$

Considering that sports represent the determined differences in the organizational learning culture at the individual and or-

ganizational level ($\chi^2 < 0.01$), their comparison was further approached (Tables 4 and 5).

Table 4. Analysis of differences in different levels of an organizational learning culture by team sports

Level of the organizational learning culture	Mean Rank			
	football	basketball	volleyball	handball
Individual	52,51	41,63	70,50 ^{1a2a}	62,54 ^{2a}
Group (team)	84,87	45,10	57,00	58,46
Organizational	48,61	42,96	69,17 ^{1a2a}	57,81

* 0.05, ** <0,01 - Kruskal Wallis test; 1, 2, 3, 4 – 1 - football, 2 - basketball, 3 - volleyball, 4 – handball; a < 0.01, b < 0.05 - Man Whitney test

At both levels, it was obtained that the respondents - managers in volleyball clubs, both at the individual and organizational level, have higher levels of organizational learning culture concerning football and basketball clubs. No statistically significant

differences were obtained at the team or group level. The individual and organizational level of a learning culture, the respondents from the examined sports, differ, showing the Man Whitney test (Table 5).

Table 5. Descriptive statistics of organizational culture (%)

Items	Football		Basketball		Volleyball		Handball		X ²	p
	Yes	No	Yes	No	Yes	No	Yes	No		
Individual-level - In my organization ...										
... people teach each other.	91,7	8,3	75,9	24,1	100	0	86,0	14	8,12	0,04
... people have been supported to improve their knowledge.	75,0	25,0	58,6	41,4	100	0	82,1	17,9	11.62	0,00
... people are rewarded for working on their training.	52,8	47,2	41,4	58,6	85,7	14,3	71,4	28,6	12,32	0,00

(Continued on next page)

(Continued from previous page)

Items	Football		Basketball		Volleyball		Handball		χ^2	p
	Yes	No	Yes	No	Yes	No	Yes	No		
... people give open and honest feedback to each other.	50,0	50,0	60,7	39,3	81,0	19,0	85,7	14,3	11,53	0,00
... when people have their point of view, they also ask what others think.	88,9	11,1	71,4	28,6	81,0	19,0	89,3	10,7	4,42	0,21
... people spend time building trust with each other.	86,1	13,9	66,7	33,3	85,7	14,3	85,7	14,3	4,96	0,17
Organizational level - My organization ...										
... assesses actual and expected performance.	91,2	8,8	89,7	10,3	100	0	92,6	7,4	2,19	0,53
... makes the lessons learned available to all.	77,9	22,1	48,3	51,7	100	0	85,2	14,8	20,06	0,00
... measures the results, time, and resources spent on training.	94,3	5,7	71,4	28,6	100	0	81,5	18,5	11,11	0,01
... "recognizes" the people who take the initiative.	61,1	38,9	62,1	37,9	90,5	9,5	88,9	11,1	11,15	0,01
... gives people control over the resources they need to work.	75,0	25,0	77,8	22,2	95,2	4,8	80,8	19,2	3,77	0,28
... supports employees who take the assessed risk.	72,2	27,8	71,4	28,6	81,0	19,0	74,1	25,9	0,68	0,87
... encouraging people to think globally.	88,2	11,8	69,0	31,0	95,2	4,8	85,2	14,8	7,25	0,06
... cooperates with other communities for the common interest.	100	0	82,8	17,2	90,5	9,5	100	0	10,65	0,01
... encourages people to find a solution through their organization.	93,9	6,1	79,3	20,7	100	0	92,6	7,4	7,28	0,06
In my organization, leaders and coaches are the ones who lead the organization.	91,4	8,6	93,1	6,9	100	0	66,7	33,3	15,04	0,00
In my organization, leaders are continually looking for opportunities to learn.	65,7	34,3	79,3	20,7	85,7	14,3	74,1	25,9	3,19	0,36
In my organization, leaders make sure that their values conduct organizational activities.	79,4	20,6	89,7	10,3	100	0	92,6	7,4	6,24	0,10

The Man Whitney test showed that statistically significant differences at the individual level, which are recognized in favor of volleyball clubs, come from better support, openness, and feedback they receive from each other in the organization, all for their career development. Otherwise, the organizational level's key differences stem from information availability, better resource management, initiative recognition, openness to cooperation, and leadership.

Discussion

Aware of the possibility that the combined knowledge of individuals, groups, and the entire organization, with clearly set norms of behavior of all its members, can improve organizational goals, researchers and managers in sports organizations pay more attention to creating such a business environment. This research examined the organizational learning culture at the individual, group, and team levels in Serbia's elite sports clubs.

The indicators of an organizational culture of learning show that the highest level of the organizational culture of learning (individual and organizational level) was determined in volleyball clubs. The obtained results indicate that employees in volleyball clubs, about other team sports, have the most outstanding ability to "preserve" what they have learned and promoted dialogue and exchange opinions among employees.

Likewise, such results can be explained through causal factors of how organizational culture affects end outcomes and organizational effectiveness. Managers in volleyball sports clubs adhere to the established direction that contains structures, systems,

technology, skills/qualities that initiate the operational culture. Other shows that volleyball clubs have a good structure in which organized groups of people function, with clear roles and activities. Such results from a group of people who work together with trust and complementarity and who have common goals greater than individual goals can be considered as expected (Senge et al., 1999). On the other hand, a group of people's skills and qualities are mainly related to leadership, i.e., its components such as communication, leadership, i.e., the function of organizing - the distribution of power.

In a broader context, given the interaction between leadership and organizational culture shown in many studies (Bas & Avolio, 1993), it should be borne in mind that the leader is the one who changes organizational culture, but that organizational culture can also determine development leadership. That is why successful leaders, with a good understanding of organizational culture, adjust the speed of changes they want to achieve. The time determinants necessary for implementing the desired changes, with the leader's patience, become an essential factor in the practical realization of leadership goals. Suppose we consider some studies that showed the highest scores in leadership characteristics in volleyball managers (Matić, 2016). In that case, work in volleyball clubs is primarily based on employees' enthusiasm and motivation, which proved to be leading in the evaluation of leadership in domestic team sports. The concept of leadership in these circumstances with modest financial resources, with a smaller number of employees, where everyone is gathered around a common goal, is easier to implement than other team sports with many employees.

The volleyball sports clubs function on specific leadership, which is considered a cascading effect. This effect by Bass (1985) emphasizes the transformational influence that leaders, mentors, and coaches have on followers gradually transferred to lower organizational levels and manifested in their other spheres of social action. It is evident that in volleyball, there is a good education of managers by the National Volleyball Federation of Serbia, as a leading body at the national level, which can lead to Lee (2019) "to initiate development-oriented reform of the winning-at-all-costs culture in sports." The success in sports results of Serbian volleyball confirms excellent organizational culture and leadership in this sport. In 2019, Serbia's senior volleyball national teams repeated the success from 2011 and won gold medals at the European Championships. Both seniors won the third European champion titles. In addition to senior gold medals, pioneers and juniors won silver medals at the Balkan Championships.

In today's business environment, organizational culture is taken seriously, and it can be stated that its importance is directly proportional to the organization's size and business success. This research showed a high level of organizational culture in volleyball clubs. Such an organizational culture level is a good prerequisite for realizing business ventures and accepting the desired innovations.

In a way, well-known, successful volleyball and other sports organizations are expected to "lead" in their organizational culture, and set new standards and trends. Creating such favorable conditions in volleyball clubs means removing obstacles to learning and encouraging and rewarding the organization's positive work characteristics (taking risks, finding a different solution to the problem, new initiatives of employees).

Acknowledgements

There are no acknowledgements.

Conflict of Interest

The authors declare that there are no conflicts of interest.

Received: 1 September 2020 | **Accepted:** 24 September 2020 | **Published:** 16 October 2020

References

- Balthazard, P., Cooke, R., & Potter, R. (2006). Dysfunctional culture, dysfunctional organization: capturing the behavioral norms that form organizational culture and drive performance. *Journal of Managerial Psychology*, 21, 709-732.
- Bass, B.M. (1985). *Leadership and performance beyond expectation*. New York: Free Press.
- Bass, B.M., & Avolio, B.J. (1993). Transformational leadership: A response to critiques. In M. M. Chemers & R. Ayman (Eds.), *Leadership theory and research: Perspectives and directions* (pp. 49-80). San Diego, CA: Academic Press.
- Britton, B. (2002). *Learning for Change: Principles and practices of learning organisations*. Sundbyberg: Swedish Mission Council.
- Cole, J., & Martin, A.J. (2018). Developing a winning sport team culture: organizational culture in theory and practice. *Sport in Society*, 21(8), 1204-1222. doi:10.1080/17430437.2018.1442197
- Colyer, S. (2000). Organizational culture in selected Western Australian sport organizations. *Journal of Sport Management*, 14(4), 321-341. doi:10.1123/jsm.14.4.321
- Denison, D.R. (1996). What is the difference between organizational culture and organizational climate? *Academy of Management Review*, 21(3), 619-654.
- Hofstede, G. (2005). *Cultures and organizations: Software of the mind*. New York: McGraw-Hill.
- Janićević, N. (2008). *Organizaciono ponašanje*. [Organizational behaviour]. Beograd: Data status.
- Jaskyte, K. (2004). Transformational leadership, organizational culture, and innovativeness in nonprofit organizations. *Nonprofit Management & Leadership*, 15, 153-168.
- Lee, Y.H. (2019). Emotional intelligence, servant leadership, and development goal orientation in athletic directors. *Sport Management Review*, 22(3), 395-406. doi:10.1016/j.smr.2018.05.003
- Lorsch, J.W., & Tierney, T.J. (2002). Creating Competitive Advantage in the Knowledge Economy. *Leader to Leader*, 26, 41-47.
- Matić, R. (2016). Primena marketing koncepta u timskim sportovima. [Application of marketing concepts in team sports]. *Doktorska disertacija*. Novi Sad: Fakultet sporta i fizičkog vaspitanja.
- Miyagawa, M. & Yoshida, K. (2010). TQM practices of Japanese-owned manufacturers in the USA and China. *International Journal of Quality and Reliability Management*, 27(7), 736-755.
- Okpara, J.O., & Wynn, P. (2008). The impact of ethical climate on job satisfaction, and commitment in Nigeria. *Journal of Management Development*, 27, 935-950.
- Pisano G., (1994). Knowledge, Integration, and the locus of learning an empirical analysis of process development. *Strategic Management*, 15(1), 85-100.
- Raduan, C.R., Naresh, K., Haslinda, A. & Goh, Y.L. (2008). Organizational Culture as a Root of Performance Improvement: Research and Recommendations. *Contemporary Management Research*, 4(1), 43-56.
- Senge, P., Kleiner, A., Roberts, C., Ross, R., Roth, G. & Smith, B. (1999) *The Dance of Change: The Challenges of Sustaining Momentum in Learning Organizations*. New York: Doubleday/Currency.
- Singh, J. (2006). The rise and decline of organizations: Can entrepreneurs' play a saviour's role? Vikalpa. *The Journal for Decision Makers*, 31, 123-127.
- Skille, E.A., & Chroni, S. (2018). Norwegian sports federations' organizational culture and national team success. *International Journal of Sport Policy and Politics*, 10(2), 321-333. doi:10.1080/19406940.2018.1425733
- Stare, A. (2012). The impact of a project organisational culture and team rewarding on the project performance. *Journal of East European Management Studies*, 1, 40-67.
- Trbojević, J., & Petrović, J. (2020). Satisfaction of basic psychological needs of young Serbian female athletes: the role of coach and teammates. *Exercise and Quality of Life*, 12(1), 37-45. doi:10.31382/eqol.200605
- Watkins, K.E., & Marsick, V.J. (1993). *Sculpting the learning organization: Lessons in the art and science of systematic change*. San Francisco, CA: Jossey-Bass.
- Watkins, K.E., & Marsick, V.J. (1996). *In action: Creating the learning organization*. Alexandria, VA: American Society for Training and Development.
- Wilderom, C.P.M., van den Berg, P.T., & Wiersma, U.J. (2012). A longitudinal study of the effects of charismatic leadership and organizational culture on objective and perceived corporate performance. *The Leadership Quarterly* 23, 835-848.
- Xie, D. (2005). Exploring organizational learning culture, job satisfaction, motivation to learn, organizational commitment, and internal service quality in a sport organization. *PhD dissertation*, Ohio State University.
- Yang, B. (2003). Identifying valid and reliable measures for dimensions of a learning culture. *Advances in Developing Human Resources*, 5(2), 152-162.
- Wright, P.M. & Boswell, W.R. (2002). Desegregating HRM: a review and synthesis of micro and macro human resource management research. *Journal of Management*, 28(3), 247-276.
- Wright, P.M. & Boswell, W.R. (2002). Desegregating HRM: a review and synthesis of micro and macro human resource management research. *Journal of Management*, 28(3), 247-276.

ORIGINAL SCIENTIFIC PAPER

Comparison of Home Advantage between Level 1 and Level 2 in Women's Football Leagues

Werlayne Leite^{1,2}, Richard Pollard³

¹Secretariat of Education of the State of Ceará (SEDUC), Fortaleza, Brazil, ²Secretariat of Education of Fortaleza (SME), Fortaleza, Brazil, ³California Polytechnic State University, Statistics Department, San Luis Obispo, USA

Abstract

Professional sports teams derive an advantage from playing at home; this phenomenon is known as home advantage. The aim of this study was to compare the magnitude of home advantage between levels 1 and 2 of women's football leagues. A total of 10 countries were included, each for the most recent 9 seasons, resulting in 31,186 matches analysed. A preliminary analysis of the results was carried out in order to assess the difference in home advantage between level 1 and level 2 for each country separately. A two-sided paired t-test was used and the effect size was reported using Cohen's d statistic. A general linear model was fitted and after adjusting for differences between seasons and between countries, the main finding was that there existed a small, yet significant difference between the home advantage of the two levels, with level 2 (mean of 55.5%) higher than at level 1 (mean of 54.6%). The results therefore suggest that in women's football there occurs the same phenomenon that has been shown to exist in women's water polo and handball, and in men's football. Some of the main factors that have been found to explain home advantage in men's football appear to operate in a slightly different way when applied to women's football. These are crowd support, familiarity, referees bias, travel effects and psychological factors.

Keywords: Home Advantage, Levels of Play, Match Location, Performance Analysis, Soccer

Introduction

Professional sports teams derive an advantage from playing at home. This has been shown to exist for a wide variety of professional sports, all over the world, and for both male and female teams (Pollard, Prieto, & Gómez, 2017). This phenomenon, known as home advantage, has been extensively researched over the last 40 years, a brief summary of which is given by Leite & Pollard (2018), with a special emphasis on football.

In football, and in other sports leagues, home advantage can be represented by the ability of the home team to attain more than 50% of all points gained in an official competition with a balanced schedule, that is one in which each team plays each other team the same number of times at home and away over the course of a league season (Courneya & Carron, 1992; Pollard, 2008a).

The worldwide presence of home advantage in football has been demonstrated by Pollard and Gómez (2014a) in an inves-

tigation that encompassed the national domestic leagues of 157 countries. Among the possible factors that lead to home advantage in football are the crowd support, travel effects, familiarity with the field of play, referee bias, territoriality and psychological factors (Pollard, 2008a; Pollard & Gómez, 2014a). It has been suggested that all these factors are likely to interact with each other and a model for their inter-relationship in football was proposed by Pollard and Pollard (2005).

Despite the extensive number of studies in men's football, research on the phenomenon of home advantage in women's football still is scarce and mainly limited to a comparison of home advantage in men's and women's football in 26 countries in Europe (Pollard & Gómez, 2014b). Home advantage in the women's leagues was found to exist, but at a significantly lower level than for men, a result that was similar to that found in a subsequent larger worldwide study of home advantage in football and other

Correspondence:

**Montenegro
Sport**

Werlayne Leite

Secretariat of Education of the State of Ceará, Av. Gen. Afonso Albuquerque Lima, S/N – Cambéa, Fortaleza-CE, 60822-325, Brazil
E-mail: werlaynestuart@yahoo.com.br

sports (Pollard, Prieto, & Gómez, 2017).

Some studies have compared the difference between levels 1 and 2 in men's football in different countries: Brazil (Almeida, Oliveira, & Silva, 2011); England (Pollard, 2006; Pollard & Pollard, 2005); France (Dosseville, 2007; Pollard, 2006); Germany (Pollard, 2006); Iran (Pollard, Armatas, & Sani, 2017); Italy (Pollard, 2006); Spain (Pollard, 2006; Sanchez, Garcia-Calvo, Leo, Pollard, & Gómez, 2009); Turkey (Seckin & Pollard, 2008). Most of these studies found a higher home advantage at level 2 than at level 1, based on a wide range of national league data going as far back as to 1892 in England. The full results of these studies are summarized by Leite & Pollard (2018) who also made a comparison of home advantage at the top two levels of the national leagues in 47 countries worldwide. Their conclusion was that home advantage was significantly more likely to be higher at level 2 than at level 1.

Only two studies could be identified in which a specific comparison of home advantage was made at different levels in women's sport. One was in Spanish water polo in which, over a four-year period starting in 2007-08, the second division (55.3%) had an overall higher home advantage than the first division (52.1%). The overall results of the study found home advantage for men to be significantly higher than for women, and home advantage at level 2 to be significantly higher than at level 1 (Prieto, Gómez, & Pollard, 2013). For the women's leagues the mean home advantage at level 1 was 52.1% compared with 55.3% at level 2. The other study involved Spanish handball, analysing the results of men's and women's leagues at two levels of play from 1997-98 for the 11 seasons ending 2007-08 (Pollard & Gómez, 2012). As with the handball report, home advantage was higher for men than for women and higher at level 2 than at level 1. The mean home advantage for the women's league were 57.8% at level 1 and 60.5% at level 2.

The vast majority of published work on home advantage is based on men's sports and to our knowledge, there have been no reports comparing home advantage at different levels of play in women's football. Hence the motivation for the present study which will compare the magnitude of home advantage between levels 1 and 2 in all countries which operate national women's football leagues at more than one level. The more general aim will be to throw new light on the phenomenon of home advantage as it applies to women's sports.

Methods

Scope

The study was developed through the analysis of the 2 main

levels of women's football in each country for which the data were available. The home and away league tables were obtained from the website (www.soccerway.com). The last nine complete seasons of the leagues in which there was a balanced schedule were analyzed. For national leagues that start and finish in a single calendar year, the seasons from 2011 to 2019 were included; for those that overlap years, the seasons 2010-11 to 2018-19 were used. A total of 10 countries were included, resulting in 31,186 matches analyzed.

Home advantage

Home advantage was quantified by the method first proposed by Pollard (1986) for calculating the advantage of playing at home, during a league in a full season for sports in which the draw is a possible result of a match. FIFA currently mandates three points for a win and one point for a draw. The advantage of playing at home is then calculated as the number of points earned by teams playing at home, expressed as a percentage of total points earned in the league, at home and away. Thus, a value of 50% represents no home advantage since the same number of points is won at home and away. The greater the value above 50%, the greater the advantage derived by the home team.

Analysis

A preliminary analysis of the results was carried out in order to assess the difference in home advantage between level 1 and level 2 for each country separately. A two-sided paired t-test was used and the effect size was reported using Cohen's d statistic. The original and still the usual interpretation of the magnitude of an effect size was used: small below 0.5, moderate between 0.5 and 0.8, and large above 0.8 (Cohen, 1988).

A general linear model was then fitted to the data using the statistical software Minitab 18. Home advantage each season was the dependent variable, with country, season and level as the three explanatory variables. This allowed the overall effect of each variable to be quantified and assessed in the presence of the other variables, with the results presented in a standard ANOVA table.

Results

The initial analysis revealed that for seven of the ten countries, home advantage was greater at level 2 than at level 1. However, only Poland had a difference with a p-value below 0.05. The effect size for this country was strong (0.82), while for France and for Turkey it was considered as moderate (above 0.50). For these three countries, home advantage at level 2 was higher than at level 1.

Table 1. Comparison of mean home advantage (%) between level 1 and level 2 over nine seasons in women's national leagues by country

Country	Level 1	Level 2	Difference	p	n ₁	n ₂	Effect size
Belgium	57.00	55.38	1.62	0.399	1512	1982	0.30
Denmark	53.01	54.00	-0.98	0.479	640	914	0.25
France	53.43	55.25	-1.82	0.143	1188	3150	0.54
Germany	53.57	54.40	-0.83	0.650	1188	2294	0.16
Iceland	52.76	53.78	-1.03	0.620	810	994	0.17
Italy	53.93	53.49	0.43	0.729	1554	4604	0.12
Norway	55.99	58.05	-2.06	0.247	1188	1144	0.42
Poland*	53.70	56.05	-2.35	0.039	1020	1591	0.82
Sweden	58.34	57.48	0.86	0.389	1166	1780	0.30
Turkey	53.88	57.23	-3.35	0.126	788	1679	0.57

Note: n₁ and n₂ are the total number of matches during the nine seasons at each level; * Country with a significant difference between level 2 and level 1 (p<0.05).

The results of the general linear model are summarized in Table 2. There was a difference between countries ($p < .001$) and also between levels ($p < .05$), but not between the seasons. The results for each variable allow for the simultaneous effects of the other variables. Thus, after adjusting for differences between the

seasons and between countries, the mean home advantage for level 1 was found to be 54.6% and the mean for level 2 was 55.5%. The proportion of the variability in the home advantage values accounted for by the combined effect of the three explanatory variables (season, country and level) was 23%.

Table 2. ANOVA table summarizing the results of fitting a general linear model to quantify the effects on home advantage of country, level of play and season

Source	df	Adj SS	Adj MS	F	p
Country	9	407.35	45.262	4.37	0.000
Level	1	40.66	40.659	3.93	0.049
Season	8	50.60	6.325	0.61	0.768
Error	161	1666.33	10.350		
Total	179	2164.94			

Discussion

The aim of this study was to compare the magnitude of home advantage between levels 1 and 2 of women's football leagues. The main finding was that there existed a small, yet significant difference between the home advantage of the two levels, with level 2 higher than at level 1. This result was based on the leagues of 10 countries over the last 9 seasons. The results therefore suggest that in women's football there occurs the same phenomenon that has been shown to exist in women's water polo and handball, and in men's football, where home advantage at level 2 usually is greater than at level 1.

This information may help anticipate match scenarios in a particular league and could be used by coaches in their match preparation and training process (Almeida & Volossovitch, 2017). In this sense, the main causes of home advantage will be discussed as possible explanations for differences found between level 1 and level 2 specifically in women's football leagues.

Crowd support

Female football players have highlighted the crowd as one of the most important causes of home advantage. They have felt that a supportive home crowd is positive for individual and team performance (George, 2015).

According to George (2015), female football players noted that a supportive home crowd was a positive factor and reduced feelings of pressure. In particular, cheering seemed to have a beneficial effect upon players as it increased their positive thinking. Interestingly, some participants in the study seemed to prefer a small crowd at home as they felt that it was more supportive and less intimidating. Additionally, these participants felt less pressure to perform in front of fewer spectators. According to Perkins (2017), if a crowd is big, intense and is near the female players and coaches, it could distract them and cause them to lose focus on the match. Thus, as women's football at level 2 certainly has a much smaller audience than at level 1, the results are plausible in accordance with the players' testimonies.

There are some other possible explanations for crowd support and its influence on home advantage at lower levels of women's football. According to Leite & Pollard (2018), the better supported teams in a country are likely to be at level 1 and a sizeable number of these supporters will travel to away matches, thus providing a more balanced level of crowd support than might be expected at level 2. Furthermore, away players at level 1 may have been better trained to cope with the atmosphere created by a hostile home crowd (Leite & Pollard, 2018).

Usually, in football matches there is a smaller audience at lower levels than at level 1. Although level 2 may have less crowd support, another possible explanation is that spectators at low-

er level games are more vocal and more hostile towards visiting teams. Smaller more intimate stadiums, with the crowd closer to the field, also may produce a more unified support in favour of the home team and create a more intimidating atmosphere even with lower attendances (Almeida et al., 2011; Leite & Pollard, 2018; Pollard et al., 2017).

The data of Almeida and Volossovitch (2017) showed that there is bigger home advantage in amateur (60.4%) and semi-professional (60.5%) level than in professional level (58.3%) in Portuguese football, even with attendances of up to over 60,000 spectators in some matches of professional level.

Familiarity

When playing at home, a team will be familiar with local conditions which include climate, altitude, and the playing surface, as well as the alignment of the pitch with familiar cues inside the stadium, among other factors. All in addition to a pre-match routine that will be the same at all home matches (Leite & Pollard, 2018).

Almeida and Volossovitch (2017) suggest one of the factors of the decrease in home advantage is that the competitive conditions have become more standardised (e.g. league structure and pitch dimensions). However, these changes probably start at level 1 and only later they are implemented at level 2. As we could verify during the data collection, level 2 women's leagues show more differences than level 1 in such factors as greater quantity and greater variation in number of teams, greater variation in league format as well as possible differences in the size of the playing field and the stadiums themselves.

Almeida et al. (2011) also suggest that the playing fields of level 2 are worse than those of level 1, and generally the teams of level 2 train in the same field in which they play, which can increase knowledge about the field and playing surface, and expand home advantage for these teams. In addition, the physical structure of the stadiums also tends to be less comfortable at level 2 than at level 1. This can inhibit away teams from preparing properly before and during the interval of matches.

Referees bias

There is overwhelming evidence that referee decisions favour the home team (Pollard, 2008a). The findings of the study by George (2015) revealed that female players perceived the referee to be a significant factor. Importantly all study participants felt that the referee's decision-making provided them with an advantage when playing at home. The participants highlighted their feeling that the referee was more likely to award arguable decisions to the home team. In the same way, the results showed that when participants played away, they felt that the referee provided

the opposition with an advantage.

Level 1 has better and more prepared referees than at level 2. Thus, it is likely they are better able to make decisions without being influenced by the noise of the home crowd (Almeida & Volossovitch, 2017). Thus, this being a relevant aspect for home advantage, the referees at level 2 (for being less prepared, technically and/or psychologically; for having a more hostile crowd in favour of the home team, and for having a crowd closer to the field due to smaller stadiums) will be more influenced by home crowd support, even if subconsciously.

According to Leite & Pollard (2018), the recent introduction of a video assistant referee (VAR) to enable certain referee decisions of important consequence to be reviewed, will in theory correct a decision seen to be clearly wrong. If subconscious referee bias influenced by the noise of the home crowd does exist, causing referees to unjustly penalize away players, or favour home players, then these decisions will be reversed and one of the effects of VAR could be to reduce home advantage. On the other hand, it is unlikely that VAR technology is yet introduced in most national women's leagues and when this occurs it is likely to be introduced initially in the major leagues at level 1.

Travel effects

For a long time, the effects of travel have been identified as a possible cause of home advantage. According to Marques (2002), the improvement of communication modes and transport decreased fatigue of players in travel to away matches. In continental European countries, which have a well-developed transport infrastructure, travel became faster, safer and more comfortable (Drummond, Drummond & Silva, 2014). However, as a consequence of lesser financial support, probably lower-level football players travel less comfortably than players in the main leagues.

As the evidence for a travel effect disadvantaging the away team is inconclusive (Pollard, 2008a), the disruption in the players' routine can be another important aspect when a team plays away. The study by George (2015) found that for female players travel, regardless of how long the time travelled, was perceived as disruptive. Female athletes expressed that they felt less physically and mentally prepared for matches when they had to travel (Vealey, 2001).

We can also speculate that the teams of level 1 have a largest support team (as well as financial support) than the teams of level 2. Thus, the organization of the away matches will be better (travelling in advance, better hotels, among other factors), leaving the players more comfortable and more focused on the match, each match being prepared in more detail.

For female athletes, the pre-match routine, when performing at home, was identified as a key type of confidence emphasising its effect on home advantage (Owen, 2009). Key aspects of preparation were identified as sleep, breakfast and relaxation time. At home, female players indicated the benefits of not travelling which allowed preparation and routine to be kept the same. In the same way, all participants said that travelling to play away was disruptive and interfered with their preparation and routine (George, 2015).

The findings of Almeida & Volossovitch (2017) indirectly suggest a negative relationship between distance travelled and the magnitude of home advantage in Portugal. In this country, amateur teams only played in their district, semi-professional teams were grouped by geographical location (e.g. south, centre and north), and professional teams played across the whole country, including the islands. Therefore, fatigue caused by long and/or uncomfortable travels does not appear to be a relevant cause. In this sense, the interruption in the players' routine can be the main aspect related to travel.

Psychological factors

Performance analysis research in team sports has shown that most teams that play on their home venues have a psychological advantage over their rivals (Campos, Stanganelli, Campos, Pasquarelli & Gómez, 2014).

The study of George (2015) sought to investigate female football players' perceptions of playing at home and away as well as examining how playing at home or away influenced their levels of confidence. The results showed that overall the female football players believed that home advantage existed and felt that they experienced a positive psychological state at home. Importantly all participants expressed feelings of greater happiness, comfort, and higher levels of confidence when playing at home.

Another probable contributor to the home advantage differences could be the players' level of expertise (Almeida & Volossovitch, 2017). Thus, players at level 1 are psychologically better prepared to face challenges and pressure. In this same sense, players less able to cope with the additional pressure involved in away matches could be weeded out in the selection process, never reaching the professional status (Pollard & Gómez, 2015).

Limitations of the study

This study contains some limitations due to the sparse data available on women's football: (i) it was not possible to include more countries, as there were no data available that met the inclusion criteria; (ii) in some countries and in some seasons, the level 2 leagues (although always having a completely balanced schedule) were played in smaller regional leagues. In addition, in some countries there was also variation in the number of teams between some seasons, although the playing schedule in all leagues was still balanced; (iii) it was not possible to quantify the attendance for leagues, since data were not available; (iv) the travel distances have not been calculated; however, none of the countries; were especially large and none involved different time zones (v) the competitive balance within in each league was not considered.

Practical and coaching application

Research on home advantage in women's sport is still limited, especially in football. This is the first study to compare different levels in women's football. Although it has been suggested that the same factors influence home advantage in men's and women's football, it was now possible to identify the main factors and the way they seem to specifically affect women. The importance of home advantage has been confirmed for teams at level 1 and, especially, for teams at level 2. Thus, coaches, managers and female players at both levels of play can better understand this phenomenon in women's football, try to make adjustments for a better adaptation, and try to minimize the adverse affects that can be found in away matches. Specific recommendations for coaches, both to exploit and minimize home advantage, have previously been proposed (Pollard, 2008b; Wolfson & Neave, 2004),

Conclusion

This study has compared the magnitude of home advantage between levels 1 and 2 of women's football leagues of 10 countries over the last 9 seasons. The main finding was that there existed a small, yet significant difference between the home advantage of the two levels, with level 2 (mean of 55.5%) higher than at level 1 (mean of 54.6%). In the context of women's football, factors that could explain this result include: (i) crowd support: spectators at level 2 being more hostile and vocal towards visiting teams; (ii) familiarity: playing fields of level 2 being worse than those of level 1, and generally the teams of level 2 training in the same field in which they play; (iii) referees bias: the referees at level 2 being less prepared and likely to be more influenced by home crowd sup-

port, even if subconsciously; (iv) travel effects: the disruption in the players' routine and the discomfort of travel being an important aspect when a team plays away, with these effects being more pronounced at level 2; (v) psychological factors: players at level 1 being psychologically better prepared to face the challenges and pressure of playing away than players at level 2.

Acknowledgements

There are no acknowledgements.

Conflict of Interest

The authors declare that there are no conflicts of interest.

Received: 19 July 2020 | **Accepted:** 14 August 2020 | **Published:** 16 October 2020

References

- Almeida, L., Oliveira, M., & Silva, C. (2011). Uma análise da vantagem de jogar em casa nas duas principais divisões do futebol profissional brasileiro. *Revista Brasileira de Educação Física e Esporte*, 25(1), 49-54.
- Almeida, C., & Volossovitch, A. (2017). Home advantage in Portuguese football: Effects of level of competition and mid-term trends. *International Journal of Performance Analysis in Sport*, 17(3), 244-255.
- Campos, F., Stanganelli, L., Campos, L., Pasquarelli, B., & Gómez, M. (2014). Performance indicators analysis at Brazilian and Italian women's volleyball leagues according to game location, game outcome, and set number. *Perceptual and Motor Skills*, 118(2), 347-361.
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences*. New York, NY: Academic Press.
- Courneya, K., & Carron, A. (1992). The home advantage in sport competitions: A literature review. *Journal of Sport & Exercise Psychology*, 14(1), 13-27.
- Dosseville, F. (2007). Influence of ball type on home advantage in French professional soccer. *Perceptual and Motor Skills*, 104(2), 347-351.
- Drummond, L., Drummond, F., & Silva, C. (2014). A vantagem em casa no futebol: comparação entre Copa Libertadores da América e UEFA Champions League. *Revista Brasileira de Educação Física e Esporte*, 28(2), 283-292.
- George, A. (2015). *The perception of home advantage and its effect on self-confidence from the viewpoint of female football players*. Unpublished undergraduate dissertation. Cardiff: Cardiff Metropolitan University.
- Leite, W., & Pollard, R. (2018). International comparison of differences in home advantage between level 1 and level 2 of domestic football league. *German Journal of Exercise and Sport Research*, 48(2), 271-277.
- Marques, A. (2002). Competitive balance in the Portuguese Premier League of professional soccer. *Journal of Economic Literature*, L19, L83.
- Owen, R. (2009). *The influence of game location, either home or away, on self-confidence in female hockey players*. Unpublished undergraduate dissertation. Cardiff: University of Wales Institute Cardiff.
- Perkins, I. (2017). *Performance and home advantage in women's collegiate basketball*. Unpublished master's thesis. Towson, MD: Goucher College.
- Pollard, R. (1986). Home advantage in soccer: a retrospective analysis. *Journal of Sports Sciences*, 4(3), 237-248.
- Pollard, R. (2006). Home advantage in soccer: variations in its magnitude and a literature review of the inter-related factors associated with its existence. *Journal of Sport Behavior*, 29(2), 169-189.
- Pollard, R. (2008a). Home advantage in football: A current review of an unsolved puzzle. *Open Sports Sciences Journal*, 1, 12-14.
- Pollard, R. (2008b). Away disadvantage: five steps for coping. *Soccer Journal*, 53(4), 53-54.
- Pollard, R., Armatas, V., & Sani, S. (2017). Home advantage in professional football in Iran – differences between teams, levels of play and the effects of climate. *International Journal of Science Culture and Sport*, 5(4), 328-339135.
- Pollard, R., & Gómez, M. (2012). Reassessment of home advantage in Spanish handball. *Perceptual and Motor Skills*, 115(3), 937-943.
- Pollard, R., & Gómez, M. (2014a). Components of home advantage in 157 national soccer leagues worldwide. *International Journal of Sport and Exercise Psychology*, 12(3), 218-233.
- Pollard, R., & Gómez, M. (2014b). Comparison of home advantage in men's and women's football leagues in Europe. *European Journal of Sport Science*, 14(S1), S77-S83.
- Pollard, R., & Gómez, M. (2015). Comparison of home advantage in college and professional team sports in the United States. *Collegium Antropologicum*, 39(3), 583-589.
- Pollard, R., Prieto, J., & Gómez, M. (2017). Global differences in home advantage by country, sport and sex. *International Journal of Performance Analysis in Sport*, 17(4), 586-599.
- Pollard, R., & Pollard, G. (2005). Home advantage in soccer: A review of its existence and causes. *International Journal of Soccer and Science*, 3(1), 28-38.
- Prieto, J., Gómez, M., & Pollard, R. (2013). Home advantage in men's and women's Spanish First and Second Division water polo leagues. *Journal of Human Kinetics*, 37(1), 137-143.
- Sanchez, P., Garcia-Calvo, T., Leo, F., Pollard, R., & Gómez, M. (2009). An analysis of home advantage in the top two Spanish professional football leagues. *Perceptual & Motor Skills*, 108(3), 789-797.
- Seckin, A., & Pollard, R. (2008). Home advantage in Turkish professional soccer. *Perceptual and Motor Skills*, 107(1), 51-54.
- Vealey, R. (2001). *Understanding and enhancing self-confidence in athletes*. In R. Singer, H. Hausenblas & C. Janelle (Eds.), *Handbook of sport psychology* (pp. 550-565) New York: Wiley.
- Wolfson, S., & Neave, N. (2004). Preparing for home and away matches. *Insight*, 8(2), 43-46.

ORIGINAL SCIENTIFIC PAPER

Water Polo Players of Montenegro and Serbia who participated in the FINA World Junior Water Polo Championship 2019 and Differences in their Body Composition

Jovan Gardasevic¹, Dusko Bjelica¹, Ivan Vasiljevic¹, Marko Aleksandrovic², Tomislav Okicic²

¹University of Montenegro, Faculty for Sport and Physical Education, Niksic, Montenegro, ²University of Nis, Faculty of Sport and Physical Education, Nis, Serbia

Abstract

This research aimed to determine the differences between the junior (U20) water polo players of national teams of Montenegro and Serbia in the anthropometric characteristics and body composition. The first sub-sample of the subjects consisted of 15 water polo players of the Serbian national team, the vice-champions of the FINA World Men's Junior Water Polo Championship in Kuwait 2019, and the second sub-sample consisted of 18 water polo players of the Montenegrin national team, who occupied the sixth position on the championship. The players were tested at the final preparations just before the World Championship in Kuwait in 2019. Anthropometric characteristics and body composition were evaluated using a battery of 11 variables: body height, body weight, triceps skinfold, biceps skinfold, skinfold of the back, abdominal skinfold, upper-leg skinfold, lower leg skinfold, body mass index, fat percentage, and muscle mass. Based on the results, it can be stated that the values of all the variables are highly similar to all of the water polo players of these two countries. The t-test found that the water polo players of the two national teams do not have statistically significant differences in the variables for assessing anthropometric characteristics and body composition. The results obtained in this research can serve as model parameters for the estimated variables for all water polo players (U20) in the World, because the players that have been analysed were the best and the most successful water polo players in their countries, and participants in the World Championship in Kuwait 2019.

Keywords: *Water Polo, Anthropometric Characteristics, Young Athletes, BMI of Athletes*

Introduction

Water polo is a popular sport worldwide. It is a highly dynamic and fast team game that, with its richness of movement, belongs to the category of polystructural sport games. Water polo is a sport characterized by numerous and various complex and dynamic kinesiological activities, which are then characterized by either cyclical or acyclical movement. It is full-contact sport, and rapidly growing sport in the World (Cecchi et al., 2019), characterized by

different swimming intensities, duelling, acceleration and deceleration (Gardasevic, Akpinar, Popovic, & Bjelica, 2019a). In water polo, top scores can be achieved only under conditions of a well-programmed training process. High quality management of the training process depends on knowing the structure of certain anthropological capabilities and water polo players' characteristics, as well as their development. Various studies have been done to establish certain principles and norms for the transformational processes

Correspondence:

**Montenegro
Sport**

J. Gardasevic
University of Montenegro, Faculty for Sport and Physical Education, Narodne omladine bb, 81400 Niksic, Montenegro
E-mail: jovan@ucg.ac.me

of the anthropological characteristics necessary for water polo, with anthropometric characteristics and body composition among them, as expected. Findings regarding anthropometric characteristics and body composition are of crucial importance for complex sports, such as water polo. The anthropometric space is defined by the longitudinal dimension of the skeleton, the transversal dimensionality of the skeleton, and the mass and volume of the body. The purpose of knowing anthropometric characteristics is to improve skills in many sports (Masanovic, Corluka, & Milosevic, 2018). The anthropometric status of top-level athletes is relatively homogeneous, depending on the sport, and can be defined as a model of athletic achievement (Gardasevic, Bjelica, Vasiljevic, & Masanovic, 2020a). Research on anthropometric characteristics and body composition among athletes of different sports indicates that athletes of different sports have specific characteristics (Bjelica, Gardasevic, Vasiljevic, Jeleskovic, & Covic, 2019; Popovic, Akpinar, Jaksic, Matic, & Bjelica, 2013), mostly because absolute size contributes a significant percentage of total variance associated with athletic success (Carvajal et al., 2012). Muscle mass improves performance in activities that require muscular strength and endurance, but also in those that require enviable aerobic ability (Rico-Sanz, 1998).

It is well known that water polo in Serbia and Montenegro have a long tradition and the best results in international competitions. Serbians and Montenegrins were the junior world champions several times. Serbia and Montenegro junior national teams are always top-ranked in Europe and the world.

It was expected that the national teams would continue with good results on the FINA World Men's Junior (U20) Water Polo Championship in Kuwait City (Kuwait) 12 - 20 December 2019,

where twenty national teams participated. It is clear that these were the best players in Serbia, and Montenegro, at age 20, and that they had many years of quality training in order to qualify to wear a representative cap. It is well known in all sports and, therefore, in water polo that long-term and intensive training is one of the critical factors that enable athletes to reach and remain at the elite representative level (Gardasevic et al., 2019a). It became interesting for researchers to determine the models of anthropometric characteristics and body composition of the water polo players who play for these two national teams to determine the differences among them.

This research aimed to determine the anthropometric characteristics and body composition of junior (U20) water polo players of national teams of Serbia and Montenegro, who participated on the FINA World Men's Water Polo Championship 2019 in Kuwait City (Kuwait). The variables between these water polo players were compared, and the possible differences between them were determined.

Methods

Participants

A sample of the subjects consists of a total of 33 water polo players, divided into two sub-samples. The first sub-sample of the subjects consisted of 15 water polo players of the national team of Serbia of an average age of 18.40 ± 1.12 , the vice-champions on the FINA World Men's Junior Water Polo Championship in Kuwait 2019. The other sub-sample consisted of 18 water polo players of the national team of Montenegro of an average age of 18.44 ± 0.98 , who occupied the sixth position on the championship in Kuwait 2019 (Table 1).

Table 1. Final rankings (12 December 2019) at the FINA World Men's Junior Water Polo Championship in Kuwait 2019

National teams	Place
Greece	1
Serbia	2
Italy	3
Croatia	4
Spain	5
Montenegro	6
USA	7
Japan	8
Hungary	9
Canada	10
Australia	11
New Zealand	12
Russia	13
South Africa	14
Egypt	15
China	16
Brazil	17
Uzbekistan	18
Iran	19
Kuwait	20

Procedure

Players of the Montenegrin national team were tested at the final preparations in Niksic (Montenegro), one week before the World Championship in Kuwait. Players of the Serbian nation-

al team were tested at the final preparations in Kragujevac (Serbia), two days before departure for the World Championship in Kuwait. All participants signed the consent form approved by the Institutional Review Board of the University of Montene-

gro, which was in accordance with the Declaration of Helsinki as amended by the World Medical Association Declaration of Helsinki (World Medical Association, 2013). Anthropometric research has been carried out with respect to the basic rules and principles related to the selection of measuring instruments and measurement techniques, standardized in accordance with the International Biological Program guidelines. For this study, eight anthropometric measures have been taken: body height, body weight, triceps skinfold, biceps skinfold, skinfold of the back, abdominal skinfold, upper leg skinfold and lower leg skinfold, and three body composition assessment variables: body mass index, fat percentage and muscle mass. An anthropometer, calliper, and measuring tape were used for anthropometric measurements. To evaluate the body composition, a Tanita body fat scale (model BC-418MA) was used. The scale is based on the principle of the indirect measurement of the body composition; a safe electrical signal is transmitted through the body via electrodes located in

the standalone unit. The Tanita Scale enables athletes to closely monitor their body weight, health condition and form with all relevant parameters.

Statistical analysis

The data obtained through the research were processed using descriptive and comparative statistical procedures. For each variable, central and dispersion parameters have been processed. The significance of the differences between the water polo players of the two national teams in the anthropometric characteristics and variables for assessing body composition was determined by t-tests, with statistical significance of $p < 0.05$.

Results

The variables for assessing anthropometric characteristics and body composition of water polo players of Serbian and Montenegrin national teams are shown in Table 2.

Table 2. Descriptive data and differences between Serbian (N=15) and Montenegrin water polo players (N=18)

Variables	National teams	Mean±SD	Mean Difference	t	Sig.
Body height	Serbia	192.12±5.49	2.52	1.14	0.260
	Montenegro	189.60±6.89			
Body weight	Serbia	90.29±8.52	1.60	0.44	0.660
	Montenegro	88.69±11.72			
Triceps skinfold	Serbia	6.54±2.24	-1.31	-1.62	0.120
	Montenegro	7.85±2.38			
Biceps skinfold	Serbia	5.45±1.32	-0.90	-1.57	0.130
	Montenegro	6.35±1.86			
Skinfold of the back	Serbia	11.90±4.16	-0.21	-0.16	0.870
	Montenegro	12.11±3.31			
Abdominal skinfold	Serbia	13.87±6.47	-2.14	-0.87	0.390
	Montenegro	16.01±7.45			
Upper leg skinfold	Serbia	13.03±3.74	-0.18	-0.13	0.900
	Montenegro	13.21±4.25			
Lower leg skinfold	Serbia	10.96±4.15	1.00	0.75	0.460
	Montenegro	9.96±3.54			
Body mass index	Serbia	24.50±2.54	-0.09	-0.11	0.910
	Montenegro	24.59±2.20			
Fat percentage	Serbia	12.69±3.89	-0.43	-0.30	0.770
	Montenegro	13.13±4.39			
Muscle mass	Serbia	44.51±3.23	1.04	0.72	0.48

Note: Mean - Arithmetic mean; SD - Standard deviation; t - values of t test; Sig. - significant difference

Based on the central and dispersion parameters of the water polo players of Serbia and Montenegro (Table 2), it can be stated that the values of all the variables are very similar to all water polo players of this two countries. There were no significant differences in variables among the water polo players of the two national teams.

Discussion

This study aimed to determine the difference in the anthropometric characteristics and body composition of the junior (U20) water polo players of the Serbian national team, who won a silver medal at the FINA World Water Polo Championship in Kuwait City (Kuwait) 12-20 December 2019, and the water polo players of the Montenegrin national team, who occupied the sixth position at the World Championship. The results were obtained using a

battery of 11 tests in the area of anthropometric characteristics and body composition. By examining the basic descriptive statistical parameters, it can be concluded that we have analysed the best selected junior age water polo players from these two countries. Similar results in their research were obtained by Kondric, Uljevic, Gabrilo, Kontić, & Sekulić (2012). It can be observed that the water polo players of two national teams are of the approximately similar mean values of the all variables analysed, which is not surprising because these are the two national teams of the same age, in countries where water polo is popular and in where water polo coaches are highly skilled. U20 water polo players have years of training experience and spend many hours in the pool each week. The results of t-test showed that the water polo players of two national teams do not differ significantly in the ana-

lysed variables, which is not surprising compared to the results achieved at the World U20 Water Polo Championship in Kuwait 2019. For all variables, some values are better for water polo players of the Serbian national team, some for those of the Montenegrin national team, although, insignificantly for statistics. All of the above mentioned indicates that water polo players of the Serbian and Montenegrin national teams have similar anthropometric parameters and body compositions. All of these players have long-term training before a greatest competitions and they are all top water polo players at the world level, so it is no surprise that there are no differences in anthropometric characteristics and body composition between them. Due to their lifestyle (constant training and sports nutrition), all top athletes take care

of body composition, this is confirmed in research Melchiorri et al. (2018) where did not get differences in body weight and body composition in 13 water polo players after a three-month training program for the Olympic Games. Use by system of bioelectrical impedance for high-level athletes involved in long and intense training periods helps to evaluate the effects of training and to prevent any decrease in the performance level of body composition (Melchiorri et al., 2018).

Given that the concentration of the best water polo players U20 is at World Championship in Kuwait 2019, the assumption is that the mean values of the analysed variables of two national teams' water polo players should be the model values for all such clubs in the world (Table 3).

Table 3. Descriptive data of all 33 water polo players

Variables	Mean±SD
Age	18.42±1.03
Body height (cm)	190.74±6.33
Body weight (kg)	89.41±10.26
Triceps skinfold (mm)	7.25±2.37
Biceps skinfold (mm)	5.94±1.67
Skinfold of the back (mm)	12.02±3.66
Abdominal skinfold (mm)	15.04±6.99
Upper leg skinfold (mm)	13.12±3.96
Lower leg skinfold (mm)	10.41±3.79
Body mass index (kg/m ²)	24.55±2.32
Fat percentage (%)	12.93±4.11
Muscle mass (kg)	43.94±4.10

Note: Mean - Arithmetic mean; SD - Standard deviation

Based on the obtained results in this research, before the start of the World Championship in Kuwait, it could not be assumed which national team would achieve a better placement. The Kuwait Championships showed that they were the national teams in which the nuances decided the final standings. For example, the Serbian national team beat the Montenegrin national team in the quarterfinals with one goal difference in the last minute of the game, thus going into the medal fight. In the finals, the Serbian national team lost to the Greek national team by a small result and thus won a silver medal. After the defeat of the Serbian national team, the Montenegrin national team fought for 5th to 8th place (winning sixth place). All this confirms that these are the best water polo players in the world under the age of 20, many of whom already play for the senior national teams.

All water polo players of the two national teams had similar levels of subcutaneous adipose tissue. Different authors state the importance of body fat as a positive fact in water polo (Platanou, 2005; Peric, Zenic, Mandic, Sekulic, & Sajber, 2012), however in other studies it is not confirmed (Vila, Manchado, Abrales, & Ferragut, 2018), and many researches showed that it is disruptive factor for athletes (Gardasevic, Bjelica, & Vasiljevic, 2020b; Masanovic, 2019; Gardasevic, Bjelica, & Vasiljevic, 2019b). Also, in previous studies of water polo players of this age, subcutaneous adipose tissue has been shown to be a disruptive factor in defence (Milanovic & Vuleta, 2013). It is well known that a low fat percentage is desirable for high physical performance in all sports. Although not every body composition characteristic is expected to play a role in optimal performance in professional sport, lower levels of body fat (that are specific to each player) are desirable for optimal performance, as body mass must be moved against

gravity (Rienzi, Drust, Reilly, Carter, & Martin, 2000; S.M. Gil, J. Gil, Ruiz, A. Irazusta, & J. Irazusta, 2007).

In addition, all the water polo players of the two national teams had similar muscle mass values; water polo is a strenuous sport that takes place in water and requires significant muscle mass. Body height is important for swimming, and long arms are important for kicks and defence; however, there were no statistically significant differences between the water polo players of the two national teams, which is perhaps surprising, considering that the Serbian national team played the final of the World U20 Championship, and the Montenegrin national team dropped out in the quarter-final of the competition. The reason for the different placement may be found in the different levels of technical and tactical preparation, and functional and psychological preparation between water polo players of the two teams. Physical preparation at such championships is essential because it is done every day, and we have not analysed it. Experience in playing deciding matches at this level of competition can be the reason for different placement. The Serbian team has the most experience, and the Montenegrin team has less experience.

The national water polo associations of Serbia and Montenegro should turn to other research studies and check the functional-motoric status, psychological preparation as well as tactical training of their players, and analyse if there are differences at water polo players that influenced the result at this world championship, and whether there is room for improvement. The values obtained in this research can be useful for coaches of these national teams for making a comparison of their players with others and prepare their training in a way that enables the reduction of adverse parameters, and raise the beneficial ones to a higher level.

el. That will surely make their water polo players even better and more successful. The results obtained in this research can serve as model parameters for the estimated variables for water polo players (U20) of all clubs in Serbia and Montenegro, because the players that have been analysed were the best and the most successful water polo players in their countries, and participants in the World Championship in Kuwait 2019.

Acknowledgments

The authors wish to thank the members of the Water Polo Association of Montenegro and Serbia for their cooperation.

Conflict of Interest

The authors declare that there is no conflict of interest.

Received: 17 July 2020 | **Accepted:** 12 August 2020 | **Published:** 16 October 2020

References

- Bjelica, D., Gardasevic, J., Vasiljevic, I., Jeleskovic, E., & Covic, N. (2019). Body Composition and Morphological Characteristics of Soccer Players in Bosnia and Herzegovina. *Kinesiology Slovenica*, 25, 5–13.
- Carvajal, W., Betancourt, H., León, S., Deturnel, Y., Martínez, M., Echevarría, I., Castillo, M.E., & Serviat, N. (2012). Kinanthropometric Profile of Cuban Women Olympic Volleyball Champions. *MEDICC Review*, 14, 16–22.
- Cecchi, N.J., Monroe, D.C., Fote, G.M., Small, S.L., & Hicks, J.W. (2019). Head impacts sustained by male collegiate water polo athletes. *PLoS ONE*, 14, e0216369. doi: 10.1371/journal.pone.0216369
- Gardasevic, J., Akpinar, S., Popovic, S., & Bjelica, D. (2019a). Increased Perceptual and Motor Performance of the Arms of Elite Water Polo Players. *Applied Bionics and Biomechanics*, 6763470. doi: 10.1155/2019/6763470
- Gardasevic, J., Bjelica, D., & Vasiljevic, I. (2019b). Morphological Characteristics and Body Composition of Elite Football Players in Montenegro. *International Journal of Morphology*, 37, 284–288.
- Gardasevic, J., Bjelica, D., Vasiljevic, I., & Masanovic, B. (2020a). Differences in body composition between young soccer players (U19) members of the best soccer clubs in Serbia, Bosnia and Herzegovina, and North Macedonia. *Pedagogy of Physical Culture and Sports*, 24(4), 175–180. doi: 10.15561/26649837.2020.0404
- Gardasevic, J., Bjelica, D., & Vasiljevic, I. (2020b). Differences in anthropometric characteristics between young soccer players (U19) members of the best soccer clubs in Montenegro, Bosnia and Herzegovina, and Kosovo. *Pedagogy of Physical Culture and Sports*, 24(1), 21–25. doi: 10.15561/26649837.2020.0103
- Gil, S.M., Gil, J., Ruiz, F., Irazusta, A., & Irazusta, J. (2007). Physiological and anthropometric characteristics of young football players according to their playing position: relevance for the selection process. *Journal of Strength and Conditioning Research*, 21, 438–445.
- Kondric, M., Uljevic, O., Gabrilo, G., Kontić, D., & Sekulić, D. (2012). General anthropometric and specific physical fitness profile of high-level junior water polo players. *Journal of Human Kinetics*, 32, 157–165. doi: 10.2478/v10078-012-0032-6
- Masanovic, B., Corluka, M., & Milosevic, Z. (2018). Comparative Study of Anthropometric Measurement and Body Composition of Junior Soccer and Handball Players from the Serbian National League. *Kinesiology Slovenica*, 24, 37–46.
- Masanovic, B. (2019). Comparative study of morphological characteristics and body composition between different team players from Serbian junior national league: soccer, handball, basketball and volleyball. *International Journal of Morphology*, 37, 612–619.
- Melchiorri, G., Viero, V., Sorge, R., Triossi, T., Campagna, A., Volpe, S.L., Lecis, D., Tancredi, V., & Andreoli, A. (2018). Body composition analysis to study long-term training effects in elite male water polo athletes. *The Journal of Sports Medicine and Physical Fitness*, 58(9), 1269–1274. doi: 10.23736/S0022-4707.17.07208-5
- Milanovic, D., & Vuleta, D. (2013). Association between morphological dimensions of water polo players and play performance indicators. *Sport Mont*, 11(37–38–39), 447–453.
- Peric, M., Zenic, N., Mandic, G. F., Sekulic, D., & Sajber, D. (2012). The reliability, validity and applicability of two sport-specific power tests in synchronized swimming. *Journal of Human Kinetics*, 32, 135–145.
- Platanou, T. (2005). On-water and dryland vertical jump in water polo players. *Journal of Sports Medicine and Physical Fitness*, 45(1), 26–31.
- Popovic, S., Akpinar, S., Jaksic, D., Matic, R., & Bjelica, D. (2013). Comparative Study of Anthropometric Measurement and Body Composition between Elite Football and Basketball Players. *International Journal of Morphology*, 31(2), 461–467.
- Rienzi, E., Drust, B., Reilly, T., Carter, J. E. L., & Martin, A. (2000). Investigation of anthropometric and work-rate profiles of elite South American international football players. *Journal of Sports Medicine and Physical Fitness*, 40, 162–169.
- Rico-Sanz, J. (1998). Body composition and nutritional assessments in soccer. *International Journal of Sport Nutrition*, 8, 113–123.
- Vila, M.H., Manchado, C., Abrales, J.A., & Ferragut, C. (2018). Predicting playing status in professional water polo players: analysis by gender. *The Journal of Sports Medicine and Physical Fitness*, 58(9), 1234–1239.
- World Medical Association (2013). World Medical Association Declaration of Helsinki: Ethical Principles for Medical Research Involving Human Subjects. *The Journal of the American Medical Association*, 310, 2191–2194

ORIGINAL SCIENTIFIC PAPER

Differences in the Quality of Movement Functionality between Judokas, Karatekas, and Non-Athletes

Amel Mekic¹, Erol Vrevic², Pavle Malovic², Safet Kapo¹, Husnija Kajmovic¹

¹University of Sarajevo, Faculty of Sport and Physical Education, Sarajevo, Bosnia and Herzegovina, ²University of Montenegro, Faculty for Sport and Physical Education, Niksic, Montenegro

Abstract

Judo and karate are polystructural acyclic sports, which require development of a great number of human abilities and characteristics. Although both are martial arts, they differ greatly in the requirements and quality of performance of individual movements. The aim of this study was to determine the differences in the quality of performing movements between karatekas, judokas and non-athletes. The respondents were 60 young people (14.1 - 14.6 years), 20 of which belonged to the group of "judokas", 20 to the group of "karatekas" and 20 to the group of "non-athletes". All individuals were physically and mentally healthy, and were fit to undergo testing. The sample of variables consisted of 7 tests, which belong to the Functional Movement Screen (FMS) method of testing the movement functionality. By using the ANOVA test, in 4 variables showed up statistically significant difference ($p < 0.05$). Group of non-athletes has the lowest values in all tests, especially in deep squat, active straight-leg raise, and rotary stability. Judokas performed the lowest values in the shoulder mobility test. The main conclusions of this paper are that training content should be specific to sports, and sports (in this case martial arts) have a positive effect on the development of the locomotor system, which could be useful for properly perform basic movements which humans use in everyday life.

Keywords: *Functional Movement Screen, Martial arts, Judo, Karate, Non-athletes*

Introduction

The impact of sports on our health has been one of the most important research topics in the field of kinesiology for a long time. The effects of engaging in various sports on physiological and morphological changes in humans are well known thanks to a great number of studies that have been carried out in recent years. However, some authors (Winberg & Gould, 1995; Saint-Phard, Van Dorsten, Marx, & York, 1999) state that the differences between athletes and non-athletes have not been fully explored. Some authors (Butt, 1987; Cox, 1994) state that athletes have some physiological characteristics that completely differentiate them from non-athletes. On the other hand, it has been proved that people who are not engaged in sport more difficulty maintain a proper body posture than athletes do (Baghbani, Woodhouse, & Gaeini, 2016). All these facts, proved by relevant research, show

the positive effects that sports have on human body.

The importance of mobility and flexibility in sports has already been proved (Bompa, 1999; Malacko, & Radjo, 2004; Milanovic, 2013; Bjelica, Georgijev, & Muratovic, 2012). This motor ability plays a major role in performance and in martial arts such as judo and karate. Although these two sports differ significantly in movement structures, both belong to acyclic polystructural sports in which the goal is the symbolic destruction of the opponent (Radjo, Kajmovic, & Kapo, 2001; Kajmovic, Radjo, & Mekic, 2011; Kapo, 2012). A high level of motor skills is necessary for the successful performance of techniques in both judo and karate, where flexibility also plays a significant role in reaching success in these disciplines. One way to diagnose the state of flexibility and possible limitations in performing the full range of motion is to screen the movement functionality.

Correspondence:

**Montenegro
Sport**

A. Mekic
University of Sarajevo, Faculty of Sport and Physical Education, Patriotske lige 41, 71000 Sarajevo, Bosnia and Herzegovina
E-mail: amelmekic21@gmail.com

Functional Movement Screen is a testing protocol used to identify constraints or asymmetries in 7 fundamental steps. These steps are of great importance to the functionality of a person's movements. Functional Movement Screen is becoming increasingly popular in the fields of sports medicine and sports performance (Janiciki, Switzler, Hayes, & Hicks-Little, 2017).

This protocol is designed to give clear and visible performances of the locomotor system by bringing a person into extreme positions where weaknesses and lack of balance become visible due to the lack of necessary mobility and motor control. The idea originated in 1995 when there was no suitable instrument for assessing movement asymmetries in normal motion.

It is noticeable that even the athletes who perform certain activities at a very high level have difficulties performing these basic movements. Functional Movement Screen test allows the trainer to begin the process of regaining movement functionality with individuals who have been diagnosed with pathology (Bardennett, Micca, DeNoyelles, Miller, Jenk, & Brooks, 2015). The goal of Functional Movement Screen test is not to identify orthopedic problems but, as a predictor of injury, to detect limitations or asymmetries in healthy individuals (Kiesel, Plisky, & Voight, 2007; Fuller et al., 2017). Tests that can predict injury in different populations (athletes

and non-athletes) are very important because of the role they can play in primary prevention (Cosio-Lima et al., 2016).

The 7 movements of Functional Movement Screen test are: Deep squat, Crossing the barrier, Lunge, Active lifting of the out-stretched leg in recumbence, Hull stability push-ups assessment, Shoulder mobility, Rotatory stability. (Cook, Burton, Hoogenboom, & Voight, 2014; Cook, Burton, Hoogenboom, & Voight, 2014a)

The aim of this research is to test the functionality of the movement of judokas, karatekas and amateur sportsmen, analyze and draw conclusions about how different sports contribute to the development of movement functionality, and present the specific effects of sports.

Methods

Data in this research was collected using the experimental method, and the research itself, according to the time criterion, is defined as transversal (one measuring point).

Three groups of subjects were involved in this study, and the total number of examinees was 60. The first subsample consisted of 20 judokas, the second subsample consisted of 20 karatekas, and the third subsample consisted of 20 non-athletes (Table 1). The age of the participants was 14.3 years $\pm$ 0.22 on average.

Table 1. Physical characteristics of the participants.

Variables	Judokas (n=20)	Karatekas (n=20)	Non-athletes (n=20)
Age	14.1	14.2	14.6
Body height	171.4	164.8	169.5
Body weight	65.7	58.6	58.5

All subjects were physically and mentally healthy and able to undergo testing and analyze the results obtained by testing movement functionality. Before the start of testing every respondent has been informed about the aim of the study. Protocol of study has been explained and every test has been demonstrated. All respondent gave their signature of approval with the possibility of giving up at every moment of testing.

Respondents from the first two groups (judokas and karatekas) have participated in the training process for at least 4 years and have at least 75% training access.

For the purposes of this study, the following 7 variables of Functional Movement Screen test method were evaluated: deep squat, hurdle step, in-line lunge, active straight-leg raise, the trunk stability push-up, rotary stability, shoulder mobility.

The quality of movement performance is evaluated with a score of 1-3, where 1 - the movement was performed completely incorrectly, 2 - the movement was performed with little difficulty and

3 - the movement was performed correctly. Each movement was evaluated 3 times, and the average value in the form of a mode (the most frequent rating) was used as the final grade (Cook et al., 2014; Cook et al., 2014a).

The statistical package SPSS v20 will be used for statistical data processing. Descriptive statistics, discriminatory measurements and a statistical significance test will be used to process the data.

Univariate analysis of variance (ANOVA) and LSD post hoc test were used to analyze differences between groups, and the significance of this test has been determined at the alpha level of $p < 0.05$.

Results

Table 2. shows the values of the parameters of descriptive statistics for the group of "judokas". It is noticeable that the skewness parameters in the 3th, 4th and 6th variables are above the normal value of this indicator. Also, the Kurtosis parameters for the 3th and 4th variables are above the normal values.

Table 2. Descriptive statistics of "judokas" group

Variables	Range	Min	Max	Mean	SD	Skewness		Kurtosis	
						Statist.	SD	Statist.	SD
Deep squat	1.00	2.00	3.00	2.50	.512	.000	.512	-2.235	.992
Hurdle step	1.00	2.00	3.00	2.30	.470	.945	.512	-1.242	.992
In-line lunge	1.00	2.00	3.00	2.90	.307	-2.888	.512	7.037	.992
Active straight-leg raise	1.00	2.00	3.00	2.90	.307	-2.888	.512	7.037	.992
The trunk stability push-up	1.00	2.00	3.00	2.55	.510	-.218	.512	-2.183	.992
Rotary stability	1.00	2.00	3.00	2.80	.410	-1.624	.512	.699	.992
Shoulder mobility	2.00	1.00	3.00	2.35	.670	-.549	.512	-.548	.992

Note: Min - the level of lowest score; Max - the level of the highest score; Mean - arithmetic mean; SD - standard deviation.

Table 3. shows the values of descriptive statistics parameters for the group of "karatekas". Speaking of data distribution, the values of Skewness for the 3th and 4th variables are slightly above

normal values (as in the group "judokas"). On the other hand, all Kurtosis values are within normal values.

Table 4. shows the values for the group of "non-athletes". It

Table 3. Descriptive statistics of “karatekas” group

Variables	Range	Min	Max	Mean	SD	Skewness		Kurtosis	
						Statist.	SD	Statist.	SD
Deep squat	1.00	2.00	3.00	2.60	.502	-.442	.512	-2.018	.992
Hurdle step	1.00	2.00	3.00	2.60	.502	-.442	.512	-2.018	.992
In-line lunge	1.00	2.00	3.00	2.80	.410	-1.624	.512	.699	.992
Active straight-leg raise	1.00	2.00	3.00	2.80	.410	-1.624	.512	.699	.992
The trunk stability push-up	1.00	2.00	3.00	2.60	.502	-.442	.512	-2.018	.992
Rotary stability	1.00	2.00	3.00	2.75	.444	-1.251	.512	-.497	.992
Shoulder mobility	.00	3.00	3.00	3.00	.000	.	.	.	.

is noticeable that the skewness parameters in the 6th variable are above the normal. The Kurtosis parameters for all variables are

within normal values.

Based on Table 5, which shows the results of the ANOVA test, it

Table 4. Descriptive statistics of “non-athletes” group

Variables	Range	Min	Max	Mean	SD	Skewness		Kurtosis	
						Statist.	SD	Statist.	SD
Deep squat	2.00	1.00	3.00	2.05	.604	-.012	.512	.189	.992
Hurdle step	1.00	2.00	3.00	2.40	.502	.442	.512	-2.018	.992
In-line lunge	1.00	2.00	3.00	2.65	.489	-.681	.512	-1.719	.992
Active straight-leg raise	1.00	2.00	3.00	2.50	.512	.000	.512	-2.235	.992
The trunk stability push-up	2.00	1.00	3.00	2.20	.695	-.292	.512	-.734	.992
Rotary stability	2.00	1.00	3.00	2.35	.587	-.212	.512	-.552	.992
Shoulder mobility	2.00	1.00	3.00	2.70	.571	-1.845	.512	2.861	.992

is noticeable that in 4 variables exist statistically significance between some groups, even at the value of $p < 0.01$. These 4 variables are: deep

squat, active straight-leg raise, rotary stability, and shoulder mobility.

Table 6. shows the results of the LSD post hoc test. This test

Table 5. Differences in tested variables analyzed by using ANOVA test

Variables		Sum	Df	Mean Square	F	Sig.
Deep squat	Between the groups	3.433	2	1.717	5.842	.001*
	Within the groups	16.750	57	.294		
	Total	20.183	59			
Hurdle step	Between the groups	.933	2	.467	1.928	.015
	Within the groups	13.800	57	.242		
	Total	14.733	59			
In-line lunge	Between the groups	.633	2	.317	1.890	.016
	Within the groups	9.550	57	.168		
	Total	10.183	59			
Active straight-leg raise	Between the groups	1.733	2	.867	4.940	.001*
	Within the groups	10.000	57	.175		
	Total	11.733	59			
The trunk stability push-up	Between the groups	1.900	2	.950	2.858	.007
	Within the groups	18.950	57	.332		
	Total	20.850	59			
Rotary stability	Between the groups	2.433	2	1.217	5.137	.001*
	Within the groups	13.500	57	.237		
	Total	15.933	59			
Shoulder mobility	Between the groups	4.233	2	2.117	8.180	.001*
	Within the groups	14.750	57	.259		
	Total	18.983	59			

Note: Sum - Measure of variation or deviation from the mean; Df - Degrees of freedom; Mean Square - Represents the variation between the sample means and is used to determine whether factors (treatments) are significant; F - Values of F test; Sig - Statistical significance; * - Statistical significance at level of $p < 0.05$.

shows more clearly where are differences between groups, and in which variables. Thus, the differences are noticeable in following

variables: deep squat, active straight-leg raise, the trunk stability push-up, rotary stability, and shoulder mobility.

Table 6. Differences between groups in tested variables analyzed by using the LSD post hoc test

Variables	(I) Group	(J) Group	Mean (I-J)	SD	Sig.	95% Confidence interval	
						Upper	Lower
Deep squat	Non-athletes	Judokas	-.450	.171	.001*	-.793	-.106
		Karatekas	-.550	.171	.001*	-.893	-.206
	Judokas	Non-athletes	.450	.171	.001*	.106	.793
		Karatekas	-.100	.171	.056	-.443	.243
	Karatekas	Non-athletes	.550	.171	.001*	.206	.893
		Judokas	.100	.171	.056	-.243	.443
Hurdle step	Non-athletes	Judokas	.100	.155	.052	-.211	.411
		Karatekas	-.200	.155	.020	-.511	.111
	Judokas	Non-athletes	-.100	.155	.052	-.411	.211
		Karatekas	-.300	.155	.006	-.611	.011
	Karatekas	Non-athletes	.200	.155	.020	-.111	.511
		Judokas	.300	.155	.006	-.011	.611
In-line lunge	Non-athletes	Judokas	-.250	.129	.006	-.509	.009
		Karatekas	-.150	.129	.025	-.409	.109
	Judokas	Non-athletes	.250	.129	.006	-.009	.509
		Karatekas	.100	.129	.044	-.159	.359
	Karatekas	Non-athletes	.150	.129	.025	-.109	.409
		Judokas	-.100	.129	.044	-.359	.159
Shoulder mobility	Non-athletes	Judokas	.350	.160	.003*	.027	.672
		Karatekas	-.300	.160	.007	-.622	.022
	Judokas	Non-athletes	-.350	.160	.003*	-.672	-.027
		Karatekas	-.650	.160	.000*	-.972	-.327
	Karatekas	Non-athletes	.300	.160	.007	-.022	.622
		Judokas	.650	.160	.000	.327	.972
Active straight-leg raise	Non-athletes	Judokas	-.400	.132	.001	-.665	-.134
		Karatekas	-.300	.132	.003	-.565	-.034
	Judokas	Non-athletes	.400	.132	.001*	.134	.665
		Karatekas	.100	.132	.045	-.165	.365
	Karatekas	Non-athletes	.300	.132	.003*	.034	.565
		Judokas	-.100	.132	.045	-.365	.165
The trunk stability push-up	Non-athletes	Judokas	-.350	.182	.006	-.715	.015
		Karatekas	-.400	.182	.003*	-.765	-.034
	Judokas	Non-athletes	.350	.182	.006	-.015	.715
		Karatekas	-.050	.182	.079	-.415	.315
	Karatekas	Non-athletes	.400	.182	.003*	.034	.765
		Judokas	.050	.182	.079	-.315	.415
Rotary stability	Non-athletes	Judokas	-.450	.153	.001*	-.758	-.141
		Karatekas	-.400	.153	.001*	-.708	-.091
	Judokas	Non-athletes	.450	.153	.001*	.141	.758
		Karatekas	.050	.153	.075	-.258	.358
	Karatekas	Non-athletes	.400	.153	.002*	.091	.708
		Judokas	-.050	.153	.075	-.358	.258

Note: * - Statistical significance at level of $p < 0.05$.

Discussion

By using the univariate analysis of variance (ANOVA), and analyzing obtained results it is evident that in 4 variables exist statistically significant difference between groups. Thus, those were following variables: deep squat, shoulder mobility, active straight-leg raise and rotary stability.

However, it is much more important to determine the differences between groups for each variable where exist statistical significance, and that was possible by using the LSD post hoc test. Thus, the quality of movement performance between karate and judokas differs statistically only in the shoulder mobility test ($p < 0.01$), in favor of karatekas. It is important to say that the structures of the movements performed by the judokas in the shoulder joint are much more complex compared to the movements performed by the karatekas. However, this fact can explain the difference between judokas and karatekas, since the levers and throws performed on this joint can leave a negative mark, which is later manifested when performing the movement functionality test in that joint. This thesis has not been scientifically confirmed, so it should be considered as a hypothesis.

Non-athletes statistically significant differ from karatekas in 4 tested variables of Functional Movement Screen test: deep squat, active straight-leg raise, the trunk stability push-up, and rotary stability. The group of karatekas has better results in all the above mentioned variables than the non-athletes group. This is proof that karate positively affects the movement functionality of these body parts.

Judokas and non-athletes statistically differ in 2 variables: active straight-leg raise and rotary stability.

When we look at the variables in which non-athletes differ from athletes (in this case karatekas and judokas), it is noticeable that this difference is primarily caused by the structure of the sports themselves. Deep squat movement presents a problem for many non-athletes, since the "neglected" and inflexible tendons and muscles do not allow proper movement. A great number of non-athletes have shortened Achilles tendons, which results in the improper performance of this movement. Also, sports such as karate and judo require athletes to have a high level of muscle and tendon flexibility, which is very significant in reducing the risk of injury. Therefore, the results of athletes in the variable active straight-leg raise are primarily due to the greater flexibility of the biceps femoris (hamstring) of those who exercise in comparison with those who do not. Shoulder mobility is a variable whose results differ from the abovementioned. In this variable, karatekas achieved the best result, followed by non-athletes, and the worst results had group of judokas. It is assumed that the shoulder joint of judokas loses its mobility by the long-term of doing this sport, because judo is a sport in which judokas are in constant contact (guard) during the shoulder girdle muscles are under relatively long and heavy load. Another possible reason is that a large number of levers and judo throws being performed over the shoulder part of the body. Although lever techniques are not allowed in competitions at this age, they are applied through the training process, which leaves a mark on the connective tissue of the shoulder girdle muscles. On the other hand, it is evident that in karate do not exist movements which in long term limit the mobility in the shoulder joint, and all exercises in this sport which are used to strengthened shoulder girdle muscles positively affect the coordination ability and performance of movements in this joint. Both karate and judo are acyclic sports, which means that the positions of the body cannot be predicted with great precision. Therefore, during these activities, the body is put into a great number of new positions, which are very unusual for non-athletes. The basis of safely performing torso rotations is a strong musculature that will support those movements. The results in

the rotary stability test show that the judokas achieved the best result. They are followed by karatekas, while the worst results were achieved by non-athletes. These results are not surprising, given that judo is a sport that requires different forms of torso rotations, while coping with external resistance during throws and falls, and a judoka is therefore expected to have the strongest muscles that perform torso rotations.

Thus, in this study, as also confirmed in study conducted by Boguszewski, Jakubowska, Adamczyk, & Białoszewski (2015), non-athletes has the lowest values in all tests. That were especially evident in deep squat, active straight-leg raise and rotary stability, while judokas had the lowest values in variable shoulder mobility, what consider usual for these type of athletes (Ciz, Štefanovský, Matejová, & Lopata, 2017; Šimenko, 2019). Although karate and judo belong to the same group of sports, this research has proved that they can have different effects on the movement functionality of individual joints (especially the shoulder mobility). Therefore, it is not the same for people (especially for children) to do any kind of sport, because it is evident that they differently affect the development of the human body. This research is also another piece of evidence that specific training content, that determines the structure of the sport, should be used for karate and judo, and that the structure of training units cannot be identical.

On the other hand, generally speaking, if the "karate" and "judo" groups are combined into one group of athletes, the benefits of exercising on the movement functionality are evident. The group of "non-athletes" achieved the worst results in almost all variables, which indicates the importance of exercise for the proper development of the locomotor system. Thus, conducting the Functional Movement Screen test method allows trainers to be aware of the importance of the implementation of individual training type, to improve the movement patterns of athletes (Boguszewski, Jakubowska, Adamczyk, & Białoszewski, 2015).

Therefore, the basic conclusions of this paper are that training content should be specific to a sport, and that doing sports (in this case martial arts) has a positive effect on proper performing basic movements which humans use in everyday life.

Acknowledgements

There are no acknowledgements.

Conflict of Interest

The authors declare that there are no conflicts of interest.

Received: 25 August 2020 | **Accepted:** 18 September 2020 | **Published:** 16 October 2020

References

- Baghbani, F., Woodhouse, L.J., & Gaeini, A.A. (2016). Dynamic Postural Control in Female Athletes and Non athletes After a Whole-Body Fatigue Protocol. *Journal of strength and conditioning research*, 30(7), 1942–1947. doi:10.1519/JSC.0000000000001275
- Bardenett, S.M., Micca, J.J., DeNoyelles, J.T., Miller, S.D., Jenk, D.T., & Brooks, G.S. (2015). FUNCTIONAL MOVEMENT SCREEN NORMATIVE VALUES AND VALIDITY IN HIGH SCHOOL ATHLETES: CAN THE FMS™ BE USED AS A PREDICTOR OF INJURY?. *International journal of sports physical therapy*, 10(3), 303–308.
- Bjelica, D., Georgiev, G., & Muratovic, A. (2012). Basic motor abilities of young handball players from Montenegro. *Sport Science*, 5(1), 71–76.
- Boguszewski, D., Jakubowska, K.L., Adamczyk, J., & Białoszewski, D. (2015). The assessment of movement patterns of children practicing karate using the Functional Movement Screen test. *Journal of combat sports and martial arts*, 6, 21–26.
- Bompa, T. (1999). *Periodization: theory and methodology training*. Champaign, IL: Human Kinetics.
- Butt, D.S. (1987). *Personality of the athlete*. New York: The psychology of sport.
- Ciz, I., Štefanovský, M., Matejová, L., & Lopata, P. (2017). Functional movement screening of elite judokas. In M. Chren, K. Péliová, & E. Sulovská (Eds.), *Nové vedecké poznatky v gymnastike, tancoch, úpoloch, fitness a silových športoch* (pp. 13–19). Bratislava.
- Cook, G., Burton, L., Hoogenboom, B.J., & Voight, M. (2014). Functional

- movement screening: the use of fundamental movements as an assessment of function - part 1. *International journal of sports physical therapy*, 9(3), 396–409.
- Cook, G., Burton, L., Hoogenboom, B.J., & Voight, M. (2014a). Functional movement screening: the use of fundamental movements as an assessment of function-part 2. *International journal of sports physical therapy*, 9(4), 549–563.
- Cosio-Lima, L., Knapik, J.J., Shumway, R., Reynolds, K., Lee, Y., Greska, E., & Hampton, M. (2016). Associations Between Functional Movement Screening, the Y Balance Test, and Injuries in Coast Guard Training. *Military medicine*, 181(7), 643–648. doi:10.7205/MILMED-D-15-00208
- Cox, R.H. (1994). *Sport psychology: concepts and applications*. Dubuque: Brown and Benchmark.
- Fuller, J.T., Chalmers, S., DeBenedictis T.S., Townsley, S., Lynagh, M., Gleeson, C., Zacharia, A., Thomson, S., & Magarey, M. (2017). High prevalence of dysfunctional, asymmetrical, and painful movement in elite junior Australian Football players assessed using the Functional Movement Screen. *Journal of Science and Medicine in Sport*, 20(2), 134–138. doi:10.1016/j.jsams.2016.05.003
- Janicki, J.J., Switzler, C.L., Hayes, B.T., & Hicks-Little, C.A. (2017). Correlation Between Ankle-Dorsiflexion and Hip-Flexion Range of Motion and the Functional Movement Screen Hurdle-Step Score. *Journal of sport rehabilitation*, 26(1), 35–41. Doi:10.1123/jsr.2015-0070
- Kajmovic, H., Radjo, I., & Mekic, A. (2011). *Judo: Analiza situacijske efikasnosti*. Sarajevo: Fakultet sporta i tjelesnog odgoja.
- Kapo, S. (2012). *Karate pronadi svoj put*. Sarajevo: Fakultet sporta i tjelesnog odgoja.
- Kiesel, K., Plisky, P.J., & Voight, M.L. (2007). Can Serious Injury in Professional Football be Predicted by a Preseason Functional Movement Screen?. *North American journal of sports physical therapy*, 2(3), 147–158.
- Malacko, J., & Radjo, I. (2004). *Tehnologija sporta i sportskog treninga*. Sarajevo: Fakultet sporta i tjelesnog odgoja.
- Milanovic, D. (2013). *Teorija treninga*. Zagreb: Kineziološki fakultet.
- Radjo, I., Kajmovic, H., & Kapo, S. (2001). *JUDO*. Sarajevo: Fakultet za fizičku kulturu.
- Saint-Phard, D., Van Dorsten, B., Marx, R.G., & York, K.A. (1999). Self-perception in Elite collegiate female gymnastics, cross-country runners and track-and-field athletes. *Mayo Clinic Proceedings*, 74(8), 770-774.
- Šimenko, J. (2019). The benefits of Functional Movement Screen in judo. *Revista de Artes Marciales Asiáticas*, 14(2s), 18-20. DOI: 10.18002/rama.v14i2s.5988
- Weinberg, D.S., & Gould, D. (1995). *Foundations of sport and exercise psychology*. Champaign, IL: Human Kinetics.

ORIGINAL SCIENTIFIC PAPER

Comparative Analysis of Anthropometric Parameters as Obesity Indicators for Sixth Grade Boys from Different Regions in Montenegro

Ivana Cerkez Zovko¹, Milena Mitrovic², Marin Corluca¹

¹University of Mostar, Faculty of Mathematics and Science Education, Mostar, Bosnia and Herzegovina, ²University of Montenegro, Faculty for Sport and Physical Education, Niksic, Montenegro

Abstract

The aim of this study was to measure the morphological characteristics and determine the nutritional levels of boys aged 11-12 years from the North, Central and Southern regions of Montenegro, as well as to compare the results, to determine whether differences in nutritional levels occur between boys living in different regions. The sample consisted of 42 sixth grade boys (11-12 years old) from Bijelo Polje (Northern Region), Podgorica (Central Region) and Budva (Southern Region). Four variables were measured in the study: body height, body weight, waist circumference and hip circumference. Basic central and dispersion descriptive parameters were calculated for all variables. Obesity levels were determined using three indices: body mass index, waist-to-hip ratio index, and waist-to-height ratio index. The results showed a fairly high percentage of obesity when it comes to all three indices, however, statistically significant differences between boys from different regions occurred only when it comes to overweight and obesity according to the waist-hip ratio index and waist-to-waist ratio index. In the first case, boys from the Northern region were statistically significantly obese than the others, and in the second case, boys from the Southern region were statistically significantly more obese than boys from the other two regions. In any case, given the rather high percentage of obesity, it is necessary to take certain measures to reduce the number of obese boys in all three regions.

Keywords: Morphology, Obesity, Differences, Sixth Grade, Boys

Uvod

Gojaznost je hronično oboljenje koje se karakteriše povećanjem masne mase tijela, u mjeri koja dovodi do narušavanja zdravlja i razvoja niza komplikacija (Branca et al., 2007; Vukušić i sar., 2015). Osnovu gojaznosti čini disbalans u kalorijskom unosu sa jedne strane i kalorijskoj potrošnji sa druge strane (WHO, 2016). Gojaznost predstavlja veliki javno zdravstveni problem XXI vijeka, a čak 124 miliona djece i adolescenata u svijetu je prekomjerno uhranjeno i gojazno (WHO, 2016). Istraživanja gojaznosti sprovedena u Crnoj Gori takođe pokazuju zabrinjavajuće rezultate. Procenat gojazne djece od 6 do 9 godina u Crnoj Gori iznosi 19.9% (Rito et al., 2019), dok je taj procenat na uzorku ispitanika djece i adolescenata od 6 do 16

godina još veći i iznosi 40% (Tarp et al., 2018).

Osnovni uzroci nastanka gojaznosti kod djece su nedovoljna fizička aktivnost i prekomjerni unos hrane (Barlow, 2007; Skinner et al., 2008; Dinarević i sar., 2011). Poznato je da djeca sve više vremena provode za računarom i ispred televizora, a sve manje u prirodi, kao i da konzumiraju brzu i nezdravu hranu i gazirane napitke, što dovodi do povećane tjelesne težine. Pretjeranim unosom nezdrave hrane i nedovoljnom fizičkom aktivnošću dolazi do disbalansa u organizmu, a rezultat toga je povećan BMI, odnosno prekomjerna tjelesna težina (Freedman et al., 2004; Jansen et al., 2008). Iz navedenog, primjećuje se da je gojaznost vrlo složen problem, koji može dovesti do niza zdravstvenih problema, kao što su poremećaj metabolizma, bolesti kardiovaskularnog i endo-

Correspondence:

**Montenegro
Sport**

M. Mitrovic
University of Montenegro, Faculty for Sport and Physical Education, Narodne omladine bb, 81400 Niksic, Montenegro
E-mail: milenam1054@gmail.com

krinog sistema, respiratornih oboljenja, kao i promjena psihičkog stanja (Strauss, 2000; Radisavljević, 2004; Jocić Stojanović, 2010; Vlajković i sar., 2015; Jocić Stojanović i sar., 2015). Takođe, studije su pokazale da djeca čija su oba roditelja gojazna imaju veći rizik za gojaznost nego ona čiji roditelji nijesu gojazni ili je gojazan samo jedan od roditelja (Power et al., 1997), kao i veću šansu da gojaznost u detinjstvu bude prisutna i u odraslom dobu (Maffei et al., 1998; Burke, & Wang, 2001; Magarey et al., 2003).

Fizička aktivnost djece, a i ishrana djece, često zavise od uslova koje im pruža sredina u kojoj žive. U Crnoj Gori postoje tri regije (Sjeverna, Centralna i Južna), koje svakako pružaju različite uslove života svojim stanovnicima. S obzirom na to, mogu se javiti razlike u motoričkim i morfološkim karakteristikama, a samim tim i u parametrima koji predstavljaju pokazatelje gojaznosti. Istraživanja sprovedena u Crnoj Gori pokazala su različite procenete gojaznosti u pojedinim regijama (Mitrović, & Dragutinović, Bacović, 2020; Malović, 2020). S obzirom na to, cilj ovog rada je izmjeriti morfološke karakteristike (tjelesna visina, tjelesna težina, obim struka, obim kukova) i utvrditi nivoe uhranjenosti dječaka uzrasta 11-12 godina iz Sjeverne, Centralne i Južne regije. Drugi cilj rada jeste upoređivanje dobijenih rezultata, kako bi utvrdili da li se javlja razlika u nivoima uhranjenosti između dječaka koji žive u različitim regijama Crne Gore. Na taj način, moguće je utvrditi u kojim je procentima gojaznost zastupljena kod dječaka uzrasta 11-12 godina u pojedinim regijama Crne Gore, kao i sprovođenje određenih mjera za smanjivanje tog broja, prije svega, zbog njenog štetnog uticaja na zdravlje.

Metod

Prema prirodi istraživanja, ovo istraživanje spada u empirijska. Prema vremenskoj usmjerenosti, istraživanje je transversalnog karaktera. Uzorak ispitanika u istraživanju činilo je 42 dječaka šestog razreda (11-12 godina) osnovnih škola iz Bijelog Polja (Sjeverna regija), Podgorice (Centralna regija) i Budve (Južna regija). Učenici su podijeljeni na 3 subuzorka, prema kriterijumu mjesta

stanovanja. Prvi subuzorak činilo je 12 dječaka iz Bijelog Polja, drugi 21 dječak iz Podgorice i treći 9 dječaka iz Budve. Odabrane varijable su izmjerene u skladu sa propisanim ISAK priručnikom, a podaci su uneseni u mjerne liste, zatim u matrice, koje su obrađene adekvatnim statističkim procedurama. Varijable mjerene u ovom istraživanju su: visina tijela, masa tijela, obim struka i obim kukova. Visina tijela je izmjerena antropometrom po Martinu na nivou tačnosti 0.01 cm, masa tijela je izmjerena pomoću digitalne vage na kojoj se očitavaju rezultati, dok su preostale 3 varijable mjerene pomoću santimetarske trake na nivou tačnosti od 0.01 cm. Pisanu saglasnost za sprovođenje mjerenja dali su direktori škola, a ono je realizovano u salama škola krajem 2019. godine. Učenici su mogli dobrovoljno da odustanu od testiranja, međutim, niko to nije učinio.

Za svaki subuzorak je izračunat indeks tjelesne mase (BMI), indeks odnosa struka i kukova (WHR) i indeks odnosa struka i visine (WtHR). Za evaluaciju rezultata za indeks tjelesne mase (BMI), koji predstavlja odnos mase tijela sa kvadratom tjelesne visine, i koji se kod djece izražava u percentilima, prema preporuci Centra za kontrolu i prevenciju bolesti, predložena je sljedeća klasifikacija: neuhranjenost (<5 percentila), normalna težina (5-85 percentila), prekomjerna težina (85-95 percentila) i gojaznost (>95 percentila). Evaluacija WHR indeksa, koji predstavlja odnos struka i kukova, urađena je prema sljedećoj skali: normalna uhranjenost (<0.9) i prekomjerna uhranjenost i gojaznost (>0.9). Evaluacija WtHR indeksa, koji predstavlja odnos struka i visine tijela, urađena je tako što su vrijednosti preko 0.5 smatrane rizikom za nastanak gojaznosti. Za sve varijable su izračunati osnovni centralni i disperzioni deskriptivni parametri. Nivoi uhranjenosti su prikazani brojčano i procentualno. Upoređivanje kategorija uhranjenosti između subuzoraka izvršeno je Hi kvadrat testom na nivou značajnosti od $p < 0.05$.

Rezultati

U Tabeli 1 prikazani su osnovni centralni i disperzioni deskriptivni parametri dječaka po regijama.

Tabela 1. Osnovni centralni i disperzioni parametri dječaka po regijama

Varijable	Subuzorak	Mean	SD
Tjelesna visina	Sjeverna regija	157.7	64.63
	Centralna regija	150.3	56.35
	Južna regija	162.3	87.01
Tjelesna težina	Sjeverna regija	46.4	9.12
	Centralna regija	45.8	9.32
	Južna regija	52.6	11.28
Obim struka	Sjeverna regija	74.0	10.45
	Centralna regija	68.4	8.52
	Južna regija	78.3	10.84
Obim kukova	Sjeverna regija	84.1	6.77
	Centralna regija	79.4	7.97
	Južna regija	89.5	10.93

Legenda: Mean – aritmetička sredina; SD – standardna devijacija

Iz Tabele 1 se može vidjeti da su najveća prosječna tjelesna visina i tjelesna masa izmjerene kod dječaka u Južnoj regiji (162.3 ± 87.01 cm; 52.6 ± 11.8 kg), a najmanja kod dječaka u Centralnoj regiji (150.3 ± 56.35 cm; 45.8 ± 9.32 kg). Identična je situacija kada se pogledaju prosječne vrijednosti obima struka i obima kukova, koje su takođe najviše kod dječaka izmjenjenih u Južnoj regiji (78.3 ± 10.84 cm; 89.5 ± 10.93), a najmanje kod dječaka izmje-

renih u Centralnoj regiji (68.4 ± 8.52 cm; 79.4 ± 7.97 cm).

U Tabeli 2 je dat brojčani i procentualni prikaz nivoa uhranjenosti dječaka iz sve tri regije prema indeksu tjelesne mase, indeksu odnosa struka i kukova i indeksu odnosa struka i tjelesne visine.

Kada je u pitanju indeks tjelesne mase (BMI), iz Tabele 2 može se vidjeti da je najviši procenat gojaznih dječaka izmjenjen u Južnoj regiji (22%), dok ih u Sjevernoj regiji nije bilo uopšte. Najviši pro-

Tabela 2. Brojčani i procentualni prikaz nivoa uhranjenosti dječaka iz sve tri regije prema BMI, WHR i WtHR

Indeksi	N	Sjeverna regija		Centralna regija		Južna regija	
		%	N	%	N	%	N
BMI	<5	1	8	0	0	0	0
	5-85	9	75	14	67	7	78
	85-95	2	17	4	19	0	0
WHR	>95	0	0	3	14	2	22
	<0.9	8	67	19	90	7	78
	>0.9	4	33	2	10	2	22
WtHR	<0.5	9	75	18	86	5	56
	>0.5	3	25	3	14	4	44

Legenda: BMI – indeks tjelesne mase, WHR – indeks odnosa struka i kukova; WtHR – indeks odnosa struka i tjelesne visine; N – broj; % - procenat

cenat prekomjerno uhranjenih dječaka izmjenen u Centralnoj regiji (19%), dok ih u Južnoj regiji nije bilo uopšte. Pregledom tabele takođe se može vidjeti da je broj neuhranjene djece veoma mali, u Sjevernoj regiji ih ima najviše (8%), dok ih u preostale dvije regije uopšte nema. Na samom kraju, procenat normalno uhranjene djece najviše je u Južnoj regiji (78%), a najmanji u Centralnoj (67%). Kada je u pitanju procenat prekomjerno uhranjenih i gojaznih dječaka zajedno, primjećuje se da ih najviše ima u Centralnoj regiji (33%), a najmanje u Sjevernoj regiji (17%)

Vrijednosti indeksa odnosa obima struka i kukova (WHR) iz

Tabele 2 pokazuju da je najviše procenat prekomjerno uhranjenih i gojaznih dječaka u Sjevernoj regiji (33%), a najmanji u Centralnoj regiji (10%).

Kada je u pitanju indeks odnosa obima struka i tjelesne visine (WtHR), njegove vrijednosti pokazuju da je najviše procenat prekomjerno uhranjenih i gojaznih dječaka u Južnoj regiji (44%), a najmanji u Centralnoj regiji (14%)

U Tabeli 3 prikazani su rezultati Hi kvadrat testa na nivou značajnosti od $p < 0.05$, kojim su upoređeni nivoi uhranjenosti prema indeksu tjelesne mase.

Tabela 3. Hi kvadrat test ($p < 0.05$) za upoređivanje nivoa uhranjenosti prema BMI

Nivo uhranjenosti	Chi Square	df	Sig.
Neuhranjenost	-	-	-
Normalna tjelesna težina	.667 ^a	1	.414
Prekomjerena tjelesna težina	2.600 ^b	2	.273
Gojaznost	.200 ^c	1	.655

Legenda: Chi Square – Pirsonov Hi kvadrat test; df – stepeni slobode; Sig. – statistička značajnost

Iz Tabele 3 može se vidjeti da ne postoje statistički značajne razlike u nivoima uhranjenosti između dječaka iz Sjeverne, Cen-

tralne i Južne regije kada je u pitanju indeks tjelesne mase.

Obzirom da je gradacija za indeks tjelesne mase četvorostepena

Tabela 4. Hi kvadrat test ($p < 0.05$) za upoređivanje nivoa uhranjenosti prema BMI

Nivo uhranjenosti	Chi Square	df	Sig.
Normalna tjelesna težina i neuhranjenost	.667 ^a	1	.414
Prekomjerena tjelesna težina i gojaznost	4.545 ^c	2	.103

a za sledeća dva indeksa dvostepena, postavilo se pitanje da li bi se pojavila statistički značajna razlika i za ovaj indeks kada bi se spojili rangovi prekomjerna uhranjenost i gojaznost, da bi se i tu napravila dvostepena gradacija. Rezultat je pokazao da ni u tom slučaju značajna razlika ne postoji između dječaka iz Sjeverne, Centralne i

Južne regije kada je u pitanju indeks tjelesne mase (Tabela 4).

U Tabeli 5 prikazani su rezultati Hi kvadrat testa na nivou značajnosti od $p < 0.05$, kojim su upoređeni nivoi uhranjenosti prema indeksu odnosa obima struka i kukova i indeksu odnosa obima struka i tjelesne visine između subzoraka.

Tabela 5. Hi kvadrat test ($p < 0.05$) za upoređivanje nivoa uhranjenosti prema WHR i WtHR

Nivo uhranjenosti	WHR			WtHR		
	Chi Square	df	Sig.	Chi Square	df	Sig.
Normalna tjelesna težina	1.000 ^a	2	.607	.200 ^a	2	.905
Prekomjerna tjelesna težina i gojaznost	7.824 ^b	2	.020	8.313 ^b	2	.016

Iz Tabele 5 može se primijetiti da ne postoje statistički značajne razlike kada je u pitanju normalna tjelesna težina kod dječaka iz Sjeverne, Centralne i Južne regije prema indeksu odnosa obima struka i kukova i indeksu odnosa obima struka i tjelesne visine. Međutim, kada je u pitanju prekomjerna tjelesna težina, prema in-

deksu odnosa obima struka i kukova i indeksu odnosa obima struka i tjelesne visine, javile su se statistički značajne razlike između dječaka iz Sjeverne, Centralne i Južne regije. Prema WHR indeksu, dječaci iz Sjeverne regije su više gojazni u odnosu na dječake iz Južne i Centralne regije, dok su prema WtHR indeksu dječaci iz

Južne regije gojazniji od dječaka iz Sjeverne i Centralne regije.

Diskusija

Rezultati dosadašnjih istraživanja pokazuju da je broj gojazne i prekomjerno uhranjene djece u stalnom porastu (Dinarevic et al., 2011; NCD Risk Factor Collaboration, 2016; 2017; Masanovic et al., 2020). Posmatrajući rezultate dobijene ovim istraživanjem, možemo zaključiti da sva tri indeksa pokazuju visok procenat gojazne i prekomjerno uhranjene djece. Međutim, ukoliko se želi saznati u kojoj regiji je procenat prekomjerno uhranjene i gojazne djece najviši dolazi, se u veliku nedoumcu jer su rezultati neusaglašeni. Indeks tjelesne mase pokazuje da najviše prekomjerno uhranjene i gojazne djece ima u Centralnoj regiji (33%), indeks odnosa obima struka i kukova pokazuje da najviše prekomjerno uhranjene i gojazne djece ima u Sjevernoj regiji (33%), dok indeks odnosa obima struka i tjelesne visine pokazuje da najviše prekomjerno uhranjene i gojazne djece ima u Južnoj regiji (44%). Rezultati su takođe neusaglašeni kada se proba zaključiti u kojoj regiji je najmanji procenat prekomjerne i uhranjene djece. Indeks tjelesne mase pokazuje da je najmanji procenat prekomjerno uhranjene i gojazne djece u Sjevernoj regiji (17%), dok preostala dva indeksa pokazuju da ih je najmanje u Centralnoj regiji (10% i 14%). Neusaglašenost rezultata takođe se primjećuje u prethodnim sprovedenim regionalnim studijama na ovu temu. Pa u svom istraživanju Mitrovic, & Dragutinovic (2020) prikazuju da su prema indeksu tjelesne mase i indeksu odnosa obima struka i tjelesne visine, djeca iz Centralne regije više gojazna u odnosu na ostale, dok su neka istraživanja pokazuju da su djeca iz Južne regije više gojazna u odnosu na druge dvije regije (Bacovic, 2020; Malovic, 2020).

Iako postoji neusaglašenost rezultata za različite indekse, kada je u pitanju normalna uhranjenost rezultati pokazuju da između dječaka koji žive u različitim regijama značajne razlike nema kada su u pitanju sva tri indeksa. Kada je u pitanju prekomjerna tjelesna težina i gojaznost, indeks tjelesne mase takođe pokazuje da između dječaka koji žive u različitim regijama značajne razlike nema. Nasuprot tome, procenat prekomjerno uhranjene i gojazne djece dobijen pomoću indeksa odnosa obima struka i kukova i indeksa odnosa obima struka i tjelesne visine pokazuju da između dječaka koji žive u različitim regijama postoji značajna razlika. Odnos struka i kukova (WHR) je pokazao da su dječaci iz Sjeverne regije značajno gojazniji u odnosu na Južnu i Centralnu regiju, dok je odnos struka i tjelesne visine (WtHR) pokazao da su dječaci iz Južne regije značajno gojazniji u odnosu na Južnu i sjevernu regiju. Dobijeni rezultati su slični rezultatima dobijenim u prethodnim istraživanjima, gdje se procenti gojaznosti u mediteranskim zemljama kreću između 20% i 40%, dok su veći u odnosu na sjeverne evropske zemlje, gdje se procenti gojazne djece kreću između 10% i 20% (Dinarevic et al., 2011). Upoređujući dobijene rezultate sa rezultatima prethodnih istraživanja gojaznosti u Crnoj Gori, gdje se procenti gojaznosti kreću između 16% i 60% (Mitrovic, & Dragutinovic, 2020; Bacovic, 2020; Malovic, 2020), možemo zaključiti da su u ovom istraživanju dobijeni nešto manji procenti gojaznosti, ali ipak dosta visoki a iznose između 10% i 44%.

Osnovni ciljevi ovog istraživanja bili su izmjeriti morfološke karakteristike (tjelesnu visinu, tjelesnu težinu, obim struka, obim kukova) dječaka uzrasta 11-12 godina iz Crne Gore, izračunati nivoe uhranjenosti pomoću tri indeksa (indeksa tjelesne mase, indeksa odnosa obima struka i kukova i indeksa odnosa obima struka i tjelesne visine), kao i utvrditi eventualne razlike koje se javljaju među dječacima iz različitih regija. Na osnovu rezultata istraživanja, može se konstatovati da su ovi procenti gojaznosti dosta zabrinjavajući, posebno ako se uzme u obzir činjenica da je gojaznost faktor nastanka mnogih hroničnih oboljenja, najčešće krvnog i srčanog sistema (Haslam et al., 2005; Gardašević, Vasi-

ljević, Bjelica, & Popović, S. (2015; Vasiljević, Bjelica, Popović, & Gardašević, 2015; Ćorluka, Bjelica, & Gardašević, 2018; Banjevic, Popovic, & Masanovic, 2020). Tome u prilog idu i predviđanja Svjetske zdravstvene organizacije da će gojaznost uskoro biti najvažniji faktor lošeg zdravstvenog stanja (Fauci et al., 2008). Posmatrajući gojaznost kao hronično stanje, koje sa jedne strane predisponira ili ubrzava nastanak brojnih oboljenja u odraslom dobu, ali već i u djetinjstvu, a sa druge strane kao stanje koje i samo po sebi doprinosi umanjuju kvaliteta života, rešavanju ovog zdravstvenog problema mora se pokloniti velika pažnja, a takođe treba nastaviti sa sprovođenjem daljih istraživanja na ovu temu, kako bi se što preciznije utvrdio broj gojazne djece i sprovele mjere za njihovo sprječavanje i dalje širenje.

Limitiranost ovog istraživanja svakako predstavlja mali broj ispitanika. Da bi se omogućila generalizacija rezultata na širu populaciju ovog uzrasta u Crnoj Gori, potrebno je sprovoditi dalja istraživanja ovog tipa. Takođe, s obzirom da različiti indeksi nivoa uhranjenosti pokazuju različite rezultate, preporučuje se njihovo dalje ispitivanje i upoređivanje, kako bi se utvrdila njihova pouzdanost.

Acknowledgements

There are no acknowledgements.

Conflict of Interest

The authors declare that there are no conflicts of interest.

Received: 1 June 2020 | **Accepted:** 16 July 2020 | **Published:** 16 October 2020

References

- Bacovic, D. (2020). Analysis of Obesity and Differences in Nutritional Status of School Children in Central and Southern Region of Montenegro. *Journal of Anthropology of Sport and Physical Education*, 4(1), 47-50.
- Banjevic, B., Popovic, S., & Masanovic, B. (2020). Body Mass Index and Body Fat Percentage of Armed Forces Personnel in Montenegro among Different Age Groups. *Iranian Journal of Public Health*, 49(5), 1010-1011.
- Barlow, S.E. (2007). Expert committee recommendations regarding the prevention, assessment, and treatment of child and adolescent overweight and obesity: summary report. *Pediatrics*, 120(Suppl. 4), 164-192.
- Branca, F., Nikogosian, H., & Lobstein, T. (2007). *Challenge of obesity in the WHO European region and the strategies for response: summary*. Copenhagen: WHO, Regional Office for Europe.
- Burke, L., & Wang, J. (2011). Treatment strategies for overweight and obesity. *Journal of Nursing Scholarship*.
- Ćorluka, M., Bjelica, D., & Gardašević, J. (2018). Sport Schools vs trends to obesity: an analysis among 5 year olds in the Makarska (Croatia). *Acta Kinesiológica*, 12(1), 111-115.
- Dinarevic S., Brankovic S., & Hasanbegovic S. (2011). Relation of diet and physical activity to obesity in children in elementary schools. *Journal of Health Sciences*, 1(1), 44-49.
- Fauci, A., Kasper, E., Longo, D., Braunwald, S., Hauser, D., Jameson, L., & Loscalzo, J. (2008). *Harrison's Principles of Internal Medicine*, 17th Edition. *Internacinal Medicine Journal*, 38(12), 932-942.
- Freedman, D.S., Khan, L.K., Serdula, M.K., Dietz, W.H., Srinivasan, S.R., & Berenson, G.S. (2005). The relation of childhood BMI to adult adiposity: the Bogalusa Heart Study. *Pediatrics*, 115(1), 22-27.
- Gardašević, J., Vasiljević, I., Bjelica, D., & Popović, S. (2015). Analysis of nutrition of boys and girls, adolescents from Montenegro. *Journal of Physical Education and Sport*, 15(4), 702-704. doi:10.7752/jpes.2015.04107
- Haslam, D.W., James, W., & Philip, T. (2005) Obesity. *Lancet*, 366(9492), 1197-209.
- Jansen, W., Raat, H., Zwanenburg, E.J., Reuvers, I., Van Walsem, R., & Brug, J. (2008). A school-based intervention to reduce overweight and inactivity in children aged 6-12 years: study design of a randomized controlled trial. *BMC Public Health*, 25(8), 257.
- Jocić Stojanović, J. (2010). *Gojaznost i asma dece školskog uzrasta*. Diplomski rad, Beograd: Medicinski fakultet.
- Jocić Stojanović, J., Živković, Z., Šumarac Dumanović, M., i Veković, V. (2015). Kvalitet života gojazne dece sa astmom. *Timočki medicinski glasnik*, 40(4), 276-280.
- Maffei, C., Talamini, G., & Tato, L. (1998). Influence of diet, physical activity and parent's obesity on children's adiposity: a four-year longitudinal

- study. *International Journal of Obesity*, 22, 758-764.
- Magarey, A.M., Daniels, L.A., Boulton, T.J., & Cockington, R.A. (2003). Predicting obesity in early adulthood from childhood and paternal obesity. *International Journal of Obesity*, 27(4), 505-513.
- Malovic, P. (2019). Anthropometric Indices as Indicators of Obesity of Children from Elementary School in Montenegro. *Journal of Anthropology of Sport and Physical Education*, 3(2), 43-47.
- Masanovic, B., Martinovic, S., Zoric G., Bacovic D., Mitrovic, M., & Vukotic, M. (2020). Trends in Body Height, Body Weight, and Body Mass Index from 1979 to 1987: An Analysis of the Young Male Population from the Municipality of Cetinje. *Journal of Anthropology of Sport and Physical Education*, 4(2), 3-7. doi: 10.26773/jaspe.200401
- Mitrovic, M., & Dragutinovic, K. (2020). Comparative Analysis of Anthropometric Parameters as Obesity Indicators for 7-8-Years-Old Children of Different Resident Status. *Journal of Anthropology of Sport and Physical Education*, 4(1), 43-46.
- NCD Risk Factor Collaboration (2016). A century of trends in adult human height. *eLife*, 5:e13410. doi: 10.7554/eLife.13410
- NCD Risk Factor Collaboration (2017). Worldwide trends in body-mass index, underweight, overweight, and obesity from 1975 to 2016: a pooled analysis of 2416 population-based measurement studies in 128.9 million children, adolescents, and adults. *The Lancet*, 390(10113), 2627-2642. doi: 10.1016/s0140-6736(17)32129-3
- Power, C., Lake J.K., & Cole, T.J. (1997). Measurement and long-term health risks of child and adolescent fatness. *International Journal of Obesity*, 21(7), 507-526.
- Radisavljević, N. (2004). Gojaznost - nova saznanja. *Opšta medicina*, 10(3-4), 143-147.
- Rito, A., Buoncristiano, M., Spinelli, A., Salanave, B., Kunešova, M., Hejgaard, T., Garcia Solano, M., Fijalkowska, A., Sturua, L., Hyska, J., Kellecher, C., Duleva, V., Musić Milanović, S., Farrugia Sant Angelo, V., Abrakhamaniva Huidumac Petrescu, C., Williams, J., Ahrens, W., & Breda, J. (2019). Association between Characteristics at Birth, Breastfeeding and Obesity in 22 Countries: The WHO European Childhood Obesity Surveillance Initiative - COSI 2015/2017. *Obesity Facts*, 12(2), 226-243.
- Skinner, A.C., Mayer, M.L., Flower, K., & Weinberger, M. (2008). Health status and health care expenditures in a nationally representative sample: how do overweight and healthy-weight children compare. *Pediatrics*, 121(2), 269-277.
- Strauss, R.S. (2000). Childhood obesity and self-esteem. *Pediatrics*, 105(1), 15.
- Tarp, J., Jarani, J., Muca, F., Spahi, A., & Grontved, A. (2018). Prevalence of overweight and obesity and anthropometric reference centiles for Albanian children and adolescents living in four Balkan nation-states. *Journal of Pediatric and Endocrinology & Metabolism*, 31(11), 1199-1206.
- Vasiljević, I., Bjelica, D., Popović, S., & Gardašević, J. (2015). Analysis of nutrition of preschool-age and younger school-age boys and girls. *Journal of Physical Education and Sport*, 15(3), 426-428. doi: 10.7752/jpes.2015.03063
- Vlajković, V., Macanović, G., Arsić, J., Jocić, I., Milovanović, D., i Arsić, D. (2015). Gojaznost kod školske dece kao faktor rizika po zdravlje. *Pons*, 12(1), 9-14.
- Vukušić, K., Filipović, D., i Milićev, S. (2015). Gojaznost. *Sport – nauka i praksa*, 5(1-2), 77-87.
- World Health Organization (2016). *Obesity and overweight*. Geneva: WHO.

ORIGINAL SCIENTIFIC PAPER

Level of Active Lifestyle and Exercise Approach among Sports-Active Female Students of The University of Sarajevo during the Covid-19 Pandemic

Izet Bajramovic¹, Srdjan Redzepagic², Dusko Bjelica³, Dragan Krivokapic³, Eldin Jeleskovic¹, Slavenko Likić¹

¹University of Sarajevo, Faculty of Sport and Physical Education, Sarajevo, Bosnia and Herzegovina, ²Université Côte d'Azur, GREDEG, Nice, France,

³University of Montenegro, Faculty for Sport and Physical Education, Niksic, Montenegro

Abstract

Closure of all educational institutions as well as various facilities for sports and recreation, due to the COVID-19 pandemic and the adoption of social movement restrictions, impaired the possibility of maintaining healthy living habits of the student population. The aim of this research was to identify changes in healthy living habits and exercise approach of sports-active female students, caused by the COVID-19 pandemic, in order to prevent possible harmful consequences on their physical fitness level. The sample of respondents were female students ($n=110$; 22.1 ± 1.69 years of age), who stated that they were active members of various sports clubs during the period before the COVID-19 pandemic. A survey questionnaire was used in the study to assess a healthy lifestyle. The analysis of the obtained answers within the survey questionnaire was evaluated in relation to the percentage values of the frequencies of these answers. The results indicate that the level of regular and planned physical exercise of sports-active female students decreased by 41.7% during the COVID-19 pandemic, and that the level of their physical condition decreased by 38.2%. Changes in eating habits were recorded in 50% of female sports-active female students, while 36.3% of them reported changes in weight gain. The largest number of sports-active female students practiced basic exercises for body shaping, relaxation and stretching (89.2%) and strength exercises with own body weight as load intensity (82.4%). Accordingly, it is also necessary to find possible ways, in order to raise awareness of the importance of maintaining their sports and healthy living habits. This primarily refers to maintaining their healthy physical condition, which is also related to their eating habits, as well as preventing the possible occurrence of undesirable health conditions. Such different approaches may apply to all future pandemic conditions, or similar life-threatening conditions.

Keywords: *Physical Activity and Exercise, Sedentary Behavior, Physical Condition, Health*

Uvod

Zatvaranje obrazovnih institucija, sportsko-rekreativnih objekata, te uvođenje mjera ograničenog kretanja stanovništva uzrokovano pandemijom novog korona virusa (COVID-19), predstavlja mogućnost narušavanja zdravih životnih navika studentske populacije. Uvođenje društveno restriktivnih mjera zbog CO-

VID-19, primoralo je studente na prilagođavanje novim uslovima života i aktivno djelovanje. Kao posljedica ograničenog kretanja pojavila se opasnost od sjedilačkog ponašanja i povećanja tjelesne neaktivnosti. Zabrinjavajuće može postati ukoliko se pandemija tjelesne neaktivnosti i sjedilačkog načina života nastavi, kao i dalji rizik za pogoršanjem nastalog stanja zbog COVID-19 (Hall, Lad-

Correspondence:

**Montenegro
Sport**

Slavenko Likić
University of Sarajevo, Faculty of Sport and Physical Education, Patriotske lige 41, 71 000 Sarajevo, Bosnia and Herzegovina
E-mail: slavenko.likic@fasto.unsa.ba

du, Phillips, Lavie, & Arena, 2020). Istraživanja su takođe ukazala na potrebu za ulaganjem dodatnih napora kako bi se poboljšale tjelesne aktivnosti žena i djevojaka u okviru obrazovnog sistema (Khalaf et al., 2013).

Poznato je da tjelesna neaktivnost može uzrokovati povećanje tjelesne težine ili određeni stepen pretilosti. Faktori rizika za povećanje tjelesne težine tokom izolacije koja je uzrokovana pandemijom COVID-19 takođe su neadekvatne prehranbene navike (Zachary et al., 2020). Moguće stresne situacije uzrokovane pandemijom novog korona virusa vjerovatno mogu dodatno pogoršati situaciju. Ranije studije su ukazale da stres ometa napore pojedinaca za većom tjelesnom aktivnošću, baš kao što negativno utiče i na druga zdravstvena ponašanja (Stults-Kolehmainen, & Sinha, 2014).

S druge strane, cjelokupni proces pokretanja aktivnosti čovjeka zasniva se na procesu motivacije (Bjelica, 2015). U skladu s tim je potrebno pronalaziti nove i alternativne puteve djelovanja u smjeru podizanja svijesti studenata, ali i djelovanja na formiranje novih zdravih životnih navika koje su povezane sa tjelesno aktivnim stilom života. Postati tjelesno aktivan i smanjiti sjedilačko ponašanje lako se postiže izbjegavanjem sjedenja tokom dužeg vremenskog perioda, upražnjavanjem tjelesnih aktivnosti kraćeg trajanja, korištenjem internetskih sadržaja vezanim za tjelesne aktivnosti, kao i mobilnih tehnologija poput telefonskih aplikacija i nosivih senzora za poticanje kretanja (Woods, et al., 2020). Dobar način je upražnjavanje vježbi prilagođenih kućnim uslovima, odnosno za koje nije potreban veliki prostor ili dodatna oprema, te koje se mogu ekonomično koristiti u svako doba dana. Pošto se ove tjelesne aktivnosti lako obavljaju kod kuće, smanjuju se ili uklanjaju poteškoće u pronalaženju objekata s odgovarajućim prostorom i specifičnom opremom.

Autori ovog istraživanja smatraju da je od velike važnosti pravovremeno evidentirati studentice sa povećanim rizikom od sjedilačkog ponašanja kako bi se moglo preventivno djelovati. Cilj istraživanja je utvrditi promjene u zdravim životnim navikama i načinu vježbanja studentica uzrokovanim pandemijom COVID-19, s ciljem sprečavanja štetnih posljedica na njihovu tjelesnu formu.

Metod

Uzorak ispitanika su činile studentice Univerziteta u Sarajevu, Bosna i Hercegovina ($n=110$; $22,1 \pm 1,69$ godina starosti). Osnovni kriterij izbora uzorka je bilo njihovo aktivno članstvo u sportskim klubovima, a u kojima su trenirale najmanje 3 puta nedjeljno prije

uvodjenja privremenih restriktivnih mjera zbog pandemije COVID-19. Anketni upitnik korišten u istraživanju je autorsko djelo profesora Fakulteta za sport i fizičko vaspitanje Univerziteta Crne Gore, a koji je prilagođen aktuelnoj situaciji izazvanoj pandemijom COVID-19.

Anketni upitnik zasnovan je na percepciji životnog stila studentica-sportistkinja tokom uvodjenja privremenih restriktivnih mjera zbog pandemije COVID-19, a uključuje skalu za samoprocjenu redovnog i planskog vježbanja (najmanje 3 puta nedjeljno) i slabljenja tjelesne forme; skalu konstruisanu za samoprocjenu prehranbenih navika i promjena u tjelesnoj težini, te skalu za samoprocjenu načina vježbanja. Podaci su prikupljeni na bazi seta sljedećih konstruisanih pitanja: 1) Da li redovno i planski vježbate?; 2) Da li osjećate slabljenje tjelesne forme?; 3) Da li osjećate promjene u povećanju količine i vrste hrane?; 4) Da li osjećate promjene u povećanju tjelesne težine?; 5) Da li koristite rekvizite i sprave za vježbanje kao što su steper, kućno vratilo, fitnes lopta, tegovi i slično?; 6) Da li radite aerobik vježbe uz muziku?; 7) Da li plešete uz muziku (sportski, akrobatski, latino, klasični ples...)?; 8) Da li praktikujete vježbanje uz uključenu video prezentaciju trenera?; 9) Da li radite vježbe za razvoj snage (sklekovi, čučnjevi, vježbe za trbušne i leđne mišiće i sl.)?; 10) Da li radite vježbe oblikovanja, istezanja i opuštanja?; 11) Da li koristite druge načine vježbanja?

Upitnik je uključivao ljestvice sa dihotomnim formatom odgovora odnosno izborom između dva, pri čemu se odgovor „Da“ bodovao sa 1; odgovor „Ne“ sa 0. Anketni upitnik je popunjen pomoću Google Forms obrazaca, koji su postavljeni na internet stranicama svih organizacionih jedinica Univerziteta u Sarajevu, u periodu 5. maj 2020. - 24. maj 2020. Dobiveni rezultati ispitivanja su se automatski izvozili u Google proračunsku tablicu. Analiza odgovora je procijenjena korištenjem Office programa Excel i SPSS 21 (SPSS Inc., Chicago, IL, USA), u odnosu na relevantne procentualne vrijednosti frekvencije dobivenih odgovora.

Rezultati

Na temelju analize frekvencija dobivenih odgovora studentica-sportistkinja Univerziteta u Sarajevu, evidentan je pad od 41,7% nivoa redovnog i planskog tjelesnog vježbanja (≥ 3 dana nedjeljno), tokom mjera zabrane kretanja uzrokovanih pandemijom COVID-19 (tabela 1). Takođe je u navedenom periodu primijetno opadanje nivo tjelesne forme za 38,2%. Percepcija promjena u načinu i količini prehrane evidentirana je kod 50% studentica-sportistkinja, dok je 36,3% studentica-sportistkinja prijavilo promjene u

Tabela 1. Percepcija nivoa tjelesne aktivnosti i prehranbenih navika studentica sportistkinja tokom pandemije COVID-19

Nivo vježbanja i tjelesne kondicije	Da %	Ne %
Redovno i plansko vježbanje	58,3	41,7
Percepcija opadanja tjelesne kondicije	61,8	38,2
Prehranbene navike i promjene u tjelesnoj težini		
Percepcija promjena u načinu i količini prehrane	50,0	50,0
Povećanje tjelesne težine	36,4	63,6

povećanju tjelesne težine (tabela 1).

Najveći broj studentica-sportistkinja je tokom mjera ograničenja kretanja uzrokovanih pandemijom COVID-19 praktikovalo vježbe oblikovanja, relaksacije i istezanja (89,2%); vježbe za razvoj snage koristeći vlastito tijelo kao intenzitet opterećenja (čučnjevi, sklekovi, vježbe za trbušnu i leđnu muskulaturu) (82,4%), ili praktikujući druge načine vježbanja (64,7%). Interesantan je utvrđeni veći procenat studentica-sportistkinja koje su kao način vježbanja praktikovale ples uz muziku (62%), vježbanje uz praćenje video

prezentacije trenera (63,3%), odnosno aerobik vježbanje uz muziku (43,8%). Manji broj studentica-sportistkinja je praktikovalo vježbanje uz korištenje rekvizita i sprava (steper, kućno vratilo, fitnes lopta, tegovi i slično) (38%) (tabela 2).

Diskusija

Evidentno je opadanje nivoa redovne tjelesne aktivnosti, kod studentica-sportistkinja koje su u ranijem periodu to činile redovno (≥ 3 dana nedjeljno). Promjene u načinu života, kao i

Tabela 2. Način vježbanja studentica-sportiskinja tokom pandemije COVID-19

Način vježbanja	Da %	Ne %
Vježbanje uz korištenje rekvizita i sprava	38,0	62,0
Aerobik vježbanje uz muziku	43,8	56,3
Ples uz muziku	62	38
Vježbanje uz video prezentaciju trenera	63,3	36,7
Vježbe snage sa vlastitim tijelom	82,4	17,6
Vježbe oblikovanja, istezanja i relaksacije	89,2	10,8
Drugi način vježbanja	64,7	35,3

navikavanje na nove uslove života, rada i učenja, uzrokuje promjene u zdravim životnim navikama. Takođe, negativne promjene u zdravstvenom ponašanju povezane su i sa povećanim psihološkim smetnjama tokom pandemije COVID-19 (Stanton et al., 2020). Posljedica tjelesne neaktivnosti može ugroziti nivo održavanja tjelesne kondicije tokom pandemije. Rezultati koji su dobiveni u nekim istraživanjima sugerišu da petonedjeljni prekid normalnog treninga može rezultirati značajnim smanjenjem brzine, snage i aerobnog kapaciteta kod sportista takmičara, zbog lošeg poštivanja propisanog režima treninga (Kovacs, Pritchett, Wickwire, Green, & Bishop, 2007). Skeletne mišice karakterizira sposobnost dinamičkog prilagođavanja na različite funkcionalne zahtjeve. U periodima nedovoljnih trenažnih podražaja nastupa mišićna neutreniranost, koju karakteriše smanjena gustoća kapilara, smanjena aktivnost oksidativnih enzima i mitohondrijskog ATP-a, te s tim u vezi smanjenjenog nivoa maksimalne kiseoničke potrošnje (VO₂max), a što je posljedica dogotrajnog mirovanja i prestanka sa trenažnim aktivnostima. Ovakve mišićne karakteristike se uobičajeno zadržavaju iznad sjedilačkih vrijednosti kod aktivnih sportista, te se brzo kroz trening mogu vratiti na svoje ranije vrijednosti (Mujika & Padilla, 2001). Ipak, važno je naglasiti da je 58,3% studentica-sportiskinja tokom pandemije COVID-19 i restriktivnih mjera o ograničavanju kretanja stanovništva nastavilo sa redovnim tjelesnim vježbanjem.

Neka istraživanja ukazuju da tjelesne aktivnosti s ciljem održavanja nivoa tjelesne kondicije u kući/stanu, te koje su realizirane uz aerobne treninge na sobnom biciklu, vježbanje sa tjelesnom težinom, plesom i slično mogu pomoći u suzbijanju štetnih fizičkih i mentalnih nuspojava uzrokovanih restriktivnim propisima o načinu zaštite života tokom pandemije COVID-19 (Hammami, Harrabi, Mohr, & Krustup, 2020). Tradicionalne vježbe za razvoj snage u kućnim uslovima: sklekovi, čučnjevi, vježbe za trbušne i leđne mišice ili slično, te vježbe oblikovanja, istezanja i opuštanja su se pokazale kao najpopularniji način vježbanja kod većeg dijela studentica-sportistkinja tokom nastalog kriznog perioda. Izbor ovakvog načina vježbanja se vjerovatno ogleda u biomehanički jednostavnim pokretima vježbi, mogućnosti da se praktikuju u kućnim uslovima, odnosno vježbama koje su studenticama-sportistkinjama od ranije poznate i koje se mogu izvoditi bez dodatne opreme i korištenjem vlastitog tijela kao opterećenja. Korištenje rekvizita i sprava za vježbanje kao što su steper, kućno vratilo, fitness lopta, tegovi, pružaju dodatne mogućnosti za doziranje opterećenja i strukturu pokreta. Ipak treba naglasiti čestu nedostupnost posebne opreme za kućno vježbanje u novonastalim okolnostima. S druge strane, nedavna istraživanja su otkrila i razlike u stavovima među ispitanicima koji su prijavili različite navike kupovine sportske opreme (Milosevic, Bjelica, & Masanovic, 2019). Moguće je da su video prezentacija trenera ili ples uz muziku mogli pružiti studenticama-sportistkinjama dodatni motiv za vježbanjem i tjelesnom aktivnošću. Ipak, potrebno je navesti da su se pored navedenih i drugi načini vježbanja praktikovali u većoj mjeri, međutim nedostatak informacija o konkretnim aktiv-

nostima otežava njihovu diskusiju.

Sudjelovanje u nekom obliku tjelesne aktivnosti ili sporta očito je temeljna komponenta održavanja zdravog načina života, te je nesumnjivo i važna javno-zdravstvena poruka onima koji borave u geografskim područjima gdje se provode restriktivne mjere samoizolacije.

S druge strane, s aktivnim sportistima je potrebno realizovati odgovarajuće trenažne aktivnosti s ciljem održavanja visokog nivoa tjelesne kondicije (Hull, Loosemore, & Schwellnus, 2020). Online treninzi s aktivnim uključivanjem trenera ili treninzi putem YouTube kanala su neminovno postali alternativa trenažnim aktivnostima tokom pandemije COVID-19, pri čemu se mogu (prema Bjelica, 2013) steći novi ili usavršavati postojeći dinamički stereotipi. Jedno od rješenja u novonastalim okolnostima pandemije, vježbanje na otvorenom može postati prihvatljiv način vježbanja. Rezultatima istraživanja (Johnsson, Ivarsson, Parker, Andersen, & Svetoft, 2019) potvrđeni su pozitivni efekti vježbanja realizovanog u teretani na otvorenom prostoru, a koji su u vezi s povećanjem opštih fitness pokazatelja, motivacije, smanjenja tjelesne težine i postotka masti, kao i smanjenja simptoma stresa.

Povećanje tjelesne težine kod sportista može biti posljedica prehrambenih navika (Mašanović, 2018). Ne čudi činjenica o povećanju tjelesne težine studentica-sportiskinja (36,3%) tokom kriznog perioda, tjelesne neaktivnosti i socijalnog distanciranja (tabela 1). Povećanja tjelesne težine stanovništva i neprimjerene prehrabene navike takođe su uočene u nedavnim istraživanjima rađenim tokom pandemije COVID-19 (Di Renzo et al., 2020; Papandreou, Arija, Aretouli, Tsilidis, & Bulló, 2020). S druge strane, italijanski studenti su izvijestili o smanjenju nivoa tjelesne aktivnosti, ali ne i prehrabnim navikama (Ferracuti, Montagna, Liguori, Orsi, & Napoli, 2020).

Ipak, nakon prihvatanja novonastalih uslova života, može uslijediti prilagođavanje i aktivno djelovanje. Od velike važnosti je pravovremeno evidentirati studentice sa povećanim rizikom od sjedilačkog ponašanja. U slučaju svih sportski aktivnih ženskih studentica, je potrebno pronaći nove alternativne pristupe sportu i vježbanju tokom pandemije Covid-19. U skladu s tim je također potrebno pronaći moguće puteve, kako bi se podigla svijest za važnost održavanja njihovih sportskih ali i zdravih životnih navika. Ovo se prije svega odnosi na održavanje njihove zdrave tjelesne kondicije, što je također povezano s njihovim navikama u ishrani, kao i sprječavanjem mogućeg nastanka nepoželjnih zdravstvenih stanja. Ovakvi različiti pristupi se mogu odnositi na sva buduća pandemična stanja, ili njima slična stanja životne ugroženosti.

Acknowledgements

There are no acknowledgements.

Conflict of Interest

The authors declare that there are no conflicts of interest.

Received: 24 July 2020 | Accepted: 20 August 2020 | Published: 16 October 2020

References

- Bergier, J., Bergier, B., & Tsos, A. (2016). Variations in Physical Activity of Male and Female Students from Different Countries. *Iranian journal of public health*, 45(5), 705–707.
- Bjelica, D. (2013). *Teorija sportskog treninga*. Podgorica: Univerzitet Crne Gore.
- Bjelica, D. (2015). *Teoretske osnove tjelesnog i zdravstvenog obrazovanja*. Podgorica-Nikšić: Fakultet za sport i fizičko vaspitanje Univerziteta Crne Gore i Crnogorska sportska akademija.
- Di Renzo, L., Gualtieri, P., Pivari, F., Soldati, L., Attinà, A., Cinelli, G., Leggeri, C., Caparello, G., Barrea, L., Scerbo, F., Esposito, E., & De Lorenzo, A. (2020). Eating habits and lifestyle changes during COVID-19 lockdown: an Italian survey. *Journal of translational medicine*, 18(1), 229. doi: 10.1186/s12967-020-02399-5
- Ferracuti, S., Montagna, M. T., Liguori, G., Orsi, G.B., & Napoli, C. (2020). Understanding Knowledge and Behaviors Related to CoViD-19 Epidemic in Italian Undergraduate Students: The EPICO Study. *International journal of environmental research and public health*, 17(10), 3481. doi: 10.3390/ijerph17103481
- Hall, G., Laddu, D.R., Phillips, S.A., Lavie, C.J., & Arena, R. (2020). A tale of two pandemics: How will COVID-19 and global trends in physical inactivity and sedentary behavior affect one another?. *Progress in cardiovascular diseases*, S0033-0620(20)30077-3. doi: 10.1016/j.pcad.2020.04.005
- Hammami, A., Harrabi, B., Mohr, M., & Krstrup, P. (2020). Physical activity and coronavirus disease 2019 (COVID-19): specific recommendations for home-based physical training. *Managing Sport and Leisure*. doi: 10.1080/23750472.2020.1757494
- Hull, J.H., Loosemore, M., & Schwellnus, M. (2020). Respiratory health in athletes: facing the COVID-19 challenge. *The Lancet. Respiratory medicine*, 8(6), 557–558. doi: 10.1016/S2213-2600(20)30175-2
- Iglesias López, M.T., Cuesta Santa Teresa, E., & Sáez Crespo, A. (2014). Estudio comparativo de hábitos entre estudiantes universitarios y preuniversitarios de la zona noroeste de Madrid [Comparative study of habits in students before and during the university in northwest area of Madrid]. *Nutricion hospitalaria*, 31(2), 966–974. doi: 10.3305/nh.2015.31.2.7703
- Johnson, U., Ivarsson, A., Parker, J., Andersen, M.B., & Svetoft, I. (2019). Connection in the fresh air: A study on the benefits of participation in an electronic tracking outdoor gym exercise programme. *Montenegrin Journal of Sports Science and Medicine*, 8(1), 61–67. doi: 10.26773/mjssm.190309
- Khalaf, A., Ekblom, Ö., Kowalski, J., Berggren, V., Westergren, A., & Al-Hazzaa, H. (2013). Female university students' physical activity levels and associated factors—a cross-sectional study in southwestern Saudi Arabia. *International journal of environmental research and public health*, 10(8), 3502–3517. doi: 10.3390/ijerph10083502
- Kovacs, M.S., Pritchett, R., Wickwire, P.J., Green, J.M., & Bishop, P. (2007). Physical performance changes after unsupervised training during the autumn/spring semester break in competitive tennis players. *British journal of sports medicine*, 41(11), 705–710. doi: 10.1136/bjism.2007.035436
- Masanovic, B. (2018). Comparative study of anthropometric measurement and body composition between junior basketball and volleyball players from Serbian national league. *Sport Mont*, 16(3), 19–24. doi: 10.26773/smj.181004
- Milosevic, Z., Bjelica, D., & Masanovic, B. (2019). Attitudes of consumers from autonomous province of vojvodina toward advertising through sport for the question: how often do consumers purchase sporting goods. *Sport Mont*, 17(1), 39–43. doi: 10.26773/smj.190207
- Mujika, I., & Padilla, S. (2001). Muscular characteristics of detraining in humans. *Medicine and Science in Sports and Exercise*, 33(8), 1297–1303. doi: 10.1097/00005768-200108000-00009
- Papandreou, C., Arija, V., Aretouli, E., Tsilidis, K.K., & Bulló, M. (2020). Comparing eating behaviours, and symptoms of depression and anxiety between Spain and Greece during the COVID-19 outbreak: Cross-sectional analysis of two different confinement strategies. *European eating disorders review : the journal of the Eating Disorders Association*, 10.1002/erv.2772. Advance online publication. doi: 10.1002/erv.2772
- Stanton, R., To, Q.G., Khalesi, S., Williams, S.L., Alley, S.J., Thwaite, T.L., Fenning, A.S., & Vandelanotte, C. (2020). Depression, Anxiety and Stress during COVID-19: Associations with Changes in Physical Activity, Sleep, Tobacco and Alcohol Use in Australian Adults. *International journal of environmental research and public health*, 17(11), 4065. doi: 10.3390/ijerph17114065
- Stults-Kolehmainen, M.A., & Sinha, R. (2014). The effects of stress on physical activity and exercise. *Sports medicine (Auckland, N.Z.)*, 44(1), 81–121. doi: 10.1007/s40279-013-0090-5
- Woods, J.A., Hutchinson, N.T., Powers, S.K., Roberts, W.O., Gomez-Cabrera, M.C., Radak, Z., Berkes, I., Boros, A., Boldogh, I., Leeuwenburgh, C., Coelho-Júnior, H.J., Marzetti, E., Cheng, Y., Liu, J., Durstine, J.L., Sun, J., & Ji, L.L. (2020). The COVID-19 pandemic and physical activity. *Sports Medicine and Health Science*, 2(2), 55–64. doi: 10.1016/j.smhs.2020.05.006
- Zachary, Z., Brianna, F., Brianna, L., Garrett, P., Jade, W., Alyssa, D., & Mikayla, K. (2020). Self-quarantine and weight gain related risk factors during the COVID-19 pandemic. *Obesity research & clinical practice*, 14(3), 210–216. doi: 10.1016/j.orcp.2020.05.004

REVIEW PAPER

Talent Identification in Youth Football: A Systematic Review

Emir Mustafovic¹, Denis Causevic², Nedim Covic², Mirza Ibrahimovic², Haris Alic², Ensar Abazovic², Semir Masic²

¹Football Club Sarajevo, Academy FC Sarajevo, Bosnia and Herzegovina, ²University of Sarajevo, Faculty of Sport and Physical Education, Bosnia and Herzegovina

Abstract

Identification of talents in football is one of the most important tasks of coaches in youth football selection. The main focus of this paper was to review the available literature on talent identification in youth football. A systematic review of PubMed, Medline, ResearchGate and Elsevier databases was performed according to PRISMA guidelines. The following keywords were used: „identification of talents“, „football talent“, „selection in football“, and „management in sport clubs“. Of 137 studies initially identified, of which 17 met the inclusion criteria of the review. Mentioned studies were analyzed and discussion was created with the emphasis on 4 segments: 1) analysis of physiological aspects and identification of talents 2) importance of specific contents for identification of talent 3) biological and chronological age as an important aspect of selection of young players 4) multidisciplinary approach to the selection of young players. It is concluded that process of identification of talents needs to be approached multidisciplinary with all contents of physiological and physical assessment and with specific football contents (SSG and TE-TA contents) respectively. Great role in performance on the field has the biological age, on which coaches need to pay attention to when evaluating quality of players.

Keywords: *Selection of footballers, Training, Coaching, Field Test*

Introduction

Identification of talents in football is one of the most important tasks of coaches of youth selections in football and that identification, in terms of chronological age, is very low (Roderick, 2006). One of the reasons why clubs are trying to identify players at such early age is the fear that that selection is done too late. Also there are two reasons: 1) development of a player 2) fear that talented player is selected by other clubs (Sæther, 2014). Earlier philosophy of identification of talents was based on monitoring of genetic predispositions of players, which can be improved by training (Hoare & Warr, 2000). In order to make a selection of players, training teams and teams which work on this issue, use different physiological and specific football indicators, which are used for identification of talented players (Vaeyens et al, 2006). Identification of players demands multidisciplinary approach because growth and development of a player, especially in adolescence, makes process of selection by standard testing of physiological and technical aspects of players, difficult. (Meylan, Cronin, Oliver, & Hughes, 2010). Still, Williams, & Reilly (2010) proved that sport science has an important role in

process of identification, monitoring and development of talents of young players and all for the purpose of development of their full potential. Work with athletes of different sports, whose abilities are developed above average, includes 3 procedures, which demand multidisciplinary, scientific and professional approach: detection and recognition of potential, orientation and selection of players, and quality development of players (Trninić, Jelaska, & Papić, 2009).

The term "talent"

Latin term „talentum“, originates from Greek word tálanton [τάλαντον], which means balance, weight, amount of money (Hoad, 1993). Different authors define talent in different ways. According to Ommundsen (2009), talent is something you have, something you are, something you can be or something you can develop. There are 2 concepts of observing talent; static and dynamic concept. Static concept observes talent as something internal; which focuses on level of performance in early age, while dynamic concept observes talent together with conative and sociological features (Güllich, 2014). Proper selection of talent in terms

Correspondence:

**Montenegro
Sport**

Čaušević Denis
University of Sarajevo, Faculty for Sport and Physical Education, Patriotske lige 41, 71000 Sarajevo, Bosnia and Herzegovina
E-mail: denis.causevic@fasto.unsa.ba

of player and management of sport organization is considered as one of the key segments for success of organization (Beechler & Woodward, 2009). Different authors link the term „talent“ with different terms that can describe its role. Some authors, the term „talent“ identify with the term „possibilities“ (Gagné, 2000; Silzer & Dowell, 2009), others identify term „talent“ with the word „capacity“ (Jericó, 2001). Stahl et al. (2007) identify this term with the term „capability“, while Ulrich (2007) identifies term „talent“ with the term „advantage“. Some authors consider talented person: a person with competences (Bethke-Langenegger, 2012; González-Cruz Martínez, Fuentes, & Pardo del Val, 2009). The term „experience“ is also related to the term „talent“ (Cheese, Thomas, & Craig, 2008), as well as the term „knowledge“ (Tansley et al., 2006). Some authors explain the term „talent“ through the term „potential“ (Tansley et al., 2007), while Stahl et al. (2007) the term „talent“ identify with the term „performance“. These different approaches to the term „talent“ show how complex is the process of identification of talent, but they also prove that criteria for selection of talent cannot be the same for all organizations and sports.

The aim of this paper was to systematically review and organise the literature on significant factors, which affect identification of football talents in young categories, and to determine criteria based on which coaches evaluate whether players have potential or not.

Methods

This paper is designed as overview using PRISMA method (Di Palma, Raiola, & Tafuri, 2016). Data base that are used for

this paper are: PubMed, Medline, ResearchGate & Elsevier. When searching papers words that were searched are: identification of talents, football talent, selection in football and management in sport clubs.

All papers designed to determine different aspects of talent identification in football (specific football aspect, anthropological aspect, psychological aspect, philosophy of the clubs, scientific aspect) were considered as a relevant for this Systematic review. Few criteria influenced acceptance of papers for further analysis: (1) papers had to be published between 2009 and 2019, older papers weren't considered (date criterion), (2) papers had to be written in English language (language criterion), (3) papers focused only on youth players were used for analysis (participants criterion), (4) papers had to be focused on talent identification in football (exposure to interest criterion), (5) papers had to be original researchs (type of publication criteria), (6) articles that didn't meet inclusion criteria's were excluded from further analysis (exclusion criteria).

Results

Scheme 1 shows the detailed process of creating systematic review. After using key-words in databases 137 publications were found. All duplicates were removed (n=26), so 111 papers were screened for title and date criteria. In this phase 42 papers were excluded. Total of 34 papers were excluded in next phase for not meeting exposure to interest criteria. After reading 35 whole papers, 18 of them were excluded for not meeting deeper type of publication criteria and language criteria.

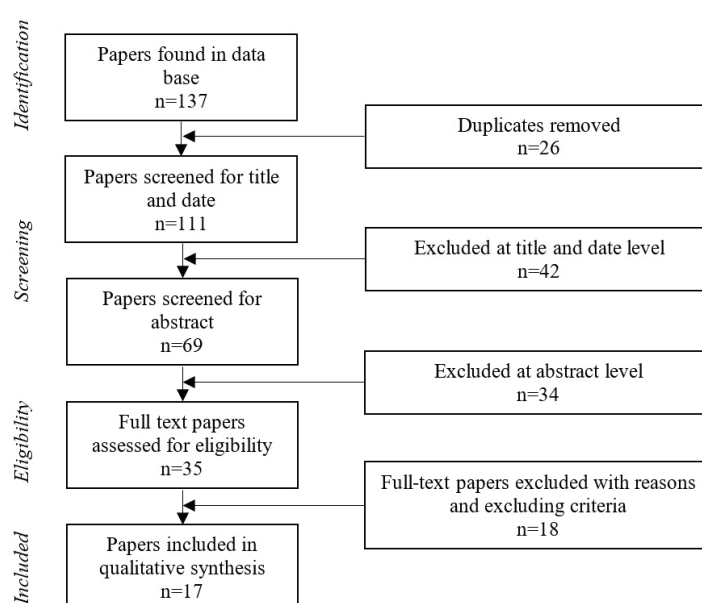
Table 1. Overview of papers with basic information which are taken in consideration during analysis of results

N	Title	Year	Authors	Magazines
1	Talent Identification in Soccer: The Role of Maturity Status on Physical, Physiological and Technical Characteristics	2010	Meylan et al.	International Journal of Sports Science and Coaching
2	Positioning and deciding: key factors for talent development in soccer	2011	Kannekens et al.	Scandinavian Journal of Medicine and Science in Sport
3	Talent identification in youth soccer	2012	Unnithan et al.	Journal of Sports Science
4	Match analysis of U9 and U10 English premier league academy soccer players using a global positioning system: relevance for talent identification and development.	2015	Goto et al.	The Journal of Strength & Conditioning Research
5	Psychometric properties of the motor diagnostics in the German football talent identification and development programme	2015	Höner et al..	Journal of Sports Science
6	The evaluation of small-sided games as a talent identification tool in highly trained pre-pubertal soccer players.	2016	Fenner et al.	Journal of Sports Medicine
7	Talent identification and selection in elite youth football: An Australian context	2016	O'Connor et al.	European Journal of Sports Science
8	Multivariate analyses of individual variation in soccer skill as a tool for talent identification and development: utilizing evolutionary theory in sports science.	2016	Willson et al.	Journal of Sports Science
9	Discriminating talent-identified junior Australian football players using a video decision-making task.	2016	Woods et al.	Journal of Sports Science
10	Two-year changes in anthropometric and motor ability values as talent identification indexes in youth soccer players.	2016	Hirose & Seki	Journal of Science and Medicine in Sport
11	Talent identification and recruitment in youth soccer: Recruiter's perceptions of the key attributes for player recruitment	2017	Larkin & O'Connor	PLoS ONE 12(4): e0175716. https://doi.org/10.1371/journal.pone.0175716

(Continued on next page)

(Continued from previous page)

N	Title	Year	Authors	Magazines
12	Progression in youth soccer: selection and identification in youth soccer players aged 13-15 years.	2017	Bidaurreazaga-Letona et al	The Journal of Strength & Conditioning Research
13	A multidimensional approach to talent identification and selection in high-level youth Australian Football players	2017	Tribolet et al	Journal of Sports Science
14	Talent identification in soccer: Physiological Aspect	2018	Dodd & Newans	Journal of Science & Medicine in Sport
15	Seasonal Body Composition Variation Amongst Elite European Professional Soccer Players: An Approach of Talent Identification	2018	Owen et al.	Journal of Human Kinetics
16	Methodological Issues in Soccer Talent Identification Research	2019	Berkamp et al.	Sports Medicine
17	Assessing the validity of a video-based decision-making assessment for talent identification in youth soccer.	2019	Bennet et al.	Journal of Science and Medicine in Sport



SCHEME 1. Elaboration of selection of papers by PRISMA methodology

Table 1 presents basic information about selected research papers. Only title, year of publication, authors and Journals were presented here. From here, it is obvious that papers were published in different scientific magazines, primary directed toward research in sport, which are placed in index data base. Six of

them were performed in Australia, 4 in United Kingdom, 2 in Netherlands and 1 in New Zealand, Germany, Japan, Spain and France. Analysis of table 2 shows that papers had different goals and that they included different segments which are significant for this paper.

Table 2. List of papers with aims and conclusions of papers

N	Authors	Aim	Conclusion
1	Meylan et al.	The aim of this paper was to create an overview on identification of talents in football, using physiological and technical tests, and to sum issues related to this process	Standard tests cannot be criterion for identification of talents because period of growth and development of young players largely affect results of tests.
2	Kannekens et al.	The aim of this study was to determine in what measure tactical knowledge affects quality and identification of talented players.	Decision-on the field and tactical knowledge are tools which influence the performance and therefore have great impact on process of identification of young player, especially midfielders.
3	Unnithan et al.	The aim of this paper was to evaluate traditional approach to identification of talents in youth football and to present newer method of identification in football.	Games on a small court can be very useful tool for identification of talents. Also, players who have better technical performance are more successful in games in smaller space.

(Continued on next page)

(Continued from previous page)

N	Authors	Aim	Conclusion
4	Goto et al.	The aim of this study was to examine profile of activities of match among elite footballers U9 and U10 and to determine whether there are differences between players which were held later by their clubs.	Players of U9 and U10 cover over 4000m per game and players who are kept in Academy cover bigger distance and in smaller speed.
5	Höner et al.	The aim of this study was to examine reliability, differential stability and validity of motor diagnostics which is conducted in whole Germany by German program for identification and development of talent and gives referential values for standardized interpretation of results of diagnostics.	Diagnostics showed satisfying factor validity with acceptable loadings on two empirical factors speed and technical skills. Results and technical skill of leading and juggling is highly different in players of different performances and therefore showed the highest criterion for validity.
6	Fenner et al.	The aim of this study was to examine physiological and technical features of pre-puberty players during multiple games in a small space (SSG) and to establish whether SSG can be used as a tool for identification of talents.	Results showed great correlation between the best players and success in SSG, perhaps because of higher rating of players, who cover bigger distance and great speed. Therefore, SSG can be used for identification of more talented footballers in pre-puberty period.
7	O'Connor et al.	The aim of this study was to identify perceptive-cognitive skills and variables of history of players, which differentiate defense and non-defense players in elite youth football (i.e. football program)	Efficiency of discriminative function is shown in 93.7% of players, which are correctly qualified, where 4 variables include 57.6% of variance. Discriminative model can give better understanding of factors that influence the selection and identification of elite model of young talents.
8	Willson et al.	The aim of this study was development of overall protocol for qualification of specific football skills, which could significantly improve identification and development of talents.	This study established simple, but overall methodology for evaluation of footballer's performance of skills, which enables coaches to quickly assess relative abilities of their players, identify perspective young players, and work on elimination of deficit of skills in players.
9	Woods et al.	The aim of this paper is to question whether the task of making decisions by viewing videos can discriminate young Australian football players, who are identified as talents, as oppose to the ones who are not identified in such manner.	This paper proved that making decisions while watching video clips fully differentiates elite (talented) players from players who are not identified in that manner, in selection U18.
10	Hirose & Seki	The aim of this study was to examine 2-year changes in antropometric variables and motor skills in elite footballers and to identify potential indexes of identification of talents.	Due to change of minimal rang in speed of sprint, in normal conditions of football game, linear speed of sprint has potential to be useful index for identification of young footballers. On the other hand, muscle strength and ability of COD can be changed during the growth, which suggests that these parameters are not useful for index of identification of talents.
11	Larkin & O'Connor	The aim of this paper was to understand abilities, which are considered to be important by coaches of young players and staff for recruitment, for identification skills of young footballers on entrance on the level of representative football in Australia, using modified Delphi method	Indicated skills and qualities are not significant in process of identification of talents, including physiological antropomertic, sociologic and several psychological attributes. It suggests that recruits of talents apply holistic multidisciplinary approach of identification of talents, with current results, which potentially give evidence at the beginning, which indicate that recruits need to consider numerous attributes during the identification of young players.
12	Bidaurrazaga-Letona et al.	The aim of this paper was to determine factors that are important for identification and selection of young footballers	Identification or promotion of players by coaches depends on indicators, which depend on the age. This paper proved that program for identification of talents is more the process of selection rather than process of promotion; selection and identification are posterior and not priority.

(Continued on next page)

(Continued from previous page)

N	Authors	Aim	Conclusion
13	Tribolet et al.	The aim of this paper was to examine differences in age of young players on high level and to examine features that influence the selection in development program on high level.	Significant differences in results of age groups, emphasizing that Academies on high level need to strive to additional selection in age over 15. Apart from this, it seems that players who were favoring for the selection in Academy on high level, matured earlier.
14	Dodd & Newans	The aim of this study was to examine physiological profile of footballer in modern age and testing and process of identification of talents should match with that data.	Using battery for physiological testing enables teams to monitor progress of their players during their development age. This enables coaches to consistently identify strength and weaknesses of players and enables players, who mature later, to be considered as potential, still.
15	Owen et al.	The aim of this study was to examine season changes in body composition in elite European footballers and to identify key position differences.	There are big variances in profiles of body composition among professional European footballers.
16	Berkamp et al.	The aim of this paper was to determine advantages and limitations of methodology, validity and utility of current research about identification of football talents.	Four methodological problems are relevant for researching identification of talents 1) operationalisation of criterion variables (efficiency which is anticipated) as a level of efficiency; 2) focusing on isolated indicators of efficiency as predictors of football performances; 3) results of limitations of volume on predictive validity of predictors, which are used in identification of talent; and 4) effect of the basic level on usefulness of procedures of identification of talent
17	Bennet et al.	The aim of this study was to examine constructive and discriminatory validity of video-based assessment of decision for identification of talents in youth football.	Coaches and athletes would be careful when interpreting data from practical, video-based evaluations in making decisions. Currently, there is limited evidence, which confirms efficiency of these evaluations for identification of talents.

Discussion

Analysis of papers which were subjected to inclusive and exclusive criteria (17 papers) showed that there is no general consensus about the term “talent”, especially about procedures of identification of talent, which is confirmed in earlier studies in overview (Gallardo-Gallardo, Dries, & González-Cruz, 2013).

Analysis of Physiological Aspects of Identification of Talents

Physiology is a segment of medicine, which is essential in sport, because effects of training are most commonly performed on physiological aspects of a man. Therefore, all segments of physiology of athletes are constantly researched, due to which different forms of testing, which will enable easier assessment of athlete's condition. Papers which write about identification of talents in football, suggest that each Academy or youth program constantly monitor different motor and physical parameters of players (Dodd & Newans, 2018). Also, programs of identification of talents need to contain batteries of tests, which are based on physical and physiological abilities of elite footballers, in order to constantly monitor their development (Dodd & Newans, 2018). However, standard procedures of testing cannot be criterion for process of identification of talents (Meylan et al., 2010; Larkin & O'Connor, 2017), and they need to be enriched with TE-TA contents and contents of making decisions (Kannekens, Elferink-Gemser, & Visscher, 2011). Since football is a team game, making decisions and collaboration of team mates based on performance of opponents, is very important psychological aspect of players, such as: emotion intelligence, motivation and making decisions (Murr, Feichtinger, Larkin, O'Connor, & Höner, 2018). Linear speed is one of the most important skills in football and therefore it is very sig-

nificant in process of identification of talent (Hirose & Saki, 2016). Also, it is not possible to determine whether young footballer is a potential, based on antropometric measures of body composition, which doesn't obstruct their performance on the field and playing football on high level (Owen et al., 2018). This attaches to the fact that mass and volume are segments of anthropometry, which is lowly genetic indigenous and can be affected by training.

Importance of Specific Contents for Identification of Talents

Since the center of football game are football actions, which are performed in offense, defense and transition, with ball or without it, and each football action includes 4 time-space components, which can never be repeated 2 times, in a same way: time, space, direction and speed (Verheijen, 2014), it shows significance of integral training in football, and choice of contents, respectively, which are specific in football game, in order to influence making decisions and speed of thinking.

It is proved that games in small spaces (SSG) have direct effect on physical performances of players, which are developed in situational conditions (Martin-Garcia et al., 2019; Giménez & Gomez, 2019). From the aspect of identification of talent, small games are rarely used during the process of selection, because there is no standardized and objective way of assessment of player's potential, through small games, yet. Still, analyzed studies prove that SSG can be very relevant indicator, which can determine whether player has potential or not and to be professional football player or not (Woods, Raynor, Bruce, & McDonald, 2016; Fenner, Iga, & Unnithan, 2016; Unnithan, White, Georgiou, Iga, & Drust, 2012), despite that their movement structure and physiological parameters do not respond fully to the game 11v11 (Giménez & Go-

mez, 2019). Technical aspect of football game is one of the most important and presents fundament of work with young players (>12 y). Although isolated technique, as such, is not fully identical to actions on the field during the games (SSG or matches), it is very important part of training process, especially in this age of a player. Technical performance of a player is not situational way of assessment of talent in young players but it is very important element of football game, which is developed in young age of players. Therefore, technical performance in combination with speed of making decisions and performances of football actions needs to be part of the process of identification and selection of young talents (Höner, Votteler, Schmid, Schultz, & Roth, 2015).

Biological and Chronological Age as an Important Aspect of Selection of Young Players

Although selection of young footballers is formed based on age, this is not a guarantee that players have equal physical and physiological abilities, due to the differences in biological age (BA). Biological and chronological age is not usually in accordance and need to be constantly monitored by coach, because they can be useful tool in explaining different specific-football phenomenon (Carvalho et al., 2019). Therefore, coaches need to monitor players and based on all analyzed segments, which include analysis of chronological and biological age, make decisions. Knowledge of biological age needs to be a very important variable in a process of selection of players (Meylan et al., 2010; Bidaurrazaga-Letona, Lekue, Amado, & Gil, 2017). It is very important to mention that if SSG is used as a test for identification of abilities and talent, respectively, one needs to have in mind that players, who play against each other, need to have similar biological and chronological age, because there is a big difference in football performances of players of a different age (Tribolet, Bennett, Watsford, M. & Fransen, 2017). Also, coaches need to be relieved of achieving results, which is not a measure of player's talent. Therefore, coaches need to be very careful with their choice of work philosophy, because it cannot be based on subjective thinking but on objective scientific facts. Coaches need to know that they, with their decisions, affect the complete process of development of young player, and that the work with young footballers is hard-working process of selection and their personal promotion (Bidaurrazaga-Letona et al., 2017). Studies proved that coaches, not considering the facts, favor physical dominant players, who bring them results. (Tribolet et al., 2017).

Multidisciplinary Approach of Selection of Young Players

In process of selection young players, it is very important to include all segments, which are directly related to the process of development of players. This doesn't mean that only one aspect of player is analysed, in order to determine whether he has potential or not, but to make a selection by consensus of different experts from different scientific disciplines. Results of physiological testing and football performance are insufficient to assess whether a player has a potential or not (Larkin & O'Connor, 2017). Coaches need to have overall and multidisciplinary approach to the selection of players, which will help in objective and real assessment of qualities of young players (Willson et al., 2016; Larkin & O'Connor, 2017; Bidaurrazaga-Letona et al., 2017). Also, video-analysis in football take an important role in selection of talents, therefore it is necessary to ask a question whether making decisions by watching videos is suitable tool for quality decisions for young players. Some researchers proved that this is a very useful and objective tool in process of selection of young players (Woods et al., 2016), but some pioneer steps are taken in that area, therefore one needs to be careful when using video analysis for assessment of quality of making decisions, especially when it comes to the selec-

tion of players (Bennet et al., 2019).

Conclusion

Process of identification of talented players at young age is very complex. There is no general rule in process of identification of talent and that area is still subject of research. This study proves that physiological tests are important part of process of selection of young players, but they cannot be measure of potential of players. Also, it is very important that all segments of football actions need to be close to conditions of the game, and that process of selection needs to be done with physiological testing and conditions of the game, through SSG and technical-tactical performance on the field. Coaches need to be aware of sensitivity of work with young categories and cannot allow that results come first. Based on earlier ways of selection, coaches need to be able to objectively observe players and to make decisions about their future, thinking about biological age of players. When testing young players, physiologically and physically, tests of genetic highly arranged abilities need to be emphasized, because they are tools for determining physical potential of players.

Acknowledgements

There are no acknowledgements.

Conflict of Interest

The authors declare that there are no conflicts of interest.

Received: 1 July 2020 | **Accepted:** 22 September 2020 | **Published:** 16 October 2020

References

- Beechler, S., & Woodward, I.C. (2009). The global "war for talent". *Journal of international management*, 15(3), 273-285.
- Bennett, K.J., Novak, A.R., Pluss, M.A., Coutts, A.J., & Fransen, J. (2019). Assessing the validity of a video-based decision-making assessment for talent identification in youth soccer. *Journal of science and medicine in sport*, 22(6), 729-734.
- Bergkamp, T.L., Niessen, A.S.M., Den Hartigh, R.J., Frencken, W.G., & Meijer, R.R. (2019). Methodological issues in soccer talent identification research. *Sports Medicine*, 49(9), 1317-1335.
- Bethke-Langenegger, P. (2012). *The differentiated workforce: Effects of categorization in talent management on workforce level*. Unpublished Diskussionspapier No 18. Zurich, SH: University of Zurich, Department of Business Administration.
- Bidaurrazaga-Letona, I., Lekue, J.A., Amado, M., & Gil, S. M. (2019). Progression in youth soccer: Selection and identification in youth soccer players aged 13–15 years. *The Journal of Strength & Conditioning Research*, 33(9), 2548-2558.
- Carvalho, H.M., Leonardi, T.J., Soares, A.L., Paes, R.R., Foster, C., & Gonçalves, C.E. (2019). Longitudinal changes of functional capacities among adolescent female basketball players. *Frontiers in physiology*, 10, 339.
- Cheese, P., Thomas, R.J., & Craig, E. (2007). *The talent powered organization: Strategies for globalization, talent management and high performance*. London: Kogan Page Publishers.
- Di Palma, D., Raiola, G., & Tafuri, D. (2016). Disability and Sport Management: a systematic review of the literature. *Journal of Physical Education and Sport*, 16(3), 785.
- Dodd, K.D., & Newans, T.J. (2018). Talent identification for soccer: Physiological aspects. *Journal of science and medicine in sport*, 21(10), 1073-1078.
- Fenner, J.S., Iga, J., & Unnithan, V. (2016). The evaluation of small-sided games as a talent identification tool in highly trained prepubertal soccer players. *Journal of Sports Sciences*, 34(20), 1983-1990.
- Gagné, F. (2000). Understanding the complex choreography of talent development through DMGT-based analysis. *International handbook of giftedness and talent*, 2, 67-79.
- Gallardo-Gallardo, E., Dries, N., & González-Cruz, T.F. (2013). What is the meaning of 'talent' in the world of work?. *Human Resource Management Review*, 23(4), 290-300.
- Giménez, J.V., & Gomez, M.A. (2019). Relationships Among Circuit Training, Small-Sided and Mini Goal Games, and Competition in Professional Soccer Players: A Comparison of On-Field Integrated Training Routines. *The Journal of Strength & Conditioning Research*, 33(7), 1887-1896.

- González Cruz, T., Martínez Fuentes, C., & Pardo del Val, M. (2009). La gestión del talento en la empresa industrial española. *Economía industrial*, (374), 21-35.
- Goto, H., Morris, J.G., & Nevill, M.E. (2015). Match analysis of U9 and U10 English premier league academy soccer players using a global positioning system: Relevance for talent identification and development. *The Journal of Strength & Conditioning Research*, 29(4), 954-963.
- Güllich, A. (2014). Selection, de-selection and progression in German football talent promotion. *European journal of sport science*, 14(6), 530-537.
- Hirose, N., & Seki, T. (2016). Two-year changes in anthropometric and motor ability values as talent identification indexes in youth soccer players. *Journal of science and medicine in sport*, 19(2), 158-162.
- Hoad, T.F. (1993). *The concise Oxford dictionary of English etymology*. Oxford: Oxford University Press.
- Hoare, D., & Warr, C.R. (2000). Talent identification and women's soccer: an Australian experience. *Journal of sports sciences*, 18(9), 751-758.
- Höner, O., Votteler, A., Schmid, M., Schultz, F., & Roth, K. (2015). Psychometric properties of the motor diagnostics in the German football talent identification and development programme. *Journal of Sports Sciences*, 33(2), 145-159.
- Jerico, P. (2001). La gestión del talento: Enfoque conceptual y empírico. *Boletín de Estudios Económicos*, 56(174), 423.
- Kannekens, R., Elferink-Gemser, M.T., & Visscher, C. (2011). Positioning and deciding: key factors for talent development in soccer. *Scandinavian journal of medicine & science in sports*, 21(6), 846-852.
- Larkin, P., & O'Connor, D. (2017). Talent identification and recruitment in youth soccer: Recruiter's perceptions of the key attributes for player recruitment. *PLOS one*, 12(4), e0175716.
- Lidor, R., Côté, J.E.A.N., & Hackfort, D. (2009). ISSP position stand: To test or not to test? The use of physical skill tests in talent detection and in early phases of sport development. *International journal of sport and exercise psychology*, 7(2), 131-146.
- Martin-Garcia, A., Castellano, J., Diaz, A.G., Cos, F., & Casamichana, D. (2019). Positional demands for various-sided games with goalkeepers according to the most demanding passages of match play in football. *Biology of Sport*, 36(2), 171.
- Meylan, C., Cronin, J., Oliver, J., & Hughes, M. (2010). Talent identification in soccer: The role of maturity status on physical, physiological and technical characteristics. *International Journal of Sports Science & Coaching*, 5(4), 571-592.
- Murr, D., Feichtinger, P., Larkin, P., O'Connor, D., & Höner, O. (2018). Psychological talent predictors in youth soccer: A systematic review of the prognostic relevance of psychomotor, perceptual-cognitive and personality-related factors. *PloS One*, 13(10), e0205337.
- O'Connor, D., Larkin, P., & Mark Williams, A. (2016). Talent identification and selection in elite youth football: An Australian context. *European journal of sport science*, 16(7), 837-844.
- Ommundsen, Y. (2009). *Hvem er talentene, må vi spesialisere tidlig og hva er en god trener-Spenningsfeltet mellom barne-og ungdomsidrett og eliteidrett. Nyere perspektiver innen idrett og idrettspedagogikk*. Oslo: Høyskoleforlaget.
- Owen, A. L., Lago-Peñas, C., Dunlop, G., Mehdi, R., Chtara, M., & Dellal, A. (2018). Seasonal body composition variation amongst elite european professional soccer players: an approach of talent identification. *Journal of human kinetics*, 62(1), 177-184.
- Martin, R. (2006) *The Work of Professional Football. A Labo-ur of Love?* London, New York: Routledge.
- Saether, S.A. (2016, 21 September). *Identification of Talent in Soccer. What Do Coaches Look For?* Retrieved from *Idrottsforum. Org website*: <http://idrottsforum.org/wp-content/uploads/2014/03/saether140319.pdf>
- Silzer, R., & Dowell, B.E. (2009). *Strategy-driven talent management: A leadership imperative*. Hoboken, NJ: John Wiley & sons.
- Stahl, G.K., Björkman, I., Farndale, E., Morris, S.S., Paauwe, J., Stiles, P., ... & Wright, P. (2007). *Global talent management: How leading multinationals build and sustain their talent pipeline*. Unpublished Working Papers No 2007/24/OB. Fontainebleau: INSEAD.
- Tansley, C., Harris, L., Stewart, K., Turner, P., Foster, C., & Williams, H. (2006). *Talent management: understanding the dimensions change agenda*. London: CIPD
- Tansley, C., Turner, P., Foster, C., Harris, L., Stewart, J., Sempik, A., & Williams, H. (2007). *Talent: Strategy, management, measurement*. London: CIPD
- Tribollet, R., Bennett, K.J., Watsford, M.L., & Fransen, J. (2018). A multidimensional approach to talent identification and selection in high-level youth Australian football players. *Journal of sports sciences*, 36(22), 2537-2543.
- Trninić, S., Jelaska, I., & Papić, V. (2009). Kinesiological, anthropological, and methodological aspects of efficacy equation in team sports games. *Acta kinesiológica*, 3(2), 7-18.
- Ulrich, D., & Smallwood, N. (2012). What is talent?. *Leader to leader*, 2012(63), 55-61. doi: 10.1002/ltl.20011
- Unnithan, V., White, J., Georgiou, A., Iga, J., & Drust, B. (2012). Talent identification in youth soccer. *Journal of sports sciences*, 30(15), 1719-1726.
- Vaeyens, R., Malina, R. M., Janssens, M., Van Renterghem, B., Bourgois, J., Vrijens, J., & Philippaerts, R. M. (2006). A multidisciplinary selection model for youth soccer: the Ghent Youth Soccer Project. *British journal of sports medicine*, 40(11), 928-934.
- Verheijen, R. (2014). *Football periodization: Always play with your strongest team*. Naarden: World Football Academy.
- Williams, A.M., & Reilly, T. (2000). Talent identification and development in soccer. *Journal of sports sciences*, 18(9), 657-667.
- Woods, C.T., Raynor, A. J., Bruce, L., & McDonald, Z. (2016). Discriminating talent-identified junior Australian football players using a video decision-making task. *Journal of sports sciences*, 34(4), 342-347.
- Wilson, R.S., James, R. S., David, G., Hermann, E., Morgan, O.J., Niehaus, A.C., ... & Smith, M.D. (2016). Multivariate analyses of individual variation in soccer skill as a tool for talent identification and development: utilising evolutionary theory in sports science. *Journal of sports sciences*, 34(21), 2074-2086.
- Williams, A.M., & Reilly, T. (2000). Talent identification and development in soccer. *Journal of sports sciences*, 18(9), 657-667.

REVIEW PAPER

Effects of the Aerobic Exercise Program with Music on the Body Composition and Subcutaneous Fat of Young Women: A Systematic Review

Miljan Hadzovic¹, Ana Lilic¹, Nikola Prvulovic¹, Predrag Ilic¹, Mima Stankovic¹

¹University of Nis, Faculty of Sport and Physical Education, Nis, Serbia

Abstract

Numerous authors recommend participation in various types of aerobic exercise, pointing out their positive effects both in terms of the prevention of various illnesses, and in the improvement of one's health status. The increasingly popular group fitness programs, as preprogrammed forms of physical exercise to music, have positive effects on body composition among women, such as a decrease in body mass and fat, and thus lead to an improvement in overall physical appearance. The aim of this research was to determine the effects of aerobic exercise to music on body composition and subcutaneous fat among young women. In order to compile data from existing research on the effects of aerobic exercise on young women, the following electronic databases were searched: PubMed, SCIndex, PEDro, J-GATE, DOAJ and Google Scholar. The systematic review was undertaken in accordance with the statement of Preferred Reports for Systematic Review and Meta-Analyses (PRISMA). Papers dating from 2006 to 2019 were analyzed. The research results confirm the positive influence of aerobic exercise to music on the parameters of body composition and subcutaneous fat among young women, irrespective of the overall duration of the exercise programs, with the conclusion that programs lasting up to 12 weeks are more effective when it comes to the decrease and maintenance of body weight, fat reduction, increase in muscle tissue, and lead to an improvement in aerobic abilities. The best effects of this type of exercise at an intensity level of 50 to 80% of maximum heart rate are primarily achieved among obese and overweight individuals.

Keywords: *Physical Activity, Dance, Body Mass Index, Body Fat Percentage*

Introduction

The most recent data from the World Health Organization (WHO) indicate that physical inactivity and an inappropriate diet lead to obesity among all age categories, which in turn represents one of the growing problems among the world population. Experts from this organization recommend physical exercise as one of the most effective means of solving the aforementioned problem (World Health Organization, 2018). It is known that maintaining body mass over a long period of time is more easily achieved through continued physical exercise than the mere reduction of one's diet (Aksović, Aleksandrović, & Jorgić, 2017). Blair, LaMonte, & Nichaman (2004) determined that reg-

ular physical exercise, in addition to maintaining normal physical weight, contributes to the improvement of quality of life, and an increase in muscle strength and endurance. Even though physical activity, which increases energy consumption and has a positive effect on the maintenance of body weight, includes various types of movement which we perform on a daily basis such as walking, climbing, various chores around the house, riding a bicycle to do our shopping and the like, in order to improve an individual's psycho-physical health, a programmed, planned and well-organized physical exercise program is required (Ostojić et al., 2009).

The implementation of an aerobic exercise program, that is,

Correspondence:

**Montenegro
Sport**

Miljan Hadzovic
University of Nis, Faculty of Sport and Physical Education, Caraojevic 10A, Nis 18000, Serbia
E-mail: miljanhadzovic@gmail.com

the activation of large muscle groups over an extended period of time, impacts the improvement and development of aerobic and cardiovascular fitness, that is, aerobic endurance (Pantelić et al., 2007; Radovanović et al., 2009). Numerous authors recommend participation in various types of aerobic exercise, pointing out the positive effects both for the prevention of various illnesses, and the improvement of one's health status (Donges, Duffield, & Drinkwater, 2010; Stasiulis, Mockiene, Vizbaraitė, & Mockius, 2010; Jorgić, Pantelić, Milanović, & Kostić, 2011; Kimura & Hozumi, 2012; Pantelić, Milanović, Sporiš, & Stojanović-Tošić, 2013; Sivvas, Batsiou, Vasoglou, & Filippou, 2015; Aksović et al., 2017). Walking, running, riding a bicycle, swimming, dance aerobics and other forms of group fitness represent the most frequently used sports-recreational activities by adults for the reduction of body weight, performed at a moderate intensity of 50% to 80% of the maximum heart rate (Đorđević, 2005). The increasingly popular group fitness programs, as programmed forms of physical exercise to music, have positive effects on body composition, the decrease in body mass, and amount of fat among women, also leading to an improvement in physical appearance (Bjelica, Gojković, & Fulurija, 2017).

The aim of the research was to determine the effects of aerobic exercise to music on body composition and subcutaneous fat among young women. After compiling the relevant data from previous experimental studies dating from 2006 to 2019, under the assumption that the implementation of aerobic exercise to music has a positive impact on the body composition of women, an evaluation of the effects was carried out.

Methods

Inclusion criteria

For an experimental study to be included in the final analysis, it had to meet certain criteria: the participants in the research were individuals not afflicted by any chronic conditions; the experimental research included females of an average age of 18 to 36, irrespective of their lifestyle; the experimental group took part in an aerobic exercise program to music during which its effects were evaluated and measurements taken of the parameters of body composition of the young women; the research was published in English. The exclusion criteria included: studies which included only male participants, studies which included female participants whose average age did not range from 18 to 36, papers not published in English.

Search strategy

In order to compile existing research on the effects of aerobic training programs among young women, the following electronic databases were searched: PubMed, SCIndeks, PEDro, J-GATE, DOAJ and Google Scholar. Papers dating from 2006 to 2019 were analyzed. The following key words were used for the data search: physical activity, dance, body mass index, body fat percentage. The titles of various studies were identified; abstracts and entire texts were then read and analyzed. This systematic review was undertaken in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (Moher, Liberati, Tetzlaff, & Altman, 2009).

Data extraction and selection

Experimental research which met the set criteria was then analyzed and presented based on the following parameters: references (the initials of the author and year of publication), the sample of participants (health status and lifestyle, age, overall number and subgroups of the participants), physical exercise program, the duration and frequency of exercise, intensity of exercise, research results.

Study quality and risk of bias

Risk of bias was evaluated according to the PRISMA recommendation and two independent reviewers assessed the risk of bias. When there was disagreement about the risk of bias a third reviewer checked the data and took the final decision on it.

Synthesis of results

By searching the electronic databases, 1112 experimental studies were identified on the set topic. Primarily 510 studies were eliminated as duplicates, then 538 papers were rejected based on abstracts, while 44 studies were excluded and based on age and gender of the subjects, or were not written in English (Graph 1). Applying the set criteria, the final analysis included 20 experimental studies (Table 1).

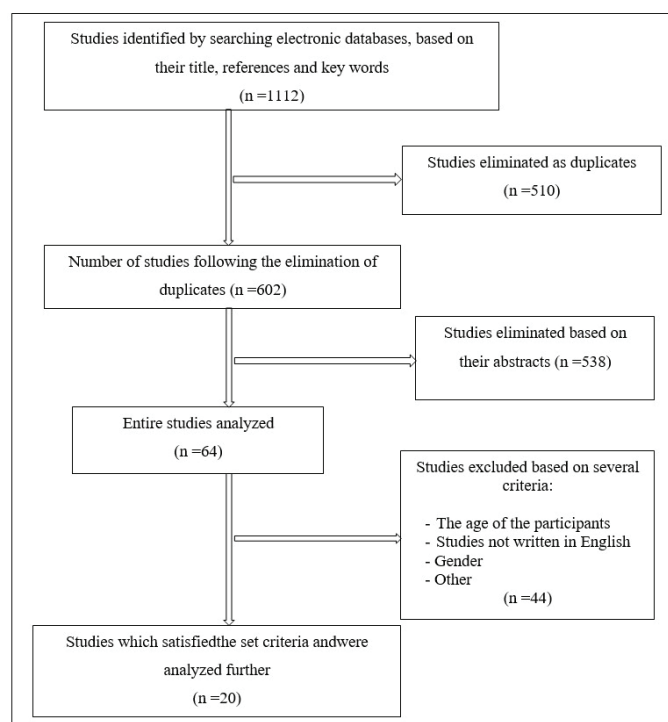
Results

In accordance with the aim of the study and the set criteria, a final analysis of 20 experimental studies was performed (Kin-Isler & Kosar, 2006; Stojiljković, Mandarić, Todorović, & Mitić, 2010; Çakmakçı, Arslan, Taşkin, & Çakmakçı, 2011; Changela, 2013; Marandi, Abadi, Esfarjani, Mojtahedi, & Ghasemi, 2013; Mathunjwa, Semple, & du Preez, 2013; Najafnia, Bararpour, Amirinejad, & Nakhaee, 2013; Nikić & Milenković, 2013; Pantelić et al., 2013; Ljubojević et al., 2014; Kostrzewa-Nowak et al., 2015; Haghjoo, Zar, & Hoseini, 2016; Delextrat, Warner, Graham, & Neupert, 2016; Kaya, Nar, Erzeybek, & Bozdoğan, 2016; Kumar, 2016; Vassilopoulou, Piperari, & Christoforou, 2017; Ativie et al., 2018; Bastug, 2018; Biswas & Bandopadhyay, 2018; Oktay, 2018). The total number of participants included in this survey was 1127. In 17 of the 20 papers, the participants were only women or 854 participants in total, while 273 male participants were represented in three studies where the groups were mixed (Kin-Isler & Kosar, 2006; Kaya et al., 2016; Bastug, 2018). The analyzed studies included young women from 10 different countries such as: Great Britain, Republic of Poland, Republic of Serbia, Bosnia and Herzegovina, Republic of Turkey, Republic of Cyprus, Islamic Republic of Iran, Republic of India, Republic of South Africa, Federal Republic of Nigeria. Although in 10 studies (Stojiljković et al., 2010; Çakmakçı et al., 2011; Changela, 2013; Marandi et al., 2013; Mathunjwa et al., 2013; Kostrzewa-Nowak et al., 2015; Haghjoo et al., 2016; Kaya et al., 2016; Vassilopoulou et al., 2017; Ativie et al., 2018) obese or overweight people were included in the exercise program, most of the analyzed studies had healthy young women with normal body weight for the respondents.

Discussion

The research results, presented in table form (Table 1), indicated that the analyzed studies mostly included programs which incorporate several types of aerobic exercise to music and a combination of various types of dance (Stojiljković et al., 2010; Changela, 2013; Marandi et al., 2013; Najafnia et al., 2013; Kumar, 2016; Ativie et al., 2018; Bastug, 2018; Biswas & Bandopadhyay, 2018), followed by studies which included various types of Zumba exercise programs (Ljubojević et al., 2014; Haghjoo et al., 2016; Delextrat et al., 2016; Kaya et al., 2016; Vassilopoulou et al., 2017; Biswas & Bandopadhyay, 2018; Oktay, 2018). The authors of four studies used "Hi-Lo" aerobics (Çakmakçı et al., 2011; Pantelić et al., 2013; Kostrzewa-Nowak et al., 2015; Biswas & Bandopadhyay, 2018), two of the studies involved independent use of "Step" aerobics (Kin-Isler & Kosar, 2006; Nikić & Milenković, 2013) and one included Tae-Bo (Mathunjwa et al., 2013).

The duration of the experimental studies differed and ranged from an interval of 24 weeks (Stojiljković et al., 2010) to eight weeks (Çakmakçı et al., 2011; Changela, 2013; Mathunjwa et al., 2013; Ljubojević et al., 2014; Haghjoo et al., 2016; Delextrat et al.,



GRAPH 1. A diagram of the course of analysis of the papers

2016; Kaya et al., 2016; Vassilopoulou et al., 2017; Ativie et al., 2018; Oktay, 2018) which is also the most frequent duration of the implemented exercise programs. The greatest weekly frequency of training was determined in 16 of the studies (Kin-Isler & Kosar, 2006; Stojiljković et al., 2010; Çakmakçı et al., 2011; Changela, 2013; Marandi et al., 2013; Mathunjwa et al., 2013; Najafnia et al., 2013; Nikić & Milenković, 2013; Pantelić et al., 2013; Ljubojević et al., 2014; Kostrzewa-Nowak et al., 2015; Haghighi et al., 2016; Delextrat et al., 2016; Kaya et al., 2016; Vassilopoulou et al., 2017; Oktay, 2018) and included three training sessions per week for a duration of 60 minutes. Kumar (2016) included exercises for a duration of 60 minutes, but five times a week, while in one study (Ativie et al., 2018) the frequency was four training sessions per week for a period of eight weeks, while the duration of the sessions increased every two weeks, ranging from 30 to 60 minutes. The longest running study included training sessions once a week, for a duration of 30 to 70 minutes. In addition, in one study there is no data on the weekly frequency and duration of the exercise (Biswas & Bandopadhyay, 2018).

The intensity of the exercise in the analyzed experimental studies was moderate to high and ranged from 50% to 90%. One study included a program with the highest intensity, 90% (Stojiljković et al., 2010). In four of the studies (Haghighi et al., 2016; Delextrat et al., 2016; Kumar, 2016; Vassilopoulou et al., 2017) the intensity was 85% of maximum heart rate (HRmax), in another four the upper limit of the exercise intensity was 80% of maximum heart rate (Nikić & Milenković, 2013; Pantelić et al., 2013; Bastug, 2018). One study (Çakmakçı et al., 2011) included a program of moderate intensity of 60-70% HRmax. The lowest intensity of exercise ranged from less than 40% HRmax (Delextrat et al., 2016;), in nine of the studies (Changela, 2013; Mathunjwa et al., 2013; Kostrzewa-Nowak et al., 2015; Nikić & Milenković, 2013; Haghighi et al., 2016; Kumar, 2016; Vassilopoulou et al., 2017; Bastug, 2018; Oktay, 2018) the minimum intensity was 50% HRmax, while in three studies (Stojiljković et al., 2010; Çakmakçı et al., 2011; Pantelić et al., 2013) the lowest exercise intensity was 60% of maximum heart rate. In two of the studies (Kin-Isler & Kosar, 2006; Marandi et al., 2013) the exercise intensity was deter-

mined based on the resting heart rate, ranging from 45 to 75% and from 60 to 80% of heart rate reserve.

The structure of the training sessions implemented in all the studies was the same, and included an introductory (warm up) part of lower intensity for a period of 5 to 10 minutes (Kin-Isler & Kosar, 2006; Çakmakçı et al., 2011; Changela, 2013; Mathunjwa et al., 2013; Najafnia et al., 2013; Nikić & Milenković, 2013; Pantelić et al., 2013; Ljubojević et al., 2014; Kostrzewa-Nowak et al., 2015; Haghighi et al., 2016; Kaya et al., 2016; Ativie et al., 2018; Oktay, 2018), followed by the main part of the training session (from 25 to 45 minutes) with higher intensity exercises, which consisted of a number of shorter choreographies with smaller breaks, and a final part (from 10 to 15 minutes) consisting of exercises meant to cool down the body, and stretching exercises.

The systematic overview of the parameters which were measured noted multiple effects of the implemented exercise programs on the body composition of young women. The effects of the aerobic exercise programs to music led to a decrease in the BMI, with a statistically significant difference in favor of the studies with longer exercise programs. The greatest statistically significant decrease ($p < 0.001$) in BMI was determined by Kumar (2016), whose program lasted for 12 weeks, with five training sessions per week of 60 minutes each, as well as in a study which lasted 24 weeks with three 60-minute training sessions a week (Stojiljković et al., 2010), followed by a study which lasted for 14 weeks with a single training session per week of 30 to 70 minutes (Bastug, 2018) and the study of Kostrzewa-Nowak et al. (2015) which lasted for 12 weeks ($p < 0.01$). The exercise programs included in studies which lasted eight weeks (Kaya et al., 2016; Ativie et al., 2018), with three and four training sessions per week, led to a statistically significant difference ($p < 0.05$) in the BMI in favor of the experimental group. Similar effects to those of the previous study were obtained by Mathunjwa et al. (2013) whose study lasted for 10 weeks, with three training sessions per week of a duration of 60 minutes.

Comparing the results of exercise on the decrease in skinfold thickness, two studies including obese or overweight participants (Mathunjwa et al., 2013; Kostrzewa-Nowak et al., 2015) showed a statistically significant difference ($p < 0.05$) in favor of the experi-

Table 1. The systematic review and characteristics of the selected research

Study (year)	Health status	Gender	Lifestyle	Participants' age (MEAN ± SD)	Sample size (n)	(BMI kg/m ²)	No. of participants	Duration Frequency (days/ weeks)	Intensity Duration of the training (min)	Type of activity	Results (r)
Kin-Isler & Kosar, 2006	Healthy individuals	F=64 M=54	Sedentary	EG (M + F) F=25.00±1.46 M=22.48±2.02 KG (M + F) F=21.00±1.42 M=22.13±1.70	n=118	X	EG=33F+27M KG=31F+27M	10 weeks 3/7	50min HRR 60–80%	EG – dance aerobics (Step) KG – no physical activity	M≠F BFP%(M) ↓ p<0.05 LBM (M) ↑ p<0.05 MSL (M) ↑ p<0.05 BW ≠ p>0.05 EG»KG (GxG) BFP% ≠ LBM ≠ MSL ≠ BW ≠ MPRBW ↑ p<0.05 EG»KG (F»M) APVJ ↑ p<0.05
Stojiljković et al., 2010	Healthy individuals with normal weight and overweight	F	Sedentary	33.6±6	n=10	25.61±3.87	EG=10	24 weeks 3/7	60min HRmax 60–90%	Omnibus aerobics (a combination of 12 types of aerobics)	EG» BW ↓ p≤0.01 BMI ↓ p≤0.01 WC ↓ p≤0.01 HC ↓ p≤0.01 WHR ≠ ON ↓ p≤0.01 OP ↓ p≤0.05 ONL ↓ p≤0.01 CCH ↓ p≤0.01 CRNC ↑ p≤0.01 SQ ↑ p≤0.01 PU ↑ p≤0.01
Çakmakçı et al., 2011	Obese individuals	F	Sedentary	EG=35.10±9.12 KG=30.27±10.85	n=55	EG=26.58±2.26 KG=26.11±3.64	EG = 29 KG = 26	8 weeks 3/7	60min HRmax 60–70%	EG – dance aerobics (Hi-Lo) KG – no physical activity	EG»KG BW ↓ p<0.05 BMI ↓ p<0.05 WC ↓ p<0.05 WHR ↓ p<0.05 SFLB ↓ p<0.05 SFLT ≠ SFLS ≠ SFLSI ≠ BMR ↓ p<0.05 BFP% ↓ p<0.05 LBM ↓ p<0.05

Study (year)	Health status	Gender	Lifestyle	Participants' age (MEAN ± SD)	Sample size (n)	(BMI kg/m ²)	No. of participants	Duration Frequency (days/ weeks)	Intensity Duration of the training (min)	Type of activity	Results (r)
Changela, 2013	Obese women	F	Sedentary	19-25	n = 20	EGa=31.93±1.47 EGb=32.19±2.57	EGa=10 EGb=10	Dance aerobics 8 weeks 3/7 Resistance training 6 weeks of alternatives	Dance aerobics 40min HRmax 50-75% Resistance training 4 sets of 10 reps 50-100% RM	Eg - dance aerobics Eg - resistance training	EGa» HRmax↑ p<0.001 SBP ↓ p<0.001 DBP ↓ p<0.05 BW ↓ ± p=0.06 BMI ↓ ± p=0.06 BFP% ↓ ± p=0.06 EGb» HRmax↓ p=0.04 SBP ↓ p<0.001 BW ↓ ± p=0.06 BMI ↓ ± p=0.06 BFP% ↓ ± p=0.06 EGa» EGb VLDL ↓ p<0.05 HDL ↑ p<0.05
Marandi et al., 2013	Obese individuals and overweight individuals	F	X	25-40	n=45	EGa=27.83±4.15 EGb=29.19±3.86 KG=30.38±4.98	EGa=15 EGb=15 KG=15	10 weeks 3/7	60min Eg 45-50% HRRmax Eg 70-75% HRRmax	Eg - dance aerobics (Low int.) Eg - dance aerobics (Moderate int.) KG - no physical activity	EGa and EGb » KG BW ↓ p<0.00 BMI ↓ p<0.00 BFP% ↓ p<0.05 FMkg ↓ p<0.03 WHR ↓ p<0.00 LBM ↑ p<0.02 HDL ↑ p<0.00
Mathunjwa et al., 2013	Obese individuals and overweight individuals	F	Sedentary	25±5	n=60	Overweight BMI>25-29.90 kg/m2 Obese BMI≥30-39.90 kg/m2	EG=60	10 weeks 3/7	60min 1-5 weeks moderate tempo 6-10 weeks high tempo	Aerobics (Tae-Bo)	EG» BW ↓ p≤0.05 BMI ↓ p≤0.05 WC ↓ p≤0.05 HC ↓ p≤0.05 ΣSFL ↓ p≤0.05
Najafnia et al., 2013	Healthy individuals	F	15 physically active individuals 15 individuals with a sedentary lifestyle	EG=26.24±5.86 KG=25.94±5.88	n=30	X	EG=15 KG=15	8 weeks 3/7	50-60min HRmax 50-75%	EG - dance aerobics (Step) KG - no physical activity	EG»KG BFP% ↓ p=0.02 FLB ↑ p=0.01 SB ↑ p=0.01 VO2max ↑ p=0.01 VIT ↑ p=0.02 BLD ↓ p=0.01 FGS ↓ p=0.02

(Continued on next page)

Study (year)	Health status	Gender	Lifestyle	Participants' age (MEAN ± SD)	Sample size (n)	(BMI kg/m ²)	No. of participants	Duration Frequency (days/ weeks)	Intensity Duration of the training (min)	Type of activity	Results (t)
Nikić & Milenković, 2013	Healthy individuals	F	X	18-25	n=24	X	EG=12 KG=12	12 weeks 3/7	60min HRmax 50–80%	EG – dance aerobics (Step) KG – no physical activity	EG»KG MS ↑ p≤0.01 CD ↑ p≤0.01 BW ↓ p≤0.01 BFP% ↓ p≤0.01 BWP% ↑ p≤0.01 MM ↑ p≤0.01 BMR ↑ p≤0.01
											EG» BW ↓ ¥ ΣSFLUB ↓ p<0.05 ΣSFL ↓ p<0.05 BFP% ↑ p<0.05 KG» BW ↓ ¥ MM% ↓ p<0.05 EG»KG ΣSFLLB ↓ p<0.05 ΣSFL ↓ p<0.05 BFP% ↓ p<0.05
Pantelić et al., 2013	Healthy individuals	F	X	EG=23.10±1.90 KG=22.70±1.80	n=59	EG=23.00±2.20 KG=21.70±1.70	EG=29 KG=30	12 weeks 3/7	60min HRmax 60–80%	EG – dance aerobics (HI-Lo) KG – no physical activity	EG» BW ↓ ¥ ΣSFLUB ↓ p<0.05 ΣSFL ↓ p<0.05 BFP% ↑ p<0.05 KG» BW ↓ ¥ MM% ↓ p<0.05 EG»KG ΣSFLLB ↓ p<0.05 ΣSFL ↓ p<0.05 BFP% ↓ p<0.05
											EG» BW ↓ p=0.02 BFP% ↓ p=0.01 BFkg ↓ p=0.00 FFM ↑ ¥ TBW ↑ ¥
Ljubojević et al., 2014	Healthy individuals	F	X	25-35	n=12	X	EG=12	8 weeks 3/7	60min	EG – Zumba exercise program	EG» BW ↓ p=0.02 BFP% ↓ p=0.01 BFkg ↓ p=0.00 FFM ↑ ¥ TBW ↑ ¥
											EGa» BW ↑ p=0.01 BMI ↑ p=0.01 VO2max ↑ p=0.05 EG6» ¥
Kostrzewa-Nowak et al., 2015	Healthy undernourished individuals, individuals with normal mass, overweight individuals	F	X	19-24	n=37	Undernourished EGa=17.90 (mean) Normal mass EGb=21.30 (mean) Overweight EGc=26.00 (mean)	EGa = 10 EGb = 12 EGc = 12	12 weeks 3/7	3 weeks 50-60% 3 weeks 55-65% 3 weeks 60-70% 3 weeks 65-75%	Dance aerobics (HI-Lo)	EGc» BW ↓ p=0.01 BMI ↓ p=0.01 FFM ↓ p=0.01 TBW ↓ p=0.05 BFP% ↓ p=0.01 ΣSFL ↓ p<0.05
											EGc» BW ↓ p=0.01 BMI ↓ p=0.01 FFM ↓ p=0.01 TBW ↓ p=0.05 BFP% ↓ p=0.01 ΣSFL ↓ p<0.05

Study (year)	Health status	Gender	Lifestyle	Participants' age (MEAN ± SD)	Sample size (n)	(BMI kg/m ²)	No. of participants	Duration Frequency (days/ weeks)	Intensity Duration of the training (min)	Type of activity	Results (t)
Haghjoo et al., 2016	Healthy overweight individuals	F	X	EG=31.50±6.97 KG=29.85±8.17	n=29	EG=27.88±4.60 KG=28.66±4.53	EG=16 KG=13	8 weeks 3/7	60min HRmax 50–85%	EG – Zumba KG – no physical activity	EG»KG BFP% ↓ p=0.001 BMI ↓ p=0.001 FMkg ↓ p=0.001 WHR ↓ p=0.004 PM p=0.92 ¥
Delextrat et al., 2016	Healthy individuals	F	X	EG=26.60±5.40 KG=27.90±6.00	n=37	X	EG=19 KG=18	8 weeks 3/7	60min HRmax 40–85%	EG – Zumba at home (DVD) KG – no physical activity	(GxT) VO2max ↑ p=0.01 EG» (GxT) VO2max ↑ +3.6%; p<0.01 KG» VO2max ± –2.4% p=0.06 EG» KG ¥ BW ↓ p>0.05 BFP% ↓ p>0.05
Kaya et al., 2016	Healthy individuals with normal weight and healthy overweight individuals	F=46 M=44	Sedentary	EGM=30.70±6.44 EGF=29.13±5.80	n=90	EGM=25.52±3.55 EGF=26.79±5.01	EGM=44 EGF=46	8 weeks 3/7	45-60min Resistance training 3 sets of 10-14 reps >40% RM Zumba training to music	EGM – resistance training EGF – Zumba	EGM » and EGF » BW ↓ p<0.05 BMI ↓ p<0.05 BMR ↓ p<0.05 BFP% ↓ p<0.05 BFkg ↓ p<0.05 FFM ↓ p<0.05 TBW ↓ p<0.05
Kumar, 2016	Healthy individuals	F	X	18–24	n=30	X	EG=15 KG=15	12 weeks 5/7	60min HRmax 50–85%	EG – Zumba and aerobics (Hi Low, Step, Floor) KG – no physical activity	EG» KG CE ↑ p<0.01 BMI ↓ p<0.001

(Continued on next page)

Study (year)	Health status	Gender	Lifestyle	Participants' age (MEAN ± SD)	Sample size (n)	(BMI kg/m ²)	No. of participants	Duration Frequency (days/ weeks)	Intensity Duration of the training (min)	Type of activity	Results (r)
Vassilopoulou et al., 2017	Healthy overweight individuals	F	Physically active	KG=27.80±4.89 EGa=28.27±4.96 EGb=27.00±5.06	n=32	KG=25.98±1.82 EGa=25.24±1.17 EGb=26.10±1.34	KG=10 EGa=11 EGb=11	8 weeks 3/7	60min HRmax 50–85%	KG – Zumba EGa – Zumba + diet + seminars EGb – Zumba + individual diet plans	KG ↓ p>0.05 BW ↓ p=0.07 BFP% ↓ p=0.59 WHR ↓ p=0.10 WC ↓ p=0.36 HC ↓ p=0.07 EGa» BW ↓ p=0.14 BFP% ↓ p=0.05 WHR ↓ p=0.59 WC ↓ p=0.07 HC ↓ p=1.00 EGb» BW ↓ p=0.04 BFP% ↓ p=0.04 WHR ↓ p=0.13 WC ↓ p=0.07 HC ↓ p=0.03
Atvie et al., 2018	Obese individuals and overweight individuals	F	X	18–30 EG=20.06±1.50 KG=21.14±1.60	n=61	EG=27.07±2.38 KG=29.12±4.63	EG=32 KG=29	8 weeks 4/7	0–2 weeks 30min 2–4 weeks 40min 4–6 weeks 50min 6–8 weeks 60min Moderate intensity 12–14 (the Borg scale 6–20)	EG – dance aerobics KG – no physical activity	EG»KG BW ↓ p<0.05 BMI ↓ p<0.05 WC ↓ p<0.05 HC ↓ p<0.05 WHR ↓ p<0.05 BFP% ↓ p<0.05 EG» MM ↑ p<0.05
Bastug, 2018	Healthy individuals	F=93 M=175	X	20.59±1.59	n=268	EG=23.30±2.75 KG=22.80±3.03	EG=129 KG=139	14 weeks 1/7	30–70min HRmax 50–80%	EG – Salsa, Zumba and Modern dance KG – no physical activity	EG»KG BW ↓ p<0.01 BMI ↓ p<0.01 FLX ↑ p<0.05 BLC ↑ p<0.001 CONC ↑ p<0.001

Study (year)	Health status	Gender	Lifestyle	Participants' age (MEAN ± SD)	Sample size (n)	(BMI kg/m ²)	No. of participants	Duration Frequency (days/ weeks)	Intensity Duration of the training (min)	Type of activity	Results (r)
Biswas et al., 2018	Healthy individuals	F	X	20–25	n=90	X	KG=30 EGa=30 EGb=30	12 weeks	X	KG – no physical activity EGa – Zumba EGb – Aerobics	EGa>KG p<0.05 BFP% ↓ p<0.05 MM% ↑ p<0.05 BMD ↑ p<0.05 EGb>KG p<0.05 BFP% ↓ p<0.05 MM% ↑ p<0.05 BMD ↑ p<0.05
Oktaç, 2018	Healthy individuals	F	Sedentary	21.30±2.29	n=20	X	EG=20	8 weeks 3/7	60min HRmax 50-60%	Zumba	EG» BW ↓ p<0.05 BFP% ↓ p<0.05 VO2max ↑ p<0.05 FLX ↑ p<0.05 SB ↑ p<0.05 SL ↑ p<0.05 DGS ↑ ‡ SBP ↓ ‡ DBP ↑ ‡ HRR ↑ ‡

M - male; F - female; X - unavailable; n - number of participants; EG-experimental group; KG- control group; BW- body weight; BFP% - body fat percentage; BFPkg - amount of fat tissue (kg); FFM - fat free mass (kg); BWP% - body water percentage; TBW - total amount of water in the body (kg); BMI - body mass index (kg/m²); RM - repetitive maximum; WHR - waist to hip ratio; BMD - bone mass density; BMR - basal metabolic rate; LBM - lean body mass (kg); MSL - leg muscle strength; SBP - systolic blood pressure; DBP - diastolic blood pressure; HDL - high density lipoproteins; VLDL - very low density lipoproteins; MPRBW - muscle power ratio to body weight (W/kg/BW); APVJ - anaerobic power of the vertical jump; PM - protein mass; MM% - muscle mass percentage; MM - muscle mass (kg); SFLB - bicep skinfold thickness (mm); SFLT - triceps skinfold thickness (mm); SFLS - subscapular skinfold thickness (mm); SFLSI - suprailiac skinfold thickness (mm); SFLUB - sum of upper body skinfold thickness (mm); SFLLB - sum of lower body skinfold thickness (mm); SFL - sum of skinfold thickness (mm); FLX - body flexibility (cm); FLB - spinal column flexibility (cm); SB - back muscle strength (kg); SL - leg muscle strength (kg); DGS - dominant grab strength (kg); BLC - body balance; CONC - concentration; EG>KG - differences in favor of the experimental group compared to the control group following the exercise program; (GxT) - significance of the effect of interaction between groups and time; (GxG) - significance of the effect of interaction between groups and gender; M±F - differences between the male and female participants; (F>M) - significant differences in favor of the female participants compared to the male ones; WC - waist circumference; HC - hip circumference; ON - upper leg circumference; OP - lower leg circumference; ONL - upper arm circumference; CCH - thoracic circumference; CRNC - crunches; SQ - squats; PU - push-ups; VO2max - maximum oxygen uptake (ml/kg/min); CE - cardiorespiratory endurance; HRmax- maximum heart rate; HRR - (heart rate reserve) resting heart rate; VIT - vitality; BLD - bewilderedness; FGS - fatigue; MS - motor skills; CD - circular dimensionality; p - level of statistical significance; ↑ - improvement/increase; ↓ - decrease; ‡ - statistical significance of moderate intensity; ‡ - lack of statistical significance; » - differences between the initial and final measurement.

mental group, compared to the control one, as did a study which included female participants with normal body weight, but which included a longer period of training (Pantelić et al., 2013). In one of the studies a statistically significant difference was noted ($p < 0.05$) only in the case of skinfold thickness of the biceps (Çakmakçi et al., 2011). A study which included overweight women (Vassilopoulou et al., 2017) showed the same value of significance, but in a group of participants with reduced food intake who adhered to a strict diet plan.

The greatest statistically significant decrease ($p \leq 0.01$) in body fat percentage (BFP%) was noted in studies where the exercise program lasted for 12 weeks (Nikić & Milenković, 2013; Kostrzewa-Nowak et al., 2015), but also in studies (Ljubojević et al., 2014; Haghjoo et al., 2016) which included a Zumba exercise program and lasted for a period of eight weeks. A statistically significant decrease in body fat percentage ($p < 0.05$) was noted in eight additional studies (Çakmakçi et al., 2011; Marandi et al., 2013; Najafnia et al., 2013; Pantelić et al., 2013; Kaya et al., 2016; Vassilopoulou et al., 2017; Ativie et al., 2018; Biswas & Bandopadhyay, 2018), where various types of aerobic exercise were used ("Hi-Lo" and "Step" aerobics, Zumba, combinations of aerobic dance exercises and modern dance). The authors of one study concluded that there is no statistically significant difference in the achieved effects on the BFP% between dance aerobics and a Zumba exercise program, indicating that the experimental group, using the various types of exercise, achieved a statistically significant decrease compared to the control group (Biswas & Bandopadhyay, 2018).

A statistically significant decrease in body weight (BW) at the $p < 0.01$ level under the influence of aerobic exercise to music was confirmed among young women in five studies which had the longest program duration (Stojiljković et al., 2010; Nikić & Milenković, 2013; Kostrzewa-Nowak et al., 2015; Bastug, 2018), and in a ten-week study which included an aerobic exercise program of low and moderate intensity (Marandi et al., 2013). In addition, in seven studies (Çakmakçi et al., 2011; Mathunjwa et al., 2013; Ljubojević et al., 2014; Kaya et al., 2016; Vassilopoulou et al., 2017; 2018; Ativie et al., 2018; Oktay, 2018) a statistically significant decrease in body weight was noted at the $p < 0.05$ level, which can be a consequence of the program duration, which lasted only eight weeks.

Three studies have determined the positive effects of exercise on the increase in muscle mass (Nikić & Milenković, 2013; Ativie et al., 2018; Biswas & Bandopadhyay, 2018). The implementation of exercise programs have led to statistically significant effects among young women ($p \leq 0.01$) in the case of parameters related to body build, that is, led to an increase in body water percentage (BWP%), an improvement of the basal metabolism (Nikić & Milenković, 2013), as well as an increase ($p < 0.05$) in bone density (Biswas & Bandopadhyay, 2018).

Studies which included exercise programs with the lowest intensity for a duration of eight weeks, consisting of Zumba exercises performed at home to the accompaniment of video recordings (Delextrat et al., 2016), were the only ones where the implementation of the program in question did not lead to visible changes in the body composition of sedentary women. This is probably a consequence of the shorter duration of the experimental program, and their authors have indicated the need to implement a different style of Zumba exercise, or a combination of exercises with a change in the participants' diet.

Based on the results, we can conclude that the implementation of an aerobic exercise program to music with an intensity of 50 to 85% of the maximum heart rate over a longer period of time, and a duration of 30 minutes in the main part of the training session, enables greater oxidation of fat that is needed to create energy, which in turn influences the reduction of body fat.

This conclusion is shared by the author Borer (2008), who stated that during low intensity aerobic exercise, most of the energy is obtained through the oxidation of fat which originates from fat depots and to a smaller extent from the glucose produced in the liver. In addition, with an increase in intensity, this relation decreases and alternative sources of energy are required. Based on the results of this study, the implementation of mild to moderate intensity exercise over an extended period of time of eight weeks can be used for the reduction of visceral, subcutaneous and total abdominal fat without a reduction of one's diet, which is in accordance of the claims of previous authors (Hrvoj, Sliško, & Šimić, 2015). In addition, the results indicate that different types of dance aerobics can be used with the aim of improving physical ability, but also as a means of impacting body composition, that is, reducing the amount of body fat, and increasing muscle mass, which is in accordance with the findings of previous studies (Blair et al., 2004; Kimura & Hozumi, 2012).

Conclusion

By analyzing the compiled studies, we can confirm the positive influence of aerobic exercise programs to music on the parameters of body composition and subcutaneous fat among young women, irrespective of the overall duration of the implemented exercise programs, with the conclusion that exercise programs which lasted 12 or more weeks had a greater impact on the reduction and maintenance of body weight, body fat reduction, increase in muscle tissue, and led to the improvement of aerobic capacity. The greatest effects were primarily noted among obese and overweight individuals who followed an exercise program of an intensity of 50 to 80% of maximum heart rate.

Finally, after summing up all the results, we can conclude that taking part in activities of moderate to high intensity accompanied to music is very useful, and thus can be recommended for continued use as a recreational activity which contributes to the decrease in the amount of fat in the body, the increase in muscle mass and amount of water in the body, as well as the maintenance of normal and a reduction of overweight body mass.

Acknowledgements

There are no acknowledgements.

Conflict of Interest

The authors declare that there are no conflicts of interest.

Received: 20 August 2020 | **Accepted:** 28 September 2020 | **Published:** 16 October 2020

References

- Aksović, N., Aleksandrović, M., & Jorgić, B. (2017). Effects of high intensive training on body composition of women. *TIMS. Acta*, 11(1), 53-64.
- Ativie, R.N., Aigbiremolen, A.A., Ohwin, P.E., Okemuo, A.J., Odigie, O.M., Agono, J., & Igweh, J.C. (2018). Modulations of 8-Week Aerobic Dance Exercise on Selected Anthropometric Indicators in Overweight and Obese Females. *Journal of Applied Life Sciences International*, 16(3), 1-8.
- Bastug, G. (2018). Examination of Body Composition, Flexibility, Balance, and Concentration Related to Dance Exercise. *Asian Journal of Education and Training*, 4(3), 210-215.
- Biswas, S.K., & Bandopadhyay, K. Effect of Aerobics and Zumba on Selected Body Composition Variables on Young Working Women. *International Journal of Physical Education and Sports*, 3(04), 34-47.
- Bjelica, B., Gojković, D., & Fulurija, D. (2017). Efekti zumba plesa na tjelesnu kompoziciju i poboljšanje zdravlja kod žena. *Sport i zdravlje*, 2(2), 5-10.
- Blair, S.N., LaMonte, M.J., & Nichaman, M.Z. (2004). The evolution of physical activity recommendations: how much is enough?. *The American journal of clinical nutrition*, 79(5), 913S-920S.
- Çakmakçi, E., Arslan, F., Taşkin, H., & Çakmakçi, O. (2011). The effects of aerobic dance exercise on body composition changes associated with weight change in sedentary women. *Selçuk University Journal of Physical Education and Sport Science*, 13(3), 298-304.
- Changela, P.K. (2013). A Study to Compare the Effect of Aerobic and Resistance Training on Cardiovascular (CVS) Fitness in Young Obese

- Sedentary Females. *International Journal of Scientific and Research Publications*, 3(2), 1-5.
- Delextrat, A.A., Warner, S., Graham, S., & Neupert, E. (2016). An 8-week exercise intervention based on Zumba improves aerobic fitness and psychological well-being in healthy women. *Journal of Physical Activity and Health*, 13(2), 131-139.
- Donges, C.E., Duffield, R., & Drinkwater, E.J. (2010). Effects of resistance or aerobic exercise training on interleukin-6, C-reactive protein, and body composition. *Medicine and science in sports and exercise*, 42(2), 304-313.
- Đorđević, A. (2005). Rekreacija kao način savremenog življenja. *Sportska medicina*, 5(2), 54-78.
- Đurašković, R. (2001). *Biologija razvoja čoveka sa medicinom sporta – praktikum*. Niš: S.I.I.C.
- Haghjoo, M., Zar, A., & Hoseini, S. (2016). Effect of 8-week Zumba training on overweight women's body composition. *Pars Journal of Medical Sciences*, 14(2), 21-29.
- Jorgić, B., Pantelić, S., Milanović, Z., & Kostić, R. (2011). The effects of physical exercise on the body composition of the elderly: A systematic review. *Facta universitatis-series: Physical Education and Sport*, 9(4), 439-453.
- Kaya, F., Nar, D., Erzeybek, M.S., & Bozdogan, Z. (2016). Among the physical fitness workouts, the efficiency of the weight training and zumba exercises on the body composition in men and women. In H.B. Gündüz, E. Masal & H. Çalışkan (Eds.), *ERPA International Congresses on Education 2016, SHS Web Conference, Volume 31* (pp.1-5). Sarajevo: EDP Sciences.
- Kimura, K., & Hozumi, N. (2012). Investigating the acute effect of an aerobic dance exercise program on neuro-cognitive function in the elderly. *Psychology of Sport and Exercise*, 13(5), 623-629.
- Kin-Isler, A., & Kosar, S.N. (2006). Effect of step aerobics training on anaerobic performance of men and women. *Journal of Strength and Conditioning Research*, 20(2), 366-371.
- Kostrzewa-Nowak, D., Nowak, R., Jastrzębski, Z., Zarębska, A., Bichowska, M., Drobnik-Kozakiewicz, I., Radzimiński, L., Leońska-Duniec, A., Ficek, K., & Cieśczyk, P. (2015). Effect of 12-week-long aerobic training programme on body composition, aerobic capacity, complete blood count and blood lipid profile among young women. *Biochemia medica: Biochemia medica*, 25(1), 103-113.
- Kumar, S. (2016). The effect of Zumba® aerobics exercise training on physical fitness variables-A study. *International Journal of Physical Education, Sports and Health* 3(5), 275-276.
- Luetgen, M., Foster, C., Doberstein, S., Mikat, R., & Porcari, J. (2012). ZUMBA®: Is the "fitness-party" a good workout?. *Journal of sports science & medicine*, 11(2), 357-358.
- Ljubojević, A., Jakovljević, V., & Popržen, M. (2014). Effects of Zumba fitness program on body composition of women. *SportLogia*, 10(1), 29-33.
- Marandi, S.M., Abadi, N.G.B., Esfarjani, F., Mojtahedi, H., & Ghasemi, G. (2013). Effects of intensity of aerobics on body composition and blood lipid profile in obese/overweight females. *International journal of preventive medicine*, 4(1), S118-S125.
- Mathunjwa, M.L., Semple, S.J., & du Preez, C. (2013). A 10-week aerobic exercise program reduces cardiometabolic disease risk in overweight/obese female African university students. *Ethnicity & disease*, 23(2), 143-148.
- Moher, D., Liberati, A., Tetzlaff, J., & Altman, D.G. (2009). Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *Annals of internal medicine*, 151(4), 264-269.
- Najafnia, Y., Bararpour, E., Amirinejad, B., & Nakhaee, H. (2013). Effects of 8-week step aerobic exercise on women's physiological characteristics, body fat percentage, and quality of life. *International Journal of Sport Studies*, 3(12), 1335-1341.
- Nieri, T., & Hughes, E. (2016). All about having fun: women's experience of Zumba fitness. *Sociology of Sport Journal*, 33(2), 135-145.
- Nikić, N., & Milenković, D. (2013). Efficiency of step aerobic program in younger women. *Acta Medica Medianae*, 52(3), 25-34.
- Oktaç, G. (2018). Investigation of effects of 8 weeks zumba exercise of women on health related physical fitness factors. *European Journal of Physical Education and Sport Science*, 4(6), 16-24.
- Ostojić, S., Stojanović, M., Veljović, D., Stojanović, M.D., Međedović, B., & Ahmetović, Z. (2009). Fizička aktivnost i zdravlje. *Tims Acta*, 3(1), 1-13.
- Pantelić, S., Kostić, R., Mikalački, M., Đurašković, R., Čokorilo, N., & Mladenović, I. (2007). The effects of a recreational aerobic exercise model on the functional abilities of women. *Factauniversitatis-series: Physical Education and Sport*, 5(1), 19-35.
- Pantelić, S., Milanović, Z., Sporiš, G., & Stojanović-Tošić, J. (2013). Effects of a twelve-week aerobic dance exercises on body compositions parameters in young women. *International Journal of Morphology*, 31(4), 1243-1250.
- Radovanović, D., Aleksandrović, M., Stojiljković, N., Ignjatović, A., Popović, T., & Marinković, M. (2009). Uticaj treninga u preadolescentnom uzrastu na kardiorespiratornu izdržljivost. *Acta Medica Medianae*, 48(1), 37-40.
- Sivvas, G., Batsiou, S., Vasoglou, Z., & Filippou, D.A. (2015). Dance contribution in health promotion. *Journal of Physical Education and Sport*, 15(3), 484-489.
- Stasiulis, A., Mockienė, A., Vizbaraitė, D., & Mockus, P. (2010). Aerobic exercise-induced changes in body composition and blood lipids in young women. *Medicina*, 46(2), 129-134.
- Sternlicht, E., Frisch, F., & Sumida, K.D. (2013). Zumba® Fitness workouts: are they an appropriate alternative to running or cycling?. *Sport Sciences for Health*, 9(3), 155-159.
- Stojiljković, S., Mandarić, S., Todorović, K., & Mitić, D. (2010). Efekti primene 'omnibus' aerobika na telesnu kompoziciju žena. *Fizička kultura*, 64(2), 59-67.
- Vassilopoulou, E., Piperari, G., & Christoforou, C. (2017). Is Zumba® Fitness Effective to Manage Overweight Without Dietary Intervention?. *Arab Journal of Nutrition and Exercise (AJNE)*, 1(3), 113-121.
- World Health Organization, (2018, 12 December). *Physical activity*. Retrieved from World Health Organization website: <https://www.who.int/news-room/fact-sheets/detail/physical-activity>

REVIEW PAPER

Special Editions of Full Prof. Dusko Bjelica, PhD: 1992-2020

Milena Mitrovic¹

¹University of Montenegro, Faculty for Sport and Physical Education, Niksic, Montenegro

Abstract

Dusko Bjelica, PhD, a full professor at the University of Montenegro, has a large number of professional and scientific publications. The main aim of this paper was to search and list his special editions (books). We found the editions by searching his biography and reports on scientific research. There were a total of 65 issues, which we divided into monographs, textbooks and scientific studies. Of the total, 38 are monographs, 25 textbooks and two scientific studies, representing the fields of sports, especially football and sport training. In his books, Professor Bjelica was most concerned with the development of football in Montenegro, sports training, physical culture, but also notable Montenegrin athletes and complex sporting phenomena, such as the bouncing of sport balls and the accuracy of hitting the ball. These numbers, as well as the various topics within these fields, show the quality of the scholarly work of the author. The classification is also significant given the abundance of Professor Bjelica works; therefore, it will enable easier literature searches and better insight into the subject matter of the books.

Keywords: Full Prof Dusko Bjelica, Monographs, Textbooks, Scientific studies

Introduction

Dusko Bjelica, PhD, is a full professor at the Faculty for Sport and Physical Education in Niksic, President of the Steering Committee of the University of Montenegro, the Sport Council, and Montenegrin Sports Academy. He performs his duties responsibly, conscientiously and accurately, putting great effort into finding the best solutions for the institutions in which he is employed. He always wants to advance and improve in the profession, as evidenced by publication numbers and scientific results. He has participated in many scientific conferences, congresses, and symposia in his home country and abroad, submitting outstanding papers on most meetings; he is also the president of a very successful conference organized by the Montenegrin Sport Academy and Faculty for Sport and Physical Education from Niksic. By the end of 2016, Professor Bjelica had released a total of 3028 bibliographic items, and their titles are represented in his bio-bibliography (Krivokapić & Drašković, 2016).

In the previous three years, the number of his published

books and scientific articles has increased. Some of these articles represent original scientific works, which are being pursued in this area for the first time. He also conducts scientific research with his associates, which contributes to the expansion of knowledge, experience and quality of teachers at the Faculty of Sports and Physical Education in Niksic. Professor Bjelica publishes his papers in domestic and foreign journals, indexed in prominent scientific databases, such as Web of Science and Scopus. Through the quality of his work, he has made a great contribution to placing Montenegrin academic sport scientific journals "Sport Mont" and "Montenegrin Journal of Sports, Science and Medicine" in one of these bases. Professor Bjelica does not neglect the newest journal of the Faculty for Sport and Physical Education, the "Journal of Anthropology of Sport and Physical Education", constantly publishing scientific articles, so the journal continues to grow and prosper. Also, according to data from Google Scholar, Professor Bjelica is the third most cited scientist at the University of Montenegro, and the first in the world in sports training, with 7363 citations.

Correspondence:

**Montenegro
Sport**

M. Mitrovic
University of Montenegro, Faculty for Sport and Physical Education, Narodne omladine bb, 81400, Niksic, Montenegro
E-Mail: milenam1054@gmail.com

In addition to a large number of published articles, Professor Bjelica has demonstrated his scientific activity through many books, contributing to the development of sports in Montenegro. While researching and analysing this area, looking for better and more efficient solutions to develop sport in the country, he has published many publications, presenting the results of his research as well as personal experiences. These publications are in the form of scientific and professional monographs, scientific studies and textbooks for higher education and faculties. It can be seen that these are very complex publications that require detailed and systematic processing of specific phenomena or events.

In monographs, Professor Bjelica paid the most attention to football, its development in Montenegro and Montenegrin athletes who have achieved significant results on the sports scene. Very complex issues, such as how balls bounce and accuracy components of the ball, which require knowledge of natural sciences, have benefitted from his involvement and research.

In addition to these publications, he has published a large number of textbooks, which make it easier for students to understand very complex processes, such as physical education and sports training. In addition to students of the Faculty of Sport and Physical Education, textbooks used by students of the Faculty of Teacher and Preschool Education, in which Professor Bjelica gave detailed explanations of the basics of physical culture and games, which is knowledge necessary for working with children of that age. Also, his books are extremely useful for coaches and players, to acquaint them better with proper physical preparation and method of conducting sports training.

It is noted that Professor Bjelica's bibliographic opus is extremely rich, so it takes a long time to review these publications. In line with this, the main goal of this work is to find and classify all books of Dusko Bjelica, PhD, which will make it easier to search his literature.

Methods

Books by Dusko Bjelica, PhD, were found by retrieving his bio-bibliography (Krivokapić, & Drašković, 2016) and profile on the Research Gate social scientific network. The books found are divided into three categories: professional and scientific monographs, scientific studies and textbooks for higher education and faculties. The monographs include books that detail one topic. In scientific studies are classified the books with the results of research specific topics, while textbooks classified books, manuals and student workshops used for the acquisition of theoretical knowledge in the universities and sports institutions. The classification of books was started from 1992, when Professor Bjelica published his first monograph, and include all years until January 2020. The total number of published monographs, scientific studies and textbooks is presented in tables. The titles of these publications are presented descriptively.

Results

Table 1 shows the total number of scientific and professional monographs, scientific studies and textbooks authored by Dusko Bjelica.

From Table 1, it is noted that professor Bjelica from 1992 until January 2020 produced a total of 65 books. Of these, most are monographs (38), followed by textbooks (25) and scientific studies (2).

In his first monographs, the author cited an analysis of the development of futsal in his country. Published monographs in the field are: "Futsal in Montenegro: 1988-1992" (Bjelica, 1992), and "Futsal in Montenegro: 1963-1993" (Bjelica, & Savićević, 1993). Since 1993, Bjelica has analysed the situation in clubs and the Football Federation of Montenegro for many years, and published the following monographs: "Yearbook of the Federation of Montenegro: 1991-1992" (Jovanović, & Bjelica, 1993a), "Yearbook of the Federation of Montenegro: 1992-1994" (Jovanović, & Bjelica, 1993b), "Yearbook of the Football Federation of Montenegro: 1993-1994" (Đurđević, & Bjelica, 1994), "Yearbook of the Football Federation of Montenegro: 1994-1995" (Krivokapić, Jovanović, & Bjelica, 1995), "Yearbook of the Football Federation of Montenegro: 1995-1996" (Krivokapić, Jovanović, & Bjelica, 1996), "Yearbook of the Football Federation of Montenegro: 1995-1996 (offprint)" (Krivokapić, Jovanović, & Bjelica, 1996), "Football Federation Of Montenegro: 1931-1996" (Jovanović, & Bjelica, 1996), "Yearbook of the Football Federation of Montenegro: 1996-1997" (Krivokapić, Jovanović, & Bjelica, 1997), "Yearbook of the Football Federation of Montenegro: 1997-1998" (Krivokapić, Jovanović, & Bjelica, 1998), "Yearbook of the Federation of Montenegro: 1998-1999" (Krivokapić, Jovanović, & Bjelica, 1999), "Register of Club Memberships of the Republic of Montenegro: Competition 1997/8" (Bjelica, 1998), "Football Referees of Montenegro" (Jovanović, Krivokapić, Radunović, & Bjelica, 1998), "Club Membership Register of the Republic of Montenegro: Competition 1998/9" (Bjelica, 1999), and "Register of Club Memberships of the Republic of Montenegro: Competition 2000-2001" (Bjelica, 2001).

Since 2002, Professor Bjelica, independently and in collaboration with other authors, has published monographs dedicated to the best Montenegrin athletes. These are: "Montenegrosport Football School: Sport Mont" (Bjelica, 2002a), "Rise Year: Montenegrosport 2001" (Bjelica, 2002b), "New Champions' Year: Montenegrosport 2002" (Bjelica, 2003a), "Montenegrosport -2002 year" (Bjelica, 2003b), "Montenegrosport karate-2002: the new champion of the Old continent" (Bjelica, 2003c), "The new champion of Old continent" (Bjelica, 2003d), "Montenegrin sports laureates: Montenegrosport 2003" (Bjelica, 2004a), "Karate Club Buducnost, Podgorica" (Bjelica, 2004a), "Who's who in Montenegrin Sport" (Bjelica, 2005a), "Laureates of Montenegrin Sport II: Montenegrosport 2007" (Bjelica, & Gavrilović, 2008), "Montenegrin Sports Laureates III: Montenegrosport 2008" (Bjelica, & Gavrilović, 2009), "Montenegrin Sports Laureates IV: Montenegrosport 2009" (Gavrilović, & Bjelica, 2010), "Montenegrin Sports Laureates V: Montenegrosport 2010" (Gavrilović, & Bjelica, 2011), "Montenegrin Sports Laureates VI: Montenegrosport 2011" (Bjelica, & Gavrilović, 2012), "Montenegrin Sport Laureates VII: Montenegrosport 2012"

Table 1. Number of monographs, scientific studies and textbooks by Full Professor Dusko Bjelica, PhD

Author	Editions			
	Monographs	Scientific studies	Textbooks	In total
Dusko Bjelica	38	2	25	65

(Bjelica, & Gavrilović, 2013), "Montenegrin Sport Laureates VIII: Montenegrosport 2013" (Bjelica, & Gavrilović, 2014), "Montenegrin Sports Laureates IX: Montenegrosport 2014" (Bjelica, & Gavrilović, 2015), "Montenegrin Sports Laureates X: Montenegrosport 2015" (Bjelica, & Gavrilović, 2016), "Montenegrin Sports Laureates XI: Montenegrosport 2016" (Bjelica, Gavrilović, & Brajković, 2017), "Montenegrin Sports Laureates XII: Montenegrosport 2017" (Bjelica, Gavrilović, & Brajković, 2018), and "Montenegrin Sports Laureates XIII: Montenegrosport 2018" (Bjelica, Gavrilović, & Brajković, 2019). Perhaps his most significant monograph, professor Bjelica, as a standalone editor, announced 2015 year. It is a professional monograph of the international character "Sport Today I" (Bjelica, 2015a).

With regards to textbooks, Professor Bjelica was mainly concerned with football, sports training, and the theoretical foundations of physical culture, physical exercise, and sports. From 1993 to the present, he, independently and in co-authorship, published the following textbooks and manuals: "Physical preparation of athletes" (Bjelica, Vujošević, & Savićević, 1993), "Attitudes of eighth grade students towards teaching Physical Education" (Bjelica, 1995), "Football Club Practicum" (Bjelica, 1996), "Attitudes of Grade VIII Students towards Physical Education in the Republic of Montenegro" (Bjelica, 1999b), "General Concepts of Sports Training (Abbreviated version)" (Bjelica, 2002c), "Attitudes of eighth grade elementary school students in Montenegro towards Physical Education teaching and extracurricular activities" (Bjelica, 2002d), "Impact of sports training on anthropomotor abilities (football cadets of Montenegro)" (Bjelica, 2004c), "Systematization of sports discipline and sports training" (Bjelica, 2005b), "Matrix-probability models in sport" (Cvetković, Bjelica, Kostić, & Kovacević, 2005), "Sports training" (Bjelica, 2006a), "Theoretical Foundations of Physical and Health Education" (Bjelica, 2006b), "Communication in Sport" (S. Bjelica, & D. Bjelica, 2006), "The theory of Physical Education and the basics of School sport" (Bjelica, & Petković, 2009), "The theoretical bases of Physical Culture" (Bjelica, & Krivokapić, 2010a), "The theory of Physical Education and the basics of school sport" (Bjelica, & Petković, 2010), "Theoretical Foundations of Physical Culture" (Bjelica, & Krivokapić, 2010b), "Sport training: Theory, Methodics, and Diagnostics" (Bjelica, & Fratrić, 2011), "Theory of game" (Bjelica, & Krivokapić, 2011), "Football: Theory-Technique-Tactics" (Bjelica, & Popović, 2012), "Communication in Sport" (Bjelica, 2012), "Sports Training Theory" (2013), "Theoretical Foundations of Physical and Health Education" (Bjelica, 2015b), "Football: Technique and Tactics" (Bjelica, 2016), "Sport Training: Theory, Methodics, and Diagnostics" (Bjelica, & Fratrić, 2018), and "Theory of Exercise and Sport" (Bjelica, & Krivokapić, 2019).

Professor Bjelica has published two scientific studies, dealing with highly complex concepts that play an important role in sports, namely: "The main components of the accuracy of kicking the ball in soccer sports" (Bjelica, 2008) and "Bouncing of sports balls" (Bjelica, 2014).

Discussion

The main goal of this paper was to review the books of Dusko Bjelica, full professor at the University of Montenegro. Searching his bio-bibliography (Krivokapić, & Drašković, 2016) and profile on the Research Gate network, a total of 65 books were found, which are classified into professional and

scientific monographs, textbooks for colleges and universities and scientific studies. Of the total, the most numerous were monographs (38), followed by textbooks (25) and scientific studies (2).

Professor Bjelica's first monograph published in 1992, which, as in the following monograph, dealt with the development of futsal in Montenegro. From 1993 to 2001, he carefully analysed the situation in Montenegrin football, thus publishing a series of yearbooks of the Football Association of Montenegro and registers of club memberships. Since 2002, to Professor Bjelica has, among other things, paid great attention to the achievements of Montenegrin athletes, and accordingly, annually publishes new editions on the awards in Montenegrin sports. As sole author and co-author, he has published a total of 19 textbooks. The first textbook was in 1993; since the year 2000, the pace of publishing has accelerated, with constantly improving editions. He was most concerned with football, sports training and the theoretical foundations of physical culture, physical exercise and sports. In addition, he published textbooks analysing communications as well as matrix-value models in sports. He wrote a textbook detailing the physical fitness of athletes, as well as a football club practicum, which can also serve specific sports institutions. His two scientific studies emerged as a result of studying very complex phenomena present in sports, such as the movement of the ball and the components of the accuracy of kicking the ball in football.

Based on the number of these publications, as well as the complexity of their topics, clear insight into the quality of professor Bjelica's work is given. All publications have a clear purpose and intent, thoroughly analyse a particular phenomenon, and are supported by the results of scientific research and experience. Describing Montenegrin sport in detail over the previous five decades, explaining the complex processes of sports training and generalizing abstract concepts of the theory of physical culture and sport, he has made a significant contribution to the development of science and sport in Montenegro. With his monographs, he undoubtedly gives great support and recognition to successful Montenegrin athletes, and the quality of his textbooks makes it easier for students to acquire knowledge. It also contributes to its scientific studies by understanding the complex phenomena that are of great importance to all those involved in sports. Given the knowledge, experience and energy they reflect, he is still working on many successful publications, completing both old and new projects. As a man of action and strong will, he will undoubtedly soon increase the number of publications. Until then, it is necessary to classify them, to facilitate the search of his literature and the topics he has with which he had engaged.

Acknowledgements

There are no acknowledgements.

Conflict of Interest

The authors declare that there are no conflicts of interest

Received: 11 May 2020 | **Accepted:** 15 July 2020 | **Published:** 16 October 2020

References

- Bjelica, D. (1992). *Futsal in Montenegro: 1988-1992* [Mali fudbal u Crnoj Gori]. Podgorica: Football Federation of Montenegro.
- Bjelica, D. (1996). *Football club practicum* [Fudbalski klupski praktikum]. Podgorica: CID.
- Bjelica, D. (1995). *Eighth Grade Students' Attitudes towards Physical Education Teaching* [Stavovi učenika osmih razreda prema nastavi fizičkog vaspita]

- tanja]. Podgorica: Association of Futsal Clubs.
- Bjelica, D. (1998). *Club Membership Register of the Republic of Montenegro: Competition 1997/98*. [Registar članstva klubova Republike Crne Gore: takmičarska 1997/98]. Podgorica: Football Federation of Montenegro.
- Bjelica, D. (1999a). *Club Membership Register of the Republic of Montenegro: Competition 1998/99*. [Registar članstva klubova Republike Crne Gore: takmičarska 1998/99]. Podgorica: Football Federation of Montenegro.
- Bjelica, D. (1999b). *Attitudes of Grade VIII Students towards Physical Education in the Republic of Montenegro* [Stavovi učenika osmih razreda prema nastavi fizičkog vaspitanja u Republici Crnoj Gori]. Podgorica: Football Federation of Montenegro.
- Bjelica, D. (2001). *Membership Register of Montenegrin Football Clubs in Competition 2000-2001* [Registar članstava crnogorskih fudbalskih klubova u takmičarskoj 2000-2001]. Podgorica: Montenegrosport.
- Bjelica, D. (2002a). *Montenegrosport Football School: Sport Mont* [Fudbalska škola Montenegrosport: Sport Mont]. Podgorica: Montenegrosport.
- Bjelica, D. (2002b). *Rise Year: Montenegrosport 2001* [Godina uspona: Montenegrosport 2001]. Podgorica: Montenegrosport.
- Bjelica, D. (2002c). *General terms of sports training* (Abbreviated version) [Osnovni pojmovi sportskog treninga (skraćena verzija)]. Podgorica.
- Bjelica, D. (2002d). *Attitudes of eighth-grade elementary school students in Montenegro towards physical education and extracurricular activities* [Stavovi učenika osmih razreda osnovnih škola u Crnoj Gori prema nastavnim i vanastavnim aktivnostima fizičkog vaspitanja]. Podgorica: Football Federation of Montenegro.
- Bjelica, D. (2003a). *New champions' year: Montenegrosport 2002 year* [Godina novih šampiona: Montenegrosport 2002. godine]. Podgorica: Montenegrosport.
- Bjelica, D. (2003b). *Montenegrosport - 2002 year* [Montenegrosport - 2002. godine]. Podgorica: Montenegrosport.
- Bjelica, D. (2003c). *Montenegrosport Karate - 2002: New Champion of the Old Continent* [Montenegrosport karate - 2002: novi šampioni Starog kontinenta]. Podgorica: Montenegrosport.
- Bjelica, D. (2003d). *New champion of the Old Continent* [Novi šampioni Starog kontinenta]. Podgorica: Montenegrosport.
- Bjelica, D. (2004a). *Montenegrin Sports Laureates: Montenegrosport 2003* [Crnogorski sportski laureati: Montenegrosport 2003]. Podgorica: Montenegrin Sports Academy.
- Bjelica, D. (2004b). *Karate Club Budućnost, Podgorica* [Karate klub Budućnost, Podgorica]. Podgorica: Montenegrosport.
- Bjelica, D. (2004c). *Impact of sports training on anthropomotor abilities: football cadets of Montenegro* [Utjecaj sportskog treninga na antropometričke sposobnosti: fudbalskih kadeta Crne Gore]. Podgorica: Montenegrin Sports Academy.
- Bjelica, D. (2005a). *Who's who in Montenegrin sports* [Ko je ko u crnogorskom sportu]. Podgorica: Montenegrin Sports Academy.
- Bjelica, D. (2005b). *Systematization of sports disciplines and sports training* [Sistematizacija sportskih disciplina i sportski trening]. Podgorica: Montenegrin Sports Academy.
- Bjelica, D. (2006a). *Sport training* [Sportski trening]. Podgorica: Montenegrin Sports Academy.
- Bjelica, D. (2006b). *Theoretical Foundations of Physical and Health Education* (First Edition) [Teorijske osnove tjelesnog i zdravstvenog obrazovanja (prvo izdanje)]. Podgorica: Montenegrin Sports Academy.
- Bjelica, D. (2008). *The main components of the accuracy of kicking the ball in football* (Scientific study) [Glavne komponente tačnosti udarca nogom po lopti u fudbalskom sportu (naučna studija)]. Podgorica: Montenegrin Sports Academy, Niksic: Faculty of Philosophy.
- Bjelica, D. (2012). *Communications in sports* [Komunikacije u sportu]. Niksic: Faculty of Sport and Physical Education.
- Bjelica, D. (2013). *Sports training theory* [Teorija sportskog treninga]. Podgorica: University of Montenegro.
- Bjelica, D. (2014). *Repulsion of sports balls* (Scientific study) [Repulzija sportskih lopti (naučna studija)]. Podgorica: Montenegrin Sports Academy, Niksic: Faculty of Sport and Physical Education.
- Bjelica, D. (2015a). *Sports Today* (First Edition) [Sport danas (prvo izdanje)]. Podgorica: Montenegrin Sports Academy, Niksic: Faculty of Sport and Physical Education.
- Bjelica, D. (2015b). *Theoretical Foundations of Physical and Health Education* (Second Edition) [Teorijske osnove tjelesnog i zdravstvenog obrazovanja (drugo izdanje)]. Niksic: Faculty of Sport and Physical Education.
- Bjelica, D., & Fratrić, F. (2011). *Sports training: Theory, Methodics, and Diagnostics* [Sportski trening: teorija, metodika i dijagnostika]. Niksic: Faculty of Sport and Physical Education.
- Bjelica, D., & Fratrić, F. (2018). *Sports training: Theory, Methodics, and Diagnostics* [Sportski trening: teorija, metodika i dijagnostika]. Podgorica: University of Montenegro.
- Bjelica, D., & Gavrilović, Z. (2008). *Montenegrin Sport Laureates II: Montenegrosport 2007* [Crnogorski sportski laureati II: Montenegrosport 2007]. Podgorica: Montenegrin Sports Academy.
- Bjelica, D., & Gavrilović, Z. (2009). *Montenegrin Sports Laureates III: Montenegrosport 2008* [Crnogorski sportski laureati III: Montenegrosport 2008]. Podgorica: Montenegrin Sports Academy.
- Bjelica, D., & Gavrilović, Z. (2012). *Montenegrin Sports Laureates VI: Montenegrosport 2011* [Crnogorski sportski laureati VI: Montenegrosport 2011]. Podgorica: Montenegrin Sports Academy.
- Bjelica, D., & Gavrilović, Z. (2013). *Montenegrin Sports Laureates VII: Montenegrosport 2012* [Crnogorski sportski laureati VII: Montenegrosport 2012]. Podgorica: Montenegrin Sports Academy.
- Bjelica, D., & Gavrilović, Z. (2014). *Montenegrin Sports Laureates VIII: Montenegrosport 2013* [Crnogorski sportski laureati VIII: Montenegrosport 2013]. Podgorica: Montenegrin Sports Academy.
- Bjelica, D., & Gavrilović, Z. (2015). *Montenegrin Sports Laureates IX: Montenegrosport 2014* [Crnogorski sportski laureati IX: Montenegrosport 2014]. Podgorica: Montenegrin Sports Academy.
- Bjelica, D., & Gavrilović, Z. (2016). *Montenegrin Sports Laureates X: Montenegrosport 2015* [Crnogorski sportski laureati X: Montenegrosport 2015]. Podgorica: Montenegrin Sports Academy.
- Bjelica, D., & Krivokapić, D. (2010a). *Theoretical foundations of Physical Culture* [Teorijske osnove fizičke kulture]. Niksic: Faculty of Sport and Physical Education, University of Montenegro.
- Bjelica, D., & Krivokapić, D. (2010b). *Theoretical foundations of Physical Culture* [Teorijske osnove fizičke kulture]. Niksic: Faculty of Sport and Physical Education, University of Montenegro.
- Bjelica, D., & Krivokapić, D. (2011). *Theory of game* [Teorija igre]. Niksic: Faculty of Sport and Physical Education, Podgorica: Montenegrin Sports Academy.
- Bjelica, D., & Krivokapić, D. (2019). *Theory of Physical training and Sports* [Teorija tjelesnog vježbanja i sporta]. Podgorica: Montenegrin Sports Academy.
- Bjelica, D., & Petković, J. (2009). *The theory of Physical Education and the basis of School sport* [Teorija fizičkog vaspitanja i osnove školskog sporta]. Niksic: Faculty of Sports and Physical Education, Podgorica: Montenegrin Sports Academy.
- Bjelica, D., & Petković, J. (2010). *The theory of Physical Education and the basis of School sport* [Teorija fizičkog vaspitanja i osnove školskog sporta]. Niksic: Faculty of Sports and Physical Education, Podgorica: Montenegrin Sports Academy.
- Bjelica, D., & Popović, S. (2012). *Football: Theory - Technique - Tactics* [Fudbal: teorija - tehnika - taktika]. Niksic: Faculty of Sport and Physical Education.
- Bjelica, D., & Popović, S. (2016). *Football: Technique and Tactics* [Fudbal: tehnika i taktika]. Podgorica: University of Montenegro.
- Bjelica, D., & Savićević, G. (1993). *Futsal in Montenegro: 1963-1993* [Mali fudbal u Crnoj Gori: 1963-1993]. Podgorica: Football Federation of Montenegro.
- Bjelica, D., Gavrilović, Z., & Brajković, T. (2017). *Montenegrin Sports Laureates XI: Montenegrosport 2016* [Crnogorski sportski laureati XI: Montenegrosport 2016]. Podgorica: Montenegrin Sports Academy.
- Bjelica, D., Gavrilović, Z., & Brajković, T. (2018). *Montenegrin Sports Laureates XII: Montenegrosport 2017* [Crnogorski sportski laureati XII: Montenegrosport 2017]. Podgorica: Montenegrin Sports Academy.
- Bjelica, D., Gavrilović, Z., & Brajković, T. (2019). *Montenegrin Sports Laureates XIII: Montenegrosport 2018* [Crnogorski sportski laureati XIII: Montenegrosport 2018]. Podgorica: Montenegrin Sports Academy.
- Bjelica, D., Vujošević, R., & Savićević, G. (1993). *Physical preparation of athletes* [Fizička priprema sportista]. Podgorica: Football Federation of Montenegro.
- Bjelica, S., & Bjelica, D. (2006). *Communications in sports* [Komunikacije u sportu]. Niksic: Faculty of Philosophy, Podgorica: Montenegrin Sports Academy.
- Cvetković, Lj., Bjelica, D., Kostić, V., & Kovačević, M. (2005). *Matrix-probability models in sport* [Matrično-verovatnosni modeli u sportu]. Podgorica: Montenegrin Sports Academy.
- Đurđević, M., & Bjelica, D. (1994). *Yearbook of the Football Association of Montenegro: 1993-1994* [Godišnjak Fudbalskog saveza Crne Gore: 1993-1994]. Podgorica: Football Federation of Montenegro.
- Gavrilović, Z., & Bjelica, D. (2010). *Montenegrin Sports Laureates IV: Montenegrosport 2009* [Crnogorski sportski laureati IV: Montenegrosport 2009]. Podgorica: Montenegrin Sports Academy.
- Gavrilović, Z., & Bjelica, D. (2011). *Montenegrin Sports Laureates V: Montenegrosport 2010* [Crnogorski sportski laureati V: Montenegrosport 2010]. Podgorica: Montenegrin Sports Academy.
- Jovanović, N., Krivokapić, P., Radunović, B., & Bjelica, D. (1998). *Football referees of Montenegro* [Fudbalske sudije Crne Gore]. Podgorica: Football Federation of Montenegro.

- Jovanović, N., & Bjelica, D. (1993a). *Yearbook of the Football Association of Montenegro: 1991-1992* [Godišnjak Fudbalskog saveza Crne Gore: 1991-1992]. Podgorica: Football Federation of Montenegro.
- Jovanović, N., & Bjelica, D. (1993b). *Yearbook of the Football Association of Montenegro: 1992-1993* [Godišnjak Fudbalskog saveza Crne Gore: 1992-1993]. Podgorica: Football Federation of Montenegro.
- Jovanović, N., & Bjelica, D. (1997). *Football Federation of Montenegro: 1931-1996* [Fudbalski savez Crne Gore: 1931-1996]. Podgorica: Football Federation of Montenegro.
- Krivokapić, P., & Drašković, N. (2017). *Prof. Dusko Bjelica, PhD - bio-bibliography* [Prof. dr Duško Bjelica – Bio-bibliografija]. Podgorica: Montenegrin Sports Academy, Niksic: Faculty of Sport and Physical Education.
- Krivokapić, P., Jovanović, N., & Bjelica, D. (1995). *Yearbook of the Football Association of Montenegro: 1994-1995* [Godišnjak Fudbalskog saveza Crne Gore: 1994-1995]. Podgorica: Football Federation of Montenegro.
- Krivokapić, P., Jovanović, N., & Bjelica, D. (1996). *Yearbook of the Football Association of Montenegro: 1995-1996* [Godišnjak Fudbalskog saveza Crne Gore: 1995-1996]. Podgorica: Football Federation of Montenegro.
- Krivokapić, P., Jovanović, N., & Bjelica, D. (1997). *Yearbook of the Football Association of Montenegro: 1996-1997* [Godišnjak Fudbalskog saveza Crne Gore: 1996-1997]. Podgorica: Football Federation of Montenegro.
- Krivokapić, P., Jovanović, N., & Bjelica, D. (1998). *Yearbook of the Football Association of Montenegro: 1997-1998* [Godišnjak Fudbalskog saveza Crne Gore: 1997-1998]. Podgorica: Football Federation of Montenegro.
- Krivokapić, P., Jovanović, N., & Bjelica, D. (1999). *Yearbook of the Football Association of Montenegro: 1998-1999* [Godišnjak Fudbalskog saveza Crne Gore: 1998-1999]. Podgorica: Football Federation of Montenegro.
- Promotion of the Book Bio-Bibliography Authored by Dusko Bjelica, PhD* (22.10.2018). Podgorica: University of Montenegro. Posted 13.1.2020 from the Web site: www.ucg.ac.me

SHORT REPORT

Independence in Learning, Education Management, and Industry 4.0: Habitat Indonesia during COVID-19

Zainal Ilmi¹, Dio Caisar Darma², Musdalifah Azis¹

¹Mulawarman University, Faculty of Economics and Business, Department of Management, Samarinda, Indonesia, ²Sekolah Tinggi Ilmu Ekonomi Samarinda, Department of Management, Samarinda, Indonesia.

Abstract

COVID-19 is the biggest pandemic of the 20th century and has hurt humans. Apart from death, this virus also has an impact on the social, economic, and educational fields. Like it or not, COVID-19 causes most schools and campuses to change their learning patterns. This paper tries to discuss freedom of learning, educational management, and challenges in industry 4.0 for Indonesia during this outbreak. With these 3 objectives, we use a literature review approach. The first conclusion is that the independence of learning in Indonesia has not been optimal so far, because in general many teachers and lecturers still use the old pattern (they are the centers of learning). The second affirmation is that education management in Indonesia is still traditionally based, meaning that the use of learning media (for example online) is not fully used. Third, in terms of Industry 4.0, education in schools and campuses, there is an obligation to take advantage of IoT with the aim that students and students can think critically, develop creativity, hone soft skills to communicate, and train to collaborate with various parties. The biggest job is under the Ministry of Education and Culture of the Republic of Indonesia, to evaluate the education curriculum and make extra policies to accelerate the civilization of education during COVID-19.

Keywords: *Learning, Education, Teachers, Lecturers, Curriculum, COVID-19, Indonesia*

Introduction

The spread of the Corona Virus Disease (COVID-19) is still causing unrest in almost all countries in the world. Although activities are slowly getting back on track, a lot has changed in daily life. The government is still making efforts to deal with the virus that causes COVID-19. The community is asked to be disciplined in implementing a protocol for preventing the transmission of this outbreak. Reports of a new type of coronavirus infection were reported in Wuhan (China) on December 31, 2019. This means that it has been around 205 days since the virus has spread. Hundreds of days on, there is no sign of a decrease in the daily infection rate, although the epicenter of the virus has shifted, from China to Europe than Latin America. In fact, in the last week, the latest data shows the record for additional new cases, the highest

in 24 hours, as many as 237,734 cases of infection as of July 18, 2020 (WHO, 2020; Darma, Wijaya, & Darma, 2020).

COVID-19 is an epidemic that can cause infectious diseases in the form of infection in the human respiratory tract caused by a virus. The COVID-19 outbreak has hit the world and Indonesia became one of the countries affected by the COVID-19 outbreak in early March 2020. The COVID-19 pandemic that has hit the world, one of which is that Indonesia has had a visible impact in various fields, including the economy, social, tourism, and especially education. The implementation of education in Indonesia during the COVID-19 pandemic has undergone several visible changes (Darma, Ilmi, Darma, & Syaharuddin, 2020; Toquero, 2020; Biswas et al., 2020).

This is a tragedy that grieves all inhabitants of the earth. All

Correspondence:

**Montenegro
Sport**

M. Azis,
Mulawarman University, Faculty of Economics and Business, Department of Management, Samarinda, 75117, Indonesia.
E-mail: musdalifah.azis@feb.unmul.ac.id

segments of human life on earth are disturbed, without exception of education. Many countries have decided to close schools and universities, including Indonesia (Aji, 2020).

The pattern of education in Indonesia

Education is an endless process and it is a process of forming basic abilities that are fundamental both in terms of intellectual power and emotional feelings directed at human nature and towards others. Therefore, the learning process is the key to the success of education so that the quality learning process requires quality service management (Sagala, 2013; Dewey, 1963).

The Indonesian government actually always pays more attention to the education sector. This is evidenced by the stipulation of several government policies in the education sector, ranging from compulsory education programs, scholarships to underprivileged people and programs that seek to improve the quality of education, and budgeting 20% of Indonesia's State Revenue and Expenditure Budget (APBN) for the education sector. However, the big question is where is the direction of Indonesian education at this time, and why then education in Indonesia is still far behind other countries in the world. In relation to "Free Learning" proclaimed by the Minister of Education and Culture of the Republic of Indonesia, understands and changes The perspective of education with the perspective of progressivism philosophy needs to be done. This is because progressivism is a school of educational philosophy that assumes that humans have unique and extraordinary abilities and can overcome various problems that threaten humans themselves. Progressivism also rejects the authoritarian style of education that has occurred in the past and the present. Authoritarian education is considered to be able to hinder in achieving good goals because it does not respect human abilities in the educational process. Whereas in education all elements are considered as the driving force to achieve progress or progress in the future. Thus, for progressivism, ideas and theories are not sufficient only to be recognized as things that exist (being), but that what exists must be looked for

in order to achieve progress (Mustaghfiroh, 2020; Muhmidayeli, 2011; Manan, 2015).

In independent learning, the teacher not only acts as a learning resource but also acts as a learning facilitator. As a learning facilitator, of course, the teacher must design a fun lesson so that students can achieve the expected learning objectives. Freedom to assess student learning with various types and forms of research instruments, free from various burdensome administrative making, independent from the politicization of the teaching profession, and free from various pressures and intimidation against them (Faisal & Martin, 2019; Dragutinovic & Mitrovic, 2019).

Industry 4.0 challenges and opportunities

In the learning process an educational ecosystem is developed which facilitates the growth and development of reasoning, character, innovation, independence, comfort, and student expertise. So freedom to learn has the effect of forming superior or quality resources to complete educational opportunities in the Industrial 4.0 era with the aim of developing the nation and the State (Yamin & Syahrir, 2020).

The desired education system is an education system that can lead to a better human civilization. To achieve this, of course, requires a leader's ability to make it happen. We know that the development of education in the world today cannot be separated from the development of the industrial revolution that is taking place in the world, because indirectly changes in the economic order also change the educational system in a country (Bennis & Nanus, 1985; Manz & Sims, 2001; Fairholm, 2013).

Internet of Things (IoT) is unwittingly widely used in various sectors. Identifying an object is done using virtual concepts to display in cyberspace. Included in educational institutions in Indonesia because it can be seen in the interactions that occur in the educational environment. Through its development, the internet can also be used for other purposes that support learning, including by using the internet for theoretical and practical learning activities (Kristanti, 2019; Prihatmoko, 2016).

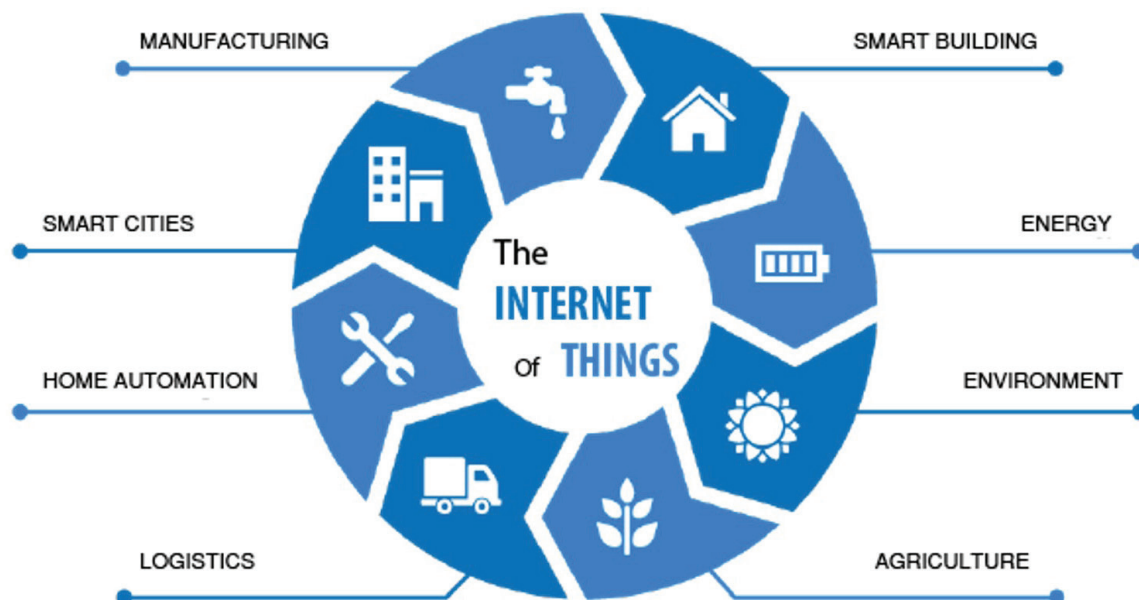


FIGURE 1. IoT model (Source: Pfeiffer, 2017)

This conversation caused upheaval for the community that would they be ready to face the golden era of globalization, the technology of the times was growing rapidly, in this position human labor would at times be marginalized if they did not have

skilled skills. For success in breaking through the shock of industry 4.0, it is not just a discourse, but there must be proof in all aspects (Maria et al., 2019).

In this way, education is used as the initial foundation or

backbone of national development. It happens in terms of one effect, which is the shaping of the human personality. The government's strategy in responding to the industrial revolution 4.0 is quite clear if it is supported by all Indonesian people. An innovative generation will be able to encourage national development in that era. These opportunities and challenges can contribute to the creation of more jobs and new technology-based investments so that a roadmap called "Making Indonesia 4.0" is formed which can create more efficient change.

Responding to the learning process

Education as a provider of skilled, effective, and productive manpower, processes potential sources and natural resources, manages organizational resources and human resource institutions because basically, development in education is a process of positive progressive change that occurs in the life of society and the nation. In various fields, be it socio-cultural, political, economic, or in other areas of life. Every change or development in the life of the nation has consequences in the education system, as well as every change or development in the world of education has consequences in the life of the nation (Awad & Barak, 2014).

Education must run in any circumstances. To reduce the number of COVID-19 spread and education activities that can run, as usual, the government has made several efforts to reduce this number, one of which is implemented in the education system in Indonesia. Teaching and learning activities have been carried out using an online system or a network system (online) since March 2020. The learning system is carried out without face-to-face but is carried out using a distance learning system. With the distance learning system, students are not required or required to come to schools or campuses to carry out learning. Many facilities are eventually applied by educators to carry out teaching and learning activities remotely. This means of distance learning cannot be avoided by the development of information and communication technology. The learning facilities include Google Meets application, Zoom, Google Classroom, Youtube, Television, and Whatsapp. All of these facilities are generated from the development of increasingly advanced information and communication technology.

However, the distance learning system does not rule out some problems in the learning process. With the implementation of this distance learning, of course, students and educators from all walks of life are required to have good internet network access. However, many areas have poor or ineffective internet access, which is one of the obstacles to carrying out teaching and learning activities properly. In addition, there are not a few students who do not get maximum learning results. Both from the subject matter and the assignments were given by educators during the COVID-19 pandemic.

Conclusion

Education is the key to human resource development. The quality of human resources is the key to the realization of the Golden Indonesia 2045, which is just and prosperous, safe, and peaceful, as well as progressive and global. Education will determine where this nation will meet its future, whether to become a large civilized nation, intelligent and ready to adapt to changing times, or become a sick giant and drown in its own problems. Losing global competition, and even randomizing various short-term interests, both from within and outside the country.

Basically, the global pandemic has impacts that can weaken human activities in general. It cannot be denied that at first many people thought that the COVID-19 period was a difficult time for mankind. However, without realizing it, there are many positive sides that we can learn from this that is currently sweeping the

world to this day. The impact that is felt is real and can be felt by everyone. However, the community cannot make problems a cause for not carrying out activities, especially in the field of education.

There are several strategic steps that schools must take during COVID-19, first, review the learning targets to be achieved, so that they are rationally aligned with the new situation and conditions in the new normal. Second, identification of the resources that need to be owned and held so that the new goals that have been set can be achieved with the availability of existing resources. Third, map the situation and condition of each teacher and student who must get ready to undertake a new learning model based on blended learning as designed. Fourth, assess the gap between need and availability to formulate strategic and operational steps that need to be taken immediately to bridge it. And finally, the execution of these steps creatively and innovatively by establishing various partnerships with external parties who care about education.

Acknowledgments

There are no Acknowledgments.

Conflicts of Interest

The author declare that there are no conflict of interest in this article.

Received: 12 August 2020 | **Accepted:** 8 September 2020 | **Published:** 16 October 2020

References

- Aji, R. H. (2020). Dampak Covid-19 pada Pendidikan di Indonesia: Sekolah, Keterampilan, dan Proses Pembelajaran. *SALAM: Jurnal Sosial & Budaya Syar-i*, 7(5), 395-402. Doi:10.15408/sjsbs.v7i5.15314
- Awad, N., & Barak, M. (2014). Sound, Waves and Communication: Students' Achievements and Motivation in Learning a STEM-Oriented Program. *Creative Education*, 5(23), 1959-1968. doi:10.4236/ce.2014.523220
- Bennis, W., & Nanus, B. (1985). *Leaders The strategies for taking charge*. New York: Harper & Row. doi:10.1002/hrm.3930240409
- Biswas, B., Roy, S. K., & Roy, F. (2020). Students Perception of Mobile Learning during COVID-19 in Bangladesh: University Student Perspective. *Aquademia*, 4(2), ep20023. doi:10.29333/aquademia/8443
- Darma, D. C., Ilmi, Z., Darma, S., & Syaharuddin, Y. (2020). COVID-19 and its Impact on Education: Challenges from Industry 4.0. *Aquademia*, 4(1), ep20025. doi:10.29333/aquademia/8453
- Darma, S., Wijaya, A., & Darma, D. C. (2020). Different Tests for the Existence of Agricultural Cooperatives in Indonesia: Before and After COVID-19. *Asia Life Sciences*, 10(3), 615-628.
- Dewey, J. (1963). *Experience and Education*. New York, NY: Macmillan.
- Dragutinovic, K., & Mitrovic, M. (2019). Differences in the attitudes of fourth grade pupils of the elementary school according to the teaching of physical education in terms of gender. *Journal of Anthropology of Sport and Physical Education*, 3(1), 29-34. doi:10.26773/jaspe.190106
- Fairholm, M. R. (2013). *Putting Your Values to Work: Becoming the Leaders Others Want to Follow*. Santa Barbara, CA: Praeger.
- Faisal, F., & Martin, S. N. (2019). Science education in Indonesia: past, present, and future. *Asia-Pacific Science Education*, 5(4), 1-29. doi:10.1186/s41029-019-0032-0
- Kristanti, N. (2019). Pengaruh Internet of Things (IoT) pada education business model: Studi kasus Universitas Atma Jaya Yogyakarta. *Jurnal Teknologi Informasi*, 13(2), 47-53. doi:10.47111/jti.v13i2.254
- Manan, M. (2015). The implementation of the right to education in Indonesia. *Indonesia Law Review*, 5(1), 51-68. doi:10.15742/ilrev.v5n1.137
- Manz, C. C., & Sims, H. P., Jr. (2001). *The new Superleadership: Leading others to lead themselves*. San Francisco, CA: Berrett-Koehler.
- Maria, S., Darma, D., Amalia, S., Hakim, Y., & Pusriadi, T. (2019). Readiness to face industry 4.0. *International Journal of Scientific & Technology Research*, 8(9), 2363-2368.
- Muhmidayeli, M. (2011). *Filsafat Pendidikan*. Bandung: Refika Aditama.
- Mustaghfiroh, S. (2020). Konsep "Merdeka Belajar": Perspektif Aliran Progresivisme John Dewey. *Jurnal Studi Guru dan Pembelajaran*, 3(1), 141-147. doi:10.30605/jsdp.3.1.2020.248
- Pfeiffer, S. (2017). The Vision of "Industrie 4.0" in the Making—a Case of Future Told, Tamed, and Traded. *Nanoethics* 11, 107-121. doi:10.1007/s11569-016-0280-3
- Prihatmoko, D. (2016). Penerapan Internet of Things (IoT) dalam pembelajaran di Unisnu Jepara. *Jurnal SIMETRIS*, 7(2), 567-574.

- doi:10.24176/simet.v7i2.769
- Sagala, S. (2013). *Etika dan Moralitas Pendidikan Peluang dan Tantangan*. Jakarta: Kencana.
- Toquero, C. M. D. (2020). Will COVID-19 Bring the Armageddon? Expanding Psychosocial Support and Environmental Care: An Autoethnographic Research. *Aquademia*, 4(2), ep20022. doi:10.29333/aquademia/8438
- World Economic Forum. (2016). *The future of jobs. Employment, skills and workforce strategy for the fourth industrial revolution*. Retrieved from: http://www3.weforum.org/docs/WEF_Future_of_Jobs.pdf (Accessed at August 6, 2020).
- World Health Organization. (2020). *Coronavirus disease (COVID-19) advice for the public*. Retrieved from: <https://www.who.int/emergencies/diseases/novel-coronavirus-2019/advice-for-public> (Accessed at July 18, 2020).
- Yamin, M., & Syahrir, S. (2020). Pembangunan pendidikan merdeka belajar (telaah metode pembelajaran). *Jurnal Ilmiah Mandala Education*, 6(1), 126-136. doi:10.36312/jime.v6i1.1121

Guidelines for Authors

Revised October 2017

*** Please use the bookmark function to navigate within the guidelines. ***

When preparing the final version of the manuscripts, either NEW or REVISED authors should strictly follow the guidelines. Manuscripts departing substantially from the guidelines will be returned to the authors for revision or, rejected.

1. UNIFORM REQUIREMENTS

1.1. Overview

The *Journal of Anthropology of Sport and Physical Education* (JASPE) applies the Creative Commons Attribution (CC BY) license to articles and other works it publishes.

There is no charge for submissions and no page charge for accepted manuscripts. However, if the manuscript contains graphics in color, note that printing in color is charged.

JASPE adopts a double-blind approach for peer reviewing in which the reviewer's name is always concealed from the submitting authors as well as the author(s)'s name from the selected reviewers.

JASPE honors six-weeks for an initial decision of manuscript submission.

Authors should submit the manuscripts as one Microsoft Word (.doc) file.

Manuscripts must be provided either in standard UK or US English or Montenegrin language. Chosen language standards should be consistent throughout the manuscripts.

Format the manuscript in A4 paper size; margins are 1 inch or 2.5 cm all around.

Type the whole manuscript double-spaced, justified alignment.

Use Times New Roman font, size eleven (11) point.

Number (Arabic numerals) the pages consecutively (centering at the bottom of each page), beginning with the title page as page 1 and ending with the Figure legend page.

Include line numbers (continuous) for the convenience of the reviewers.

Apart from chapter headings and sub-headings avoid any kind of formatting in the main text of the manuscripts.

1.2. Type & Length

JASPE publishes following types of papers:

Original scientific papers are the results of empirically- or theoretically-based scientific research, which employ scientific methods, and which report experimental or observational aspects of anthropology of sport and physical education from five major fields of anthropology: cultural, global, biological, linguistic and medical. Descriptive analyses or data inferences should include rigorous methodological structure as well as sound theory. Your manuscript should include the following sections: Introduction, Methods, Results, and Discussion.

☒ Open Submissions

☒ Indexed

☒ Peer Reviewed

Original scientific papers should be:

- Up to 3000 words (excluding title, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References);
- A structured abstract of less than 250 words;
- Maximum number of references is 30;
- Maximum combined total of 6 Tables/Figures.

Review papers should provide concise in-depth reviews of both established and new areas, based on a critical examination of the literature, analyzing the various approaches to a specific topic in all aspects of anthropology of sport and physical education from five major fields of anthropology: cultural, global, biological, linguistic and medical.

☒Open Submissions

☒Indexed

☒Peer Reviewed

Review papers should be:

- Up to 6000 words (excluding title, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References);
- A structured abstract of less than 250 words;
- Maximum number of references is 100.

Editorials are written or commissioned by the editors, but suggestions for possible topics and authors are welcome. It could be peer reviewed by two reviewers who may be external or by the Editorial Board.

☐Open Submissions

☒Indexed

☒Peer Reviewed

Editorials should be:

- Up to 1000 words (excluding title, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References);
- A structured abstract of less than 250 words;
- Maximum number of references is 10.

Short reports of experimental work, new methods, or a preliminary report can be accepted as two page papers. Your manuscript should include the following sections: Introduction, Methods, Results, and Discussion.

☒Open Submissions

☒Indexed

☒Peer Reviewed

Short reports should be:

- Up to 1500 words (excluding title, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References);
- A structured abstract of less than 250 words;
- Maximum number of references is 15.

Peer review - fair review provides authors who feel their paper has been unfairly rejected (at any journal) the opportunity to share reviewer comments, explain their concerns, and have their paper reviewed for possible publication in JASPE.

☒Open Submissions

☒Indexed

☐Peer Reviewed

Peer review - fair review should be:

- Up to 1500 words (excluding title, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References);
- A structured abstract of less than 250 words;
- Maximum number of references is 15.

Invited papers and award papers include invited papers from authors with outstanding scientific credentials. Nomination of invited authors is at the discretion of the JASPE editorial board. JASPE also publishes award papers selected by the scientific committee of the publisher's conferences.

☐Open Submissions

☒Indexed

☐Peer Reviewed

Invited papers and award papers should be:

- Up to 3000 words (excluding title, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References);
- A structured abstract of less than 250 words;
- Maximum number of references is 30;
- Maximum combined total of 6 Tables/Figures.

1.3. Submission

JASPE only accepts electronic submission to the e-mail of the Journal Office: **jaspe@ac.me**.

Submitted material includes:

- A manuscript prepared according to the Guidelines for the Authors;
- A signed form that states the study was not previously published, nor has been submitted simultaneously for consideration of publication elsewhere, that states that all of the authors are in agreement with submission of the manuscript to JASPE, and that, for studies that use animal or human individuals, authors must include information regarding their institution's ethics committee, and which identifies the official approval number;
- A signed form that there is no conflict of interest.

Name the files according to the family name of the first author. Authors submitting revised versions of the manuscript can use the identification number of their manuscript as provided by the Journal Office. *See example:*

- ✓ FAMILY NAME-manuscript.doc – (main manuscript file)
- ✓ FAMILY NAME-statement.PDF – (authorship statement)
- ✓ FAMILY NAME-declaration.PDF – (declaration of potential conflict of interest)
- ✓ FAMILY NAME-fig1.tiff – (Figure 1)

1.4. Peer Review Process

A manuscript submitted for publication will be submitted to the review process as long as it fits the following criteria:

- The study was not previously published, nor has been submitted simultaneously for consideration of publication elsewhere;
- All persons listed as authors approved its submission to JASPE;
- Any person cited as a source of personal communication has approved the quote;
- The opinions expressed by the authors are their exclusive responsibility;
- The author signs a formal statement that the submitted manuscript complies with the directions and guidelines of JASPE.

The editors-in-chief and associate editors will make a preliminary analysis regarding the appropriateness, quality, originality and written style/grammar of the submitted manuscript. The editors reserve the right to request additional information, corrections, and guideline compliance before they submit the manuscript to the ad-hoc review process.

JASPE uses ad-hoc reviewers, who volunteer to analyze the merit of the study. Typically, one or two expert reviewers are consulted in a double-blind process. Authors are notified by e-mail when their submission has been accepted (or rejected). Minor changes in the text may be made at the discretion of the editors-in-chief and/or associate editors. Changes can include spelling and grammar in the chosen language, written style, journal citations, and reference guidelines. The author is notified of changes via email. The final version is available to the author for his or her approval before it is published.

1.5. Open Access License and Publisher Copyright Policies



JASPE applies the Creative Commons Attribution (CC BY) license to articles and other works it publishes. If author(s) submit its paper for publication by JASPE, they agree to have the CC BY license applied to their work. Under this Open Access license, the author(s) agree that anyone can reuse their article in whole or part for any purpose, for free, even for commercial purposes. Anyone may copy, distribute, or reuse the content as long as the author(s) and original source are properly cited. This facilitates freedom in re-use and also ensures that JASPE content can be mined without barriers for the needs of research. On the other hand, the author(s) may use content owned by someone else in their article if they have written permission to do so. If the manuscript contains content such as photos, images, figures, tables, audio files, videos, et cetera, that the author(s) do not own, JASPE will require them to provide it with proof that the owner of that content has given them written permission to use it, and has approved of the CC BY license being applied to their content. Otherwise, JASPE will ask the author(s) to remove that content and/or replace it with other content that you own or have such permission to use. JASPE provides a form the author(s) can use to ask for and obtain permission from the owner.

In addition, the author(s) may freely use the content they previously published in a paper through another publisher and they own the rights to that content; however, that's not necessarily the case and it depends on the license that covers the other paper. Some publishers allow free and unrestricted reuse of article content they own, such as under the CC BY

license. Other publishers use licenses that allow reuse only if the same license is applied by the person or publisher reusing the content. If the article was published under a CC BY license or another license that allows free and unrestricted use, the author(s) may use the content in the submitted manuscript provided that the author(s) give proper attribution, as explained above. If the content was published under a more restrictive license, the author(s) must ascertain what rights they have under that license. JASPE advises the author(s) not to include any content in the submitted manuscript which they do not have rights to use, and always give proper attribution.

The editors of JASPE consider plagiarism to be a serious breach of academic ethics. Any author who practices plagiarism (in part or totality) will be suspended for six years from submitting new submissions to JASPE. If such a manuscript is approved and published, public exposure of the article with a printed mark (“plagiarized” or “retracted”) on each page of the published file, as well as suspension for future publication for at least six years, or a period determined by the editorial board. Third party plagiarized authors or institutions will be notified, informing them about the faulty authors. Plagiarism will result in immediate rejection of the manuscript.

JASPE only publishes studies that have been approved by an institutional ethics committee (when a study involves humans or animals). Fail to provide such information prevent its publication. To ensure these requirements, it is essential that submission documentation is complete. If you have not completed this step yet, go to JASPE website and fill out the two required documents: Declaration of Potential Conflict of Interest and Authorship Statement. Whether or not your study uses humans or animals, these documents must be completed and signed by all authors and attached as supplementary files in the originally submitted manuscript.

1.6. After Acceptance

After the manuscript has been accepted, authors will receive a PDF version of the manuscripts for authorization, as it should look in printed version of JASPE. Authors should carefully check for omissions. Reporting errors after this point will not be possible and the Editorial Board will not be eligible for them.

Should there be any errors, authors should report them to the Office e-mail address jaspe@ac.me. If there are not any errors authors should also write a short e-mail stating that they agree with the received version.

1.7. Code of Conduct Ethics Committee of Publications



JASPE is hosting the Code of Conduct Ethics Committee of Publications of the COPE (the Committee on Publication Ethics), which provides a forum for publishers and Editors of scientific journals to discuss issues relating to the integrity of the work submitted to or published in their journals.

2. MANUSCRIPT STRUCTURE

2.1. Title Page

The first page of the manuscripts should be the title page, containing: title, type of publication, running head, authors, affiliations, corresponding author, and manuscript information. *See example:*

Analysis of Dietary Intake and Body Composition of Female Athletes over a Competitive Season

Original Scientific Paper

Diet and Body Composition of Female Athletes

Svetlana Nepocatych¹, Gytis Balilionis¹, Eric K. O'Neal²

¹Elon University, Department of Exercise Science¹, Elon, NC 27215

²University of North Alabama, Department of Health, Physical Education and Recreation, Florence, AL 35632

Corresponding author:

S. Nepocatych

Elon University

Department of Exercise Science

100 Campus Dr.

2525 CB

Elon, NC 27244

United States

E-mail: snepocatych@elon.edu

Word count: 2,946

Word count: 4259

Abstract word count: 211

Number of Tables: 3

2.1.1. Title

Title should be short and informative and the recommended length is no more than 20 words. The title should be in Title Case, written in uppercase and lowercase letters (initial uppercase for all words except articles, conjunctions, short prepositions no longer than four letters etc.) so that first letters of the words in the title are capitalized. Exceptions are words like: “and”, “or”, “between” etc. The word following a colon (:) or a hyphen (-) in the title is always capitalized.

2.1.2. Type of publication

Authors should suggest the type of their submission.

2.1.3. Running head

Short running title should not exceed 50 characters including spaces.

2.1.4. Authors

The form of an author's name is first name, middle initial(s), and last name. In one line list all authors with full names separated by a comma (and space). Avoid any abbreviations of academic or professional titles. If authors belong to different institutions, following a family name of the author there should be a number in superscript designating affiliation.

2.1.5. Affiliations

Affiliation consists of the name of an institution, department, city, country/territory (in this order) to which the author(s) belong and to which the presented / submitted work should be attributed. List all affiliations (each in a separate line) in the order corresponding to the list of authors. Affiliations must be written in English, so carefully check the official English translation of the names of institutions and departments.

Only if there is more than one affiliation, should a number be given to each affiliation in order of appearance. This number should be written in superscript at the beginning of the line, separated from corresponding affiliation with a space. This number should also be put after corresponding name of the author, in superscript with no space in between.

If an author belongs to more than one institution, all corresponding superscript digits, separated with a comma with no space in between, should be present behind the family name of this author.

In case all authors belong to the same institution affiliation numbering is not needed.

Whenever possible expand your authors' affiliations with departments, or some other, specific and lower levels of organization.

2.1.6. Corresponding author

Corresponding author's name with full postal address in English and e-mail address should appear, after the affiliations. It is preferred that submitted address is institutional and not private. Corresponding author's name should include only initials of the first and middle names separated by a full stop (and a space) and the last name. Postal address should be written in the following line in sentence case. Parts of the address should be separated by a comma instead of a line break. E-mail (if possible) should be placed in the line following the postal address. Author should clearly state whether or not the e-mail should be published.

2.1.7. Manuscript information

All authors are required to provide word count (excluding title page, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References), the Abstract word count, the number of Tables, and the number of Figures.

2.2. Abstract

The second page of the manuscripts should be the abstract and key words. It should be placed on second page of the manuscripts after the standard title written in upper and lower case letters, bold.

Since abstract is independent part of your paper, all abbreviations used in the abstract should also be explained in it. If an abbreviation is used, the term should always be first written in full with the abbreviation in parentheses immediately after it. Abstract should not have any special headings (e.g., Aim, Results...).

Authors should provide up to six key words that capture the main topics of the article. Terms from the Medical Subject Headings (MeSH) list of Index Medicus are recommended to be used.

Key words should be placed on the second page of the manuscript right below the abstract, written in italic. Separate each key word by a comma (and a space). Do not put a full stop after the last key word. *See example:*

Abstract

Results of the analysis of

Key words: *spatial memory, blind, transfer of learning, feedback*

2.3. Main Chapters

Starting from the third page of the manuscripts, it should be the main chapters. Depending on the type of publication main manuscript chapters may vary. The general outline is: Introduction, Methods, Results, Discussion, Acknowledgements (optional), Conflict of Interest (optional), and Title, Author's Affiliations, Abstract and Key words must be in English (for both each chosen language of full paper). However, this scheme may not be suitable for reviews or publications from some areas and authors should then adjust their chapters accordingly but use the general outline as much as possible.

2.3.1. Headings

Main chapter headings: written in bold and in Title Case. *See example:*

✓ **Methods**

Sub-headings: written in italic and in normal sentence case. Do not put a full stop or any other sign at the end of the title. Do not create more than one level of sub-heading. *See example:*

✓ *Table position of the research football team*

2.3.2 Ethics

When reporting experiments on human subjects, there must be a declaration of Ethics compliance. Inclusion of a statement such as follow in Methods section will be understood by the Editor as authors' affirmation of compliance: "This study was approved in advance by [name of committee and/or its institutional sponsor]. Each participant voluntarily provided written informed consent before participating." Authors that fail to submit an Ethics statement will be asked to resubmit the manuscripts, which may delay publication.

2.3.3 Statistics reporting

JASPE encourages authors to report precise p-values. When possible, quantify findings and present them with appropriate indicators of measurement error or uncertainty (such as confidence intervals). Use normal text (i.e., non-capitalized, non-italic) for statistical term "p".

2.3.4. 'Acknowledgements' and 'Conflict of Interest' (optional)

All contributors who do not meet the criteria for authorship should be listed in the 'Acknowledgements' section. If applicable, in 'Conflict of Interest' section, authors must clearly disclose any grants, financial or material supports, or any sort of technical assistances from an institution, organization, group or an individual that might be perceived as leading to a conflict of interest.

2.4. References

References should be placed on a new page after the standard title written in upper and lower case letters, bold.

All information needed for each type of must be present as specified in guidelines. Authors are solely responsible for accuracy of each reference. Use authoritative source for information such as Web of Science, Medline, or PubMed to check the validity of citations.

2.4.1. References style

JASPE adheres to the American Psychological Association 6th Edition reference style. Check "American Psychological Association. (2009). Concise rules of APA style. American Psychological Association." to ensure the manuscripts conform to this reference style. Authors using EndNote® to organize the references must convert the citations and bibliography to plain text before submission.

2.4.2. Examples for Reference citations

One work by one author

- ✓ In one study (Reilly, 1997), soccer players
- ✓ In the study by Reilly (1997), soccer players
- ✓ In 1997, Reilly's study of soccer players

Works by two authors

- ✓ Duffield and Marino (2007) studied
- ✓ In one study (Duffield & Marino, 2007), soccer players
- ✓ In 2007, Duffield and Marino's study of soccer players

Works by three to five authors: cite all the author names the first time the reference occurs and then subsequently include only the first author followed by et al.

- ✓ First citation: Bangsbo, Iaia, and Krstrup (2008) stated that
- ✓ Subsequent citation: Bangsbo et al. (2008) stated that

Works by six or more authors: cite only the name of the first author followed by et al. and the year

- ✓ Krstrup et al. (2003) studied
- ✓ In one study (Krstrup et al., 2003), soccer players

Two or more works in the same parenthetical citation: Citation of two or more works in the same parentheses should be listed in the order they appear in the reference list (i.e., alphabetically, then chronologically)

- ✓ Several studies (Bangsbo et al., 2008; Duffield & Marino, 2007; Reilly, 1997) suggest that

2.4.3. Examples for Reference list

Journal article (print):

- Nepocaty, S., Balilionis, G., & O'Neal, E. K. (2017). Analysis of dietary intake and body composition of female athletes over a competitive season. *Montenegrin Journal of Sports Science and Medicine*, 6(2), 57-65. doi: 10.26773/mjssm.2017.09.008
- Duffield, R., & Marino, F. E. (2007). Effects of pre-cooling procedures on intermittent-sprint exercise performance in warm conditions. *European Journal of Applied Physiology*, 100(6), 727-735. doi: 10.1007/s00421-007-0468-x
- Krstrup, P., Mohr, M., Amstrup, T., Rysgaard, T., Johansen, J., Steensberg, A., Bangsbo, J. (2003). The yo-yo intermittent recovery test: physiological response, reliability, and validity. *Medicine and Science in Sports and Exercise*, 35(4), 697-705. doi: 10.1249/01.MSS.0000058441.94520.32

Journal article (online; electronic version of print source):

- Williams, R. (2016). Krishna's Neglected Responsibilities: Religious devotion and social critique in eighteenth-century North India [Electronic version]. *Modern Asian Studies*, 50(5), 1403-1440. doi:10.1017/S0026749X14000444

Journal article (online; electronic only):

- Chantavanich, S. (2003, October). Recent research on human trafficking. *Kyoto Review of Southeast Asia*, 4. Retrieved November 15, 2005, from <http://kyotoreview.cseas.kyoto-u.ac.jp/issue/issue3/index.html>

Conference paper:

- Pasadilla, G. O., & Milo, M. (2005, June 27). *Effect of liberalization on banking competition*. Paper presented at the conference on Policies to Strengthen Productivity in the Philippines, Manila, Philippines. Retrieved August 23, 2006, from <http://siteresources.worldbank.org/INTPHILIPPINES/Resources/Pasadilla.pdf>

Encyclopedia entry (print, with author):

- Pittau, J. (1983). Meiji constitution. In *Kodansha encyclopedia of Japan* (Vol. 2, pp. 1-3). Tokyo: Kodansha.

Encyclopedia entry (online, no author):

- Ethnology. (2005, July). In *The Columbia encyclopedia* (6th ed.). New York: Columbia University Press. Retrieved November 21, 2005, from <http://www.bartleby.com/65/et/ethnolog.html>

Thesis and dissertation:

- Pyun, D. Y. (2006). *The proposed model of attitude toward advertising through sport*. Unpublished Doctoral Dissertation. Tallahassee, FL: The Florida State University.

Book:

Borg, G. (1998). *Borg's perceived exertion and pain scales*: Human kinetics.

Chapter of a book:

Kellmann, M. (2012). Chapter 31-Overtraining and recovery: Chapter taken from Routledge Handbook of Applied Sport Psychology ISBN: 978-0-203-85104-3 *Routledge Online Studies on the Olympic and Paralympic Games* (Vol. 1, pp. 292-302).

Reference to an internet source:

Agency. (2007). Water for Health: Hydration Best Practice Toolkit for Hospitals and Healthcare. Retrieved 10/29, 2013, from www.rcn.org.uk/newsevents/hydration

2.5. Tables

All tables should be included in the main manuscript file, each on a separate page right after the Reference section.

Tables should be presented as standard MS Word tables.

Number (Arabic) tables consecutively in the order of their first citation in the text.

Tables and table headings should be completely intelligible without reference to the text. Give each column a short or abbreviated heading. Authors should place explanatory matter in footnotes, not in the heading. All abbreviations appearing in a table and not considered standard must be explained in a footnote of that table. Avoid any shading or coloring in your tables and be sure that each table is cited in the text.

If you use data from another published or unpublished source, it is the authors' responsibility to obtain permission and acknowledge them fully.

2.5.1. Table heading

Table heading should be written above the table, in Title Case, and without a full stop at the end of the heading. Do not use suffix letters (e.g., Table 1a, 1b, 1c); instead, combine the related tables. *See example:*

✓ **Table 1.** Repeated Sprint Time Following Ingestion of Carbohydrate-Electrolyte Beverage

2.5.2. Table sub-heading

All text appearing in tables should be written beginning only with first letter of the first word in all capitals, i.e., all words for variable names, column headings etc. in tables should start with the first letter in all capitals. Avoid any formatting (e.g., bold, italic, underline) in tables.

2.5.3. Table footnotes

Table footnotes should be written below the table.

General notes explain, qualify or provide information about the table as a whole. Put explanations of abbreviations, symbols, etc. here. General notes are designated by the word *Note* (italicized) followed by a period.

✓ *Note.* CI: confidence interval; Con: control group; CE: carbohydrate-electrolyte group.

Specific notes explain, qualify or provide information about a particular column, row, or individual entry. To indicate specific notes, use superscript lowercase letters (e.g. ^{a,b,c}), and order the superscripts from left to right, top to bottom. Each table's first footnote must be the superscript ^a.

✓ ^aOne participant was diagnosed with heat illness and n = 19.^bn = 20.

Probability notes provide the reader with the results of the tests for statistical significance. Probability notes must be indicated with consecutive use of the following symbols: * † ‡ § ¶ || etc.

✓ *P<0.05, †p<0.01.

2.5.4. Table citation

In the text, tables should be cited as full words. *See example:*

- ✓ Table 1 (first letter in all capitals and no full stop)
- ✓ ...as shown in Tables 1 and 3. (citing more tables at once)
- ✓ ...result has shown (Tables 1-3) that... (citing more tables at once)
- ✓in our results (Tables 1, 2 and 5)... (citing more tables at once)

2.6. Figures

On the last separate page of the main manuscript file, authors should place the legends of all the figures submitted separately.

All graphic materials should be of sufficient quality for print with a minimum resolution of 600 dpi. JASPE prefers TIFF, EPS and PNG formats.

If a figure has been published previously, acknowledge the original source and submit a written permission from the copyright holder to reproduce the material. Permission is required irrespective of authorship or publisher except for documents in the public domain. If photographs of people are used, either the subjects must not be identifiable or their pictures must be accompanied by written permission to use the photograph whenever possible permission for publication should be obtained.

Figures and figure legends should be completely intelligible without reference to the text.

The price of printing in color is 50 EUR per page as printed in an issue of JASPE.

2.6.1. Figure legends

Figures should not contain footnotes. All information, including explanations of abbreviations must be present in figure legends. Figure legends should be written below the figure, in sentence case. *See example:*

- ✓ **Figure 1.** Changes in accuracy of instep football kick measured before and after fatigued. SR – resting state, SF – state of fatigue, * $p > 0.01$, † $p > 0.05$.

2.6.2. Figure citation

All graphic materials should be referred to as Figures in the text. Figures are cited in the text as full words. *See example:*

- ✓ Figure 1
 - × figure 1
 - × Figure 1.
 - ✓ ...exhibit greater variance than the year before (Figure 2). Therefore...
 - ✓ ...as shown in Figures 1 and 3. (citing more figures at once)
 - ✓ ...result has shown (Figures 1-3) that... (citing more figures at once)
 - ✓in our results (Figures 1, 2 and 5)... (citing more figures at once)

2.6.3. Sub-figures

If there is a figure divided in several sub-figures, each sub-figure should be marked with a small letter, starting with a, b, c etc. The letter should be marked for each subfigure in a logical and consistent way. *See example:*

- ✓ Figure 1a
- ✓ ...in Figures 1a and b we can...
- ✓ ...data represent (Figures 1a-d)...

2.7. Scientific Terminology

All units of measures should conform to the International System of Units (SI).

Measurements of length, height, weight, and volume should be reported in metric units (meter, kilogram, or liter) or their decimal multiples.

Decimal places in English language are separated with a full stop and not with a comma. Thousands are separated with a comma.

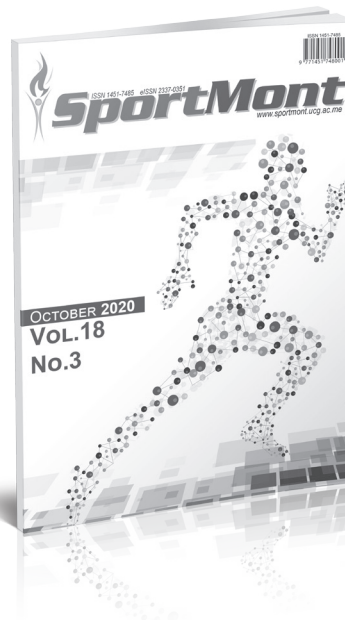
Percentage	Degrees	All other units of measure	Ratios	Decimal numbers
✓ 10%	✓ 10°	✓ 10 kg	✓ 12:2	✓ 0.056
× 10 %	× 10 °	× 10kg	× 12 : 2	× .056
Signs should be placed immediately preceding the relevant number.				
✓ 45±3.4	✓ p<0.01	✓ males >30 years of age		
× 45 ± 3.4	× p < 0.01	× males > 30 years of age		

2.8. Latin Names

Latin names of species, families etc. should be written in italics (even in titles). If you mention Latin names in your abstract they should be written in non-italic since the rest of the text in abstract is in italic. The first time the name of a species appears in the text both genus and species must be present; later on in the text it is possible to use genus abbreviations. See example:

✓ First time appearing: *musculus biceps brachii*

Abbreviated: *m. biceps brachii*



Publication date: Winter issue – February 2021
Summer issue – June 2021
Autumn issue – October 2021

University of Montenegro

Cetinjska br.2
81 000 Podgorica
Montenegro

E-mail:
rektor@ac.me
rektorat@ac.me

Phone: + 382 20 414 255
Fax: + 382 20 414 230

Web:
www.ucg.ac.me



The University of Montenegro is the leading higher education and research institution in Montenegro. It is a public institution, established by the state, operating as a unique legal entity represented by the Rector. It is an integrated university organized on the model of the most European universities. Organizational units are competent for provision of study programmes, scientific-research and artistic work, use of allocated funds and membership in professional associations.

Since its foundation, the University of Montenegro has continuously been conducting reforms in the area of education and research, while since 2003 in line with the trends in EHEA. After adoption of the Bologna Declaration, University of Montenegro organized systematic preparation of documents aligned with it. Already in 2003, the experimental teaching programme started and today, all studies are organised in line with the Bologna principles. During the last two years systematic reforms of the University's study programmes have been conducted in order to harmonize domestic higher education system with European standards and market needs to highest extent.

The University of Montenegro has unique academic, business and development objectives. It comprises 19 faculties and two research institutes. The seat of the UoM is in Podgorica, the capital city, while university units are located in eight Montenegrin towns. The University support services and centers (advisory services, accounting department, international cooperation, career orientation) are located in the Rectorate.

Academic community of University of Montenegro is aware of the importance of its functioning for further development of the state and wider region. It has been so far, and will be in the future, the leader in processes of social and cultural changes, along with the economic development.

In the aspect of attaining its mission, University of Montenegro is oriented towards the priority social needs of the time in which it accomplishes its mission; open for all the students and staff exclusively based on their knowledge and abilities; dedicated to preservation of multicultural and multi-ethnic society in Montenegro; entrepreneurial in stimulating social and economic application of supreme achievements within the scope of its activities.

In 2015/16 there were a total of 1.192 employees at UoM, 845 of which were engaged in teaching. In the same year there were 20.236 students registered at all three cycles of studies.

Internationalization is high on the agenda of UoM priorities, thus it has participated in a number of international projects – over 50 projects funded under the Tempus programme, over 15 Erasmus Mundus Action 2 projects for student mobility, a number of projects under FP7 funding scheme or IPA supported projects, Erasmus + capacity building and International credit mobility projects and other.

For more information about University of Montenegro, please visit our website www.ucg.ac.me or send e-mail to pr.centar@ac.me.





USEFUL CONTACTS

Editorial enquiries and journal proposals:

Dusko Bjelica
Stevo Popovic
Editors-in-Chief
Email: office@mjssm.me

Selcuk Akpinar
Executive Editor
Email: office@mjssm.me

Marketing enquiries:

Jovan Gardasevic
Marketing Manager
Email: administration@mjssm.me

Sports Science and Medicine Journals from Montenegrin Sports Academy

We have expanded the quality of our journals considerably over the past years and can now claim to be the market leader in terms of breadth of coverage.

As we continue to increase the quality of our publications across the field, we hope that you will continue to regard MSA journals as authoritative and stimulating sources for your research. We would be delighted to receive your comments and suggestions, mostly due to the reason your proposals are always welcome.

Look Inside!



Sport Mont Journal

Editors-in-Chief: **Dusko Bjelica**, Montenegro; **Zoran Milosevic**, Serbia
Managing Editor: **Jovan Gardasevic**, Montenegro

Volume 18, 2020, 3 issues per year; Print ISSN: 1451-7485, Online ISSN: 2337-0351

Sport Mont Journal is a scientific journal that provides: Open-access and freely accessible online; Fast publication time; Peer review by expert, practicing researchers; Post-publication tools to indicate quality and impact; Community-based dialogue on articles; Worldwide media coverage. SMJ is published three times a year, in February, June and October of each year. SMJ publishes original scientific papers, review papers, editorials, short reports, peer review - fair review, as well as invited papers and award papers in the fields of Sports Science and Medicine, as well as it can function as an open discussion forum on significant issues of current interest.

www.sportmont.ucg.ac.me



Montenegrin Journal of Sports Science and Medicine

Editors-in-Chief: **Dusko Bjelica**, Montenegro; **Stevo Popovic**, Montenegro
Executive Editor: **Selcuk Akpinar**, Turkey
Associate Editors: **Mehmet Uygur**, USA; **Catalina Casaru**, USA; and **Predrag Bozic**, Serbia

Volume 9, 2020, 2 issues per year; Print ISSN: 1800-8755, Online ISSN: 1800-8763

Montenegrin Journal of Sports Science and Medicine (MJSSM) is published biannually, in September and March of each year. MJSSM publishes original scientific papers, review papers, editorials, short reports, peer review - fair review, as well as invited papers and award papers in the fields of Sports Science and Medicine, as well as it can function as an open discussion forum on significant issues of current interest. MJSSM covers all aspects of sports science and medicine; all clinical aspects of exercise, health, and sport; exercise physiology and biophysical investigation of sports performance; sport biomechanics; sports nutrition; rehabilitation, physiotherapy; sports psychology; sport pedagogy, sport history, sport philosophy, sport sociology, sport management; and all aspects of scientific support of the sports coaches from the natural, social and humanistic side.

www.mjssm.me



MONTENEGRIN **J**OURNAL
OF **S**PORTS **S**CIENCE
AND **M**EDICINE



SSN 1800-8755

CALL FOR CONTRIBUTIONS

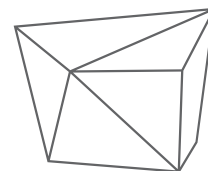
- Open-access and freely accessible online;
- Fast publication time;
- Peer review by expert, practicing researchers;
- Post-publication tools to indicate quality and impact;
- Community-based dialogue on articles;
- Worldwide media coverage.

MJSSM covers all aspects of sports science and medicine; all clinical aspects of exercise, health, and sport; exercise physiology and biophysical investigation of sports performance; sport biomechanics; sports nutrition; rehabilitation, physiotherapy; sports psychology; sport pedagogy, sport history, sport philosophy, sport sociology, sport management; and all aspects of scientific support of the sports coaches from the natural, social and humanistic side.

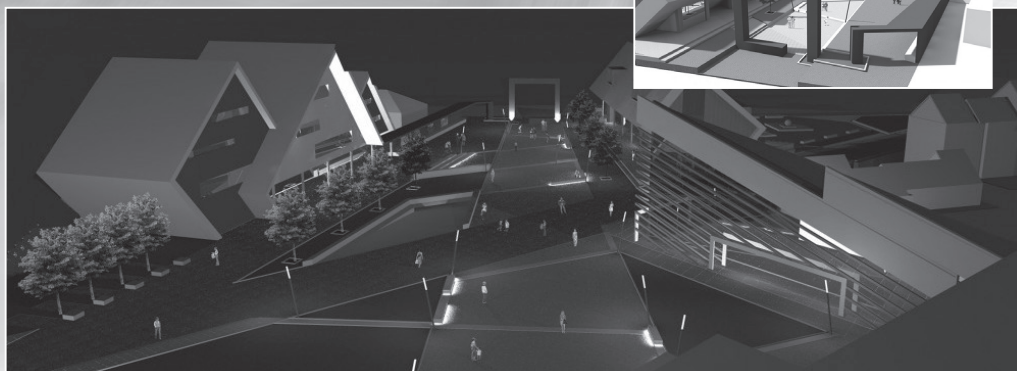
Prospective authors should submit manuscripts for consideration in Microsoft Word-compatible format. For more complete descriptions and submission instructions, please access the Guidelines for Authors pages at the MJSSM website: <http://www.mjssm.me/?sekcija=page&p=51>. Contributors are urged to read MJSSM's guidelines for the authors carefully before submitting manuscripts. Manuscripts submissions should be sent in electronic format to office@mjssm.me or contact following Editors:

Dusko BJELICA, Editor-in Chief – dbjelica@ucg.ac.me
 Stevo POPOVIC, Editor-in Chief – stevop@ucg.ac.me
 Selçuk AKPINAR, Executive Editor – sakpinar@nevsehir.edu.tr

Publication date: Spring issue – March 2021
Autumn issue – September 2021



The goal of establishment of our institution is the education highly qualified professional cadre based on the best knowledge of the theory and practice in the world, and its application to the development and implementation of plans and projects in the space - as a basic condition for the quality valorization, programming, management and protection of natural and inherited built environment. In this way conceptualized school forms internationally experts in all areas of creativity - in the field of urban planning, architecture, construction and design - which includes the ability to create useful objects, architectural forms of all categories, urban and vacant space at different levels. Such qualified cadre are the spiritus movens of development of culture and technology in the modern world.



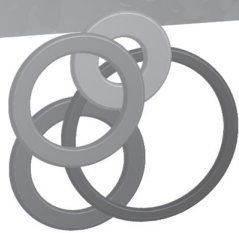
Bulevar Džordža Vašingtona bb
81000 Podgorica, Montenegro
Tel +382 20 269 262
Fax +382 20 81 269 317

*We follow the highest academic
and professional standards*

arhitektura@ac.me
www.ucg.ac.me/af



**EVROPSKA
GRAĐANSKA
PERSPEKTIVA**



**Montenegrin Journal
of Sports Science and Medicine**
www.mjssm.me

ISSN 1800-8755



9 771800 875006

SEPTEMBER 2020



Vol.9

No.2

Vol.9

No.2



**Faculty for sport
and physical education
NIKŠIĆ**

Phone: + 382 40 235 204; Fax: + 382 40 235 207, +382 40 235 200
E-mail: fakultetzasportnk@ac.me; Web: www.ucg.ac.me/sport

Znanje i zdravlje!



Univerzitet Crne Gore

UNIVERZITET CRNE GORE INSTITUT ZA BIOLOGIJU MORA



University of Montenegro – Institute for marine biology is located in Kotor, Montenegro. Since its establishment in 1961, the Institute performed comprehensive research of the marine and coastal area, which has its wide impact to the environmental protection, pollution-prevention and practical application. Core competencies of the Institute are focused on research in the fields of marine conservation, ichthyology and marine fisheries, marine chemistry, aquaculture, plankton research, neuro and eco-physiology. The main research area is investigating and protection of Adriatic sea with special interest of South Adriatic area. Institute for marine biology have a wide range of international cooperation with Marine research institutions and Universities all over Mediterranean area through a numerous EU funded scientific projects.

All over the year Institute is looking to hire a young students from the field of general biology, marine biology, marine chemistry, molecular biology or similar disciplines on voluntary basis to work with us. We need opportunity for international internship or MSc or PhD thesis that could be performed on Institute in our 5 different labs: Fisheries and ichthyology, Aquaculture, Marine chemistry, Plankton and sea water quality and Benthos and marine conservation.

Every year Institute organize several summer schools and workshop for interested students, MSc and PhD candidates. From 01-05 July 2019 we will organize Summer school "Blue Growth: emerging technologies, trends and opportunities" in frame of InnoBlueGrowth Project who is financed by Interreg Med programme. Through the specific theme courses, workshops and working labs offered – covering different areas of the blue economy – the Summer School aims at encouraging young people involvement in blue economy sectors by offering high-quality technical knowledge and fostering their entrepreneurial spirit. The Summer School will facilitate fruitful exchanges and a stronger understanding among a variety of actors coming from different Mediterranean countries with diverse profiles, including representatives from the academia, the public and private sectors, but also potential funders and investors. These activities will count on specific team building activities for participants as well to reinforce interpersonal skills and foster cohesion among blue academia and sectors.

If You are interested apply on the following link: <https://www.ucg.ac.me/objava/blog/1221/objava/45392-ljetnja-skola-plavi-rast-nove-tehnologije-trendovi-i-mogucnosti>

University of Montenegro – Institute for marine biology

Dobrota bb, P.o. box 69. 83550 Kotor, Montenegro

ibmk@ucg.ac.me

+38232334569

www.ucg.ac.me/ibm

Faculty of Economics *University of Montenegro*

The Faculty of Economics celebrated its 57th anniversary this year, and it is the oldest higher education institution in the country. Since its establishment, 8,630 students graduated at our Faculty.

Today, Faculty of Economics is a largely interdisciplinary institution, characterized by expressed dynamism in its work. Employees at the Faculty are dedicated to constant improvements and enhancements, all in accordance with the needs brought by the changes.

We provide our students with the best theoretical and practical knowledge, enabling them to develop critical spirit in approaching economic phenomena and solving concrete problems in daily work. From September 2017, at the Faculty, the new generation will start a 3 + 2 + 3 study, which will improve the quality of studying.



Development of Faculty of Economics in the coming period will follow the vision of development of the University of Montenegro, pursuing full achievement of its mission

Comprehensive literature, contemporary authors and works have always been imperative in creation of new academic directions at Faculty of Economics, which will form the basis of our future.

Faculty and its employees are dedicated to developing interest in strengthening the entrepreneurial initiative, creative and interdisciplinary approach among young people, using modern teaching and research methods. In this regard, the Faculty has modern textbooks and adequate IT technology, which supports the objectives set.





UNIVERSITY OF MONTENEGRO FACULTY OF MECHANICAL ENGINEERING Podgorica

www.ucg.ac.me/mf



Mechanical engineering studies in Montenegro started during the school year 1970/71. On April 15th, within the Technical Faculty, the Department of Mechanical Engineering was formed. The Department of Mechanical Engineering of the Technical Faculty was transformed in 1978 into the Faculty of Mechanical Engineering, within the University "Veljko Vlahović". Since 1992 the Faculty of Mechanical Engineering is an autonomous University unit of the University of Montenegro. It is situated in Podgorica.

The University of Montenegro is the only state university in the country, and the Faculty of Mechanical Engineering is the only faculty in Montenegro from the field of mechanical engineering.

Activities of the Faculty of Mechanical Engineering can be divided into three fields: teaching, scientific-research work and professional work.

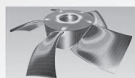
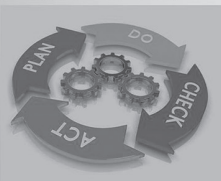
Two study programmes were accredited within the Faculty of Mechanical Engineering:

- Academic study programme MECHANICAL ENGINEERING
- Academic study programme ROAD TRAFFIC

The study programmes are realised according to the Bologna system of studies in accordance to the formula 3+2+3.

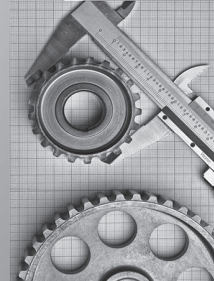
On the study program Mechanical Engineering it is possible to study next modules:

- Mechanical Engineering – Production
- Applied Mechanics and Construction
- Energetics
- Energy Efficiency
- Mechatronics
- Quality



At the Faculty of Mechanical Engineering, as organisational units, there are centres and laboratories through which scientific-re search and professional work is done:

- Centre for Energetics
- Centre for Vehicles
- Centre for Quality
- Centre for Construction Mechanics
- Centre for Traffic and Mechanical Engineering Expertise
- Centre for transport machines and metal constructions
- 3D Centre
- Didactic Centre – Centre for Automation and Mechanomics training
- European Information and Innovation Centre
- Cooperation Training Centre
- Laboratory for Metal Testing
- Laboratory for Turbulent Flow Studies
- Laboratory for Vehicle Testing
- Laboratory for Attesting of Devices on the Technical Examination Line



**CRNOGORSKI OLIMPIJSKI KOMITET
MONTENEGRIN OLYMPIC COMMITTEE**

CIP – Каталогизација у публикацији
Национална библиотека Црне Горе, Цетиње

ISSN 2536-569X
COBISS.CG-ID 33826832

FIEP Congress

MSA Conference 2021

18th Annual Scientific Conference and 16th FIEP European Congress Sport, Physical Education, Physical Activity and Health: Contemporary perspectives

<http://www.csakademija.me/conference/>



8th - 11th April 2021,
Dubrovnik - Croatia